

BASIC MATHEMATICS

Student's book

Senior 5

Version edited in 2026

Copyright

© 2026 Rwanda Basic Education Board, Kigali.

All rights reserved.

This book is the property of Rwanda Basic Education Board.

Credit should be given to REB when the source of this book is quoted.

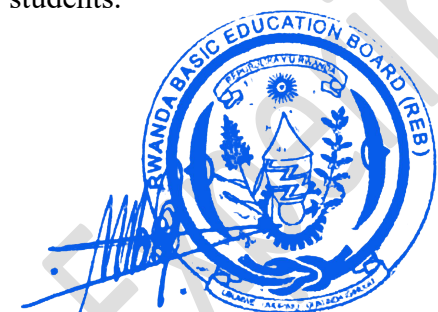
Experimental version

FOREWORD

Dear student,

The Rwanda Basic Education Board (REB) is honored to present the Basic Mathematics book for Senior five (S5). This Textbook is designed to support the effective implementation of the Competence-Based Curriculum (CBC), ensuring learner-centered teaching practices that align with Rwanda's vision of empowering learners to reach their full potential, actively participate in society, and pursue meaningful careers.

As Rwanda advances towards building a knowledge-based economy, education plays a pivotal role. Teachers are central to this transformation, guiding students to connect classroom knowledge with real-life experiences, thus fostering individual growth and national development. This student book serves as a comprehensive resource for teaching Basic Mathematics at Senior five in the following pathways: **Languages, Art and Humanities**. The subject is an integral part of the CBC, aimed at equipping students with essential knowledge, skills, values, and attitudes necessary for becoming responsible and competent citizens. Through this subject, students are encouraged to develop problem-solving abilities and contribute positively to their communities. In pursuit of quality education, the Government of Rwanda emphasizes the importance of using teaching and learning materials that are aligned with the curriculum. Although this guide provides answers to all student activities, teachers are encouraged to independently work through the exercises to better evaluate and understand students' responses. I sincerely thank everyone who contributed to the refinement of this textbook. Special recognition goes to the REB team who coordinated the entire process. We thank all partners, lecturers, teachers, and individuals who were involved in the process from the start to the end. All feedback was highly valued and will help improve learning outcomes of our students.



Dr. MBARUSHIMANA Nelson
Director General

ACKNOWLEDGEMENT

I would like to extend my sincere appreciation to everyone who contributed to the refinement of the Mathematics student's book for Senior Five (S5). This achievement would not have been possible without the active involvement of various stakeholders.

Special thanks go to Rwanda Basic Education Board (REB) staff, subject experts, university lecturers, and teachers for their valuable work in the adaptation of this textbook.

I am also grateful to all our education development partners whose support was instrumental throughout the process.



MURUNGI Joan

Head of CTLR department

Table of Contents

UNIT 1: INTRODUCTION TO LOGICS.....	vii
1.1. Simple statement	2
1.2. Compound Statements.....	4
1.3. Truth values and truth tables	1
1.4. Negation “not”	3
1.5. Conjunction “and”	5
1.6. Disjunction “or”	7
1.7. Conditional statement “if ... then.....	10
1.8. Converse and contra-positive of a conditional statement	13
1.9. Inverse of a conditional statement.....	15
1.10. Bi-conditional statement “if and only if”	16
1.11. Tautology and contradiction	19
1.12. Universal quantifier “for all”	20
1.13. Existence quantifier “there exists”	21
1.14. Negation of quantifiers	22
1.15. Hypothetical syllogism.....	24
1.16. Affirming the Antecedent.....	26
1.17. Denying the Consequent.....	27
End of unit assessment.....	28
UNIT 2:POINTS , LINES AND GEOMETRIC SHAPES IN 2D	30
2.1. Cartesian coordinates of a point.....	31
2.2. Distances between two points.....	33
2.3. Midpoint of a line segment	35
2.4. Vector in 2D.....	37
2.5. Dot product of vectors in 2 D	4
2.6. Properties of dot product in 2D	5
2.8. Equation of a straight line given 2 points	3
2.10. Position of a point to the line.....	8
2.11. Position of two lines	11
2.12. Condition of parallelism	13
2.13. Angles between two lines	15
2.14. Condition of perpendicularity	17
2.16. Perimeter and area of geometric shapes in 2D: Square, Triangle, Rectangle.....	3

2.17. Perimeter and area of geometric shapes in 2D: Parallelogram, Pentagon and Hexagon	5
END OF UNIT ASSESSMENT	7
UNIT 3: GRAPHS AND FUNCTIONS	9
3.1. Generalities on numerical functions.....	11
3.2. Constant function and identity functions	2
3.3. Monomial and polynomial functions.....	4
3.4. Rational and irrational functions	5
3.5. Domain of definition of polynomial functions	7
3.6. Domain of definition of rational functions.....	12
3.7. Domain of definition of irrational functions.....	17
3.8. Parity of a function (odd or even).....	20
3.9. Addition and subtraction of functions	24
3.11. Composite function	2
3.12. Inverse of a function	3
3.13. Graphical representation and interpretation of linear function	3
3.14. Graphical representation and interpretation of quadratic function.....	3
3.16. Price as function of quantity demanded and the cost function.....	8
3.17 The Revenue function and the profit function	1
UNIT 4: NUMBER PATTERNS OR SEQUENCES	2
4.1. Definition of sequences.....	3
4.3. n^{th} term of an arithmetic sequence	9
4.4. Arithmetic mean.....	11
4.5. Sum of n first terms of an arithmetic sequence.....	12
4.6. Geometric sequence	17
4.7. N^{th} term of a geometric sequence.....	18
4.8. Geometric mean	21
4.9. Sum of n first terms of a geometric sequence.....	22
4.10. Application of sequences in solving real life problems	25
END OF UNIT ASSESSMENT	29
BIBLIOGRAPHY.....	31



UNIT 1: INTRODUCTION TO LOGICS

Key unit competence: Use Mathematical logic as a tool of reasoning and decision making in daily life.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding:

- Describe a simple statement, compound statement, and their truth values.
- Explain the function of logical connectives: Negation, Conjunction, Disjunction, Conditional, and Bi-conditional.
- Describe the concepts of converse, inverse, and contrapositive.
- Identify tautologies and contradictions. State the meaning of universal and existential quantifiers.
- Outline the structure of a hypothetical syllogism and its valid forms (affirming the antecedent, denying the consequent).

Skills:

- Differentiate between simple and compound statements.
- Construct truth tables for all logical connectives and compound statements.
- Convert everyday language statements into logical notation and vice versa.
- Analyze a truth table to classify a statement as a tautology, contradiction, or contingency.
- Apply rules to negate quantified statements correctly.
- Demonstrate the validity of a hypothetical syllogism using logical principles.
- Evaluate real-world arguments to identify valid reasoning or fallacies.

Attitudes and values:

- Show carefulness in distinguishing between factual statements and opinions.
- Value precision in the use of logical language (e.g., “**and**” vs “**or**”).
- Appreciate the role of clear definitions and logical structure in avoiding misunderstandings.
- Develop a critical mindset towards unsubstantiated claims

Introduction

Consider how you make decisions every single day. You might tell yourself, "If it is raining, then I will take an umbrella." Or perhaps, "I will go to the movies if it is Saturday and I have finished my homework." These everyday thoughts follow a specific structure of reasoning. We use logic constantly without realizing it. In mathematics and computer science, we formalize this thinking to ensure our conclusions are valid. We look at statements that are clearly either true or false and analyze how they connect to form complex arguments. Just as a builder needs a strong foundation to build a house, a mathematician needs logic to build proofs. Logic is a branch of mathematics that deals with reasoning and the study of valid arguments.



Introductory activity

Read the following situation carefully and answer to the given questions: During a discussion in class, a student says:

"If it is raining, then the ground is wet."

Another student says: "It is raining today."

Then a third student concludes: "Therefore, the ground is wet." The teacher asks the class if this conclusion is correct.

Questions:

- 1) Identify the statements in the situation above.
- 2) Do you think the conclusion "the ground is wet" is correct? Explain your answer.
- 3) Can you think of another situation in daily life where someone uses similar reasoning?
- 4) Why is it important to check whether reasoning is correct before making a conclusion?

1.1. Simple statement



Activity 1.1

From the following expressions, give your answer by using "**true**" or "**false**".

- 1) Every integer greater than 1 is positive.
- 2) Kampala is in Rwanda.
- 3) How old are you?
- 4) Every liquid is water.
- 5) Write down the names of the president of Uganda.
- 6) $1 - x^2 = 0$.
- 7) Rwanda is an African country or Rwanda is a member of Commonwealth.



Content Summary

A sentence which is either true or false but not both simultaneously is named **statement** or **proposition**. In the context of logic, a proposition or a statement is the sentence in the grammatical sense conveying a situation which is neither imperative, interrogative nor exclamatory. A **simple statement** is a sentence that expresses a complete idea and can be **true** or **false**, but not both at the same time. Such sentences are also called **propositions** in mathematical logic.

Examples of Simple Statements

- 1) Kigali is the capital city of Rwanda. (*True*)
- 2) $5 + 3 = 8$. (*True*)
- 3) The Earth is flat. (*False*)
- 4) 10 is an even number. (*True*)

Each of these sentences can be judged as **true** or **false**, therefore they are **simple statements**. Some sentences cannot be called statements because they **do not have truth values**.

Examples

- 1) Close the door. (Command)
- 2) What is your name? (Question)
- 3) Wow! That is beautiful. (Exclamation)

These sentences cannot be true or false, so they are not statements. The truth value of a statement shows whether the statement is True (T) or False (F).

Example

Statement	Truth Value
$2 + 2 = 4$	True
7 is an even number	False
7 is an odd number	True

The logical statements are required to have a definite **truth-value**, or to be either true or false, but never both, and to always have the same truth value. The two truth values of a proposition are true and false and are denoted by the symbols **T** and **F** respectively.

Occasionally, **true** and **false** are also denoted by the symbols **1** and **0** respectively.



Application activity 1.1

- 1) Read each sentence carefully. If it is a statement (proposition), decide whether it is true or false and write T or F. If it is not a statement (proposition), write NS (Not a Statement):
 - a) $9+4=13$
 - b) Stop talking

- c) The sun rises in the East
- d) Where do you live?
- e) 12 is divisible by 3
- f) Close the window
- g) 15 is multiple of 5

2) Choose the correct answer from the options below:

(i) Which of the following is a simple statement?

- A. Close the door
- B. $5 + 3 = 8$
- C. What is your name?
- D. Wow! That is amazing

(ii) Which sentence can not be called a statement?

- A. 10 is an even number
- B. Rwanda is in Africa
- C. Stop shouting
- D. 7 is a prime number

(iii) Which of the following sentences is true?

- A. The Earth is flat
- B. $2 + 2 = 4$
- C. Close the window
- D. What is your age?

3) Define a simple statement in your own words.

4) Write three examples of simple statements from your daily life.

5) Explain why “Stop talking” and “Where do you live?” are not statements.

1.2. Compound Statements



Activity 1.2

Read the following sentences and answer to questions:

- 1) Kigali is the capital city of Rwanda.
- 2) 5 is greater than 9.
- 3) Bwenge reads a Mathematics book and a newspaper.
- 4) A student is in class or in the library.

Questions:

- (i) Which sentences express one idea only?
- (ii) Which sentences combine two ideas?
- (iii) What words connect the ideas in sentences 3 and 4?



Content Summary

A **compound proposition** is a proposition formed by connecting two or more propositions or by negating a single proposition. The words and phrases (or symbols) used to form compound propositions are called **connectives (logical operators)**.

Common logical connectors include: “**Conjunction (and)**”, “**Disjunction (or)**”, “**Negation (not)**”, “**Conditional /Implication (if...then...)**” and “**Biconditional (iff)**”. Small letters are often used to represent statements such as $p, q, r, \dots$

Example 1

Simple statement:

p : Today is Monday.

q : We have a Mathematics lesson.

Compound statement: Today is Monday **and** we have a Mathematics lesson.

Example 2

Simple statement:

p : A student is in class.

q : A student is in the playground.

Compound statement: A student is in class **or** in the playground.

Example 3

Simple statement:

p : You study hard.

q : You will pass the exam.

Compound Statement: **If** you study hard, **then** you will pass the exam.

Example 4

p : The light is on

q : The switch is up.

Compound Statement: The light is on **if and only if** the switch is up.



Application activity 1.2

- 1) State whether the following statements are True (T) or False (F).
 - a) 4 is an even number and 9 is greater than 5.
 - b) Rwanda is in Africa and 3 is greater than 10.
 - c) 8 is less than 12 or 20 is less than 15.
- 2) Choose the correct answer.
 - (i) A compound statement is formed by:
 - A. Joining two or more statements
 - B. Writing numbers only
 - C. Writing one sentence only
 - D. Drawing diagrams
 - (ii) Which word is used to connect statements?

A. Book B. And C. Table D. School
- 3) Give two examples of compound statements from daily life.

1.3. Truth values and truth tables



Activity 1.3

Read the following statements and answer to questions:

- 1) Rwanda is in Africa.
- 2) 10 is less than 5.
- 3) $7 + 3 = 10$.
- 4) The sun rises in the west.

Questions

- (i) Which statements are true?
- (ii) Which statements are false?
- (iii) How do you know whether a statement is true or false?



Content Summary

The way we will define compound statements is to list all the possible combinations of the truth-values (abbreviated **T** or **1** and **F** or **0**) of the simple statements (that are being combined into a compound statement) in a table, called a **truth table**. The name of each statement is at the top of a column of the table. Let us find the number of combinations of truth values we can consider when we have a compound statement with n distinct simple statements.

Example 1

One proposition p has two truth values ($2^1 = 2$ possible combinations), the truth table is

p		p
T	Or	1
F		0

Example 2

Suppose we are given two mathematical statements, named p and q , new mathematical statements that incorporate p and q are called **compound statements**. Their truth values will be determined solely by the truth-values of p and of q .

For two propositions p and q , we have $2^2 = 4$ possible combinations of truth values, the truth table is:

p	q		p	q
T	T		1	1
T	F	or	1	0
F	T		0	1
F	F		0	0

Any proposition can be represented by a truth table. It shows truth values for all combinations of its constituent variables

Example 3

Let us take the proposition r involving 2 true statements p and q where

p : I am at home

q : It is raining.

One combination of statements is “*I am at home and it is raining*” (T).

Others are the following: “*I am at home and it is not raining*” (F), “*I am not at home and it is raining*” (F) and “*I am not at home and it is not raining*” (F).

All these results are summarized in the following table:

All possible combinations of truth values of 2 propositions p and q		Truth values of compound proposition r
p	q	r
True	True	True
True	False	False
False	True	False
False	False	False

A proposition can involve any number of variables; each row corresponds to a possible combination of variables. With n variables the truth table has: $n+1$ columns (one for each of the n variables and one for the compound expression); it has also 2^n rows plus a

header. Therefore, if the compound statement contains n distinct simple statements, we will consider 2^n possible combinations of truth values in order to obtain the truth table.



Application activity 1.3

- 1) Decide whether the following statements are True (T) or False (F):
 - a) A compound statement is formed by combining two or more simple statements.
 - b) The truth value of a compound statement depends only on the truth values of its simple statements.
 - c) If there are two propositions, there are three possible combinations of their truth values.
 - d) If a compound statement contains n distinct simple statements, the truth table has 2^n rows.
- 2) Choose the correct answer
 - (i) A compound statement is:
 - A. A single idea expressed in a sentence
 - B. A statement formed by joining two or more simple statements
 - C. A question
 - D. A number equation
 - (ii) If p : “It is raining” and q : “I am at home”, the compound statement “ p AND q ” is true when:
 - A. It is raining or I am at home
 - B. It is raining and I am at home
 - C. It is raining only
 - D. I am at home only
- 3) Consider these statements: p : I am at school and q : It is raining.
 - a) Construct the truth table for: p AND q
 - b) Construct the truth table for: p OR q

1.4. Negation “not”



Activity 1.4

Imagine you are given some mathematical statements. Your task is to think of the opposite statement (negation) and determine if it is true or false.

- 1) 7 is an even number.

- 2) The sum of 5 and 3 is 8.
- 3) 12 is divisible by 5.
- 4) A triangle has 4 sides.

Task for Students

- (i) Work in pairs or small groups.
- (ii) For each statement, write the negation.
- (iii) Determine whether the negated statement is true or false.



Content Summary

The negation of a proposition is what is asserted when that proposition is denied. The negation of a statement is the opposite of the given mathematical statement. The negation of a statement p is made by introducing the word “not”, denoted by prefixing the statement p and it has opposite truth value of the statement p . The negation of a statement p is denoted by $\neg p$ or $\bar{p}$ or $\sim p$. From this definition, it follows that the negation of a true statement is a false while the negation of false statement is true; simply if p is true, then $\neg p$ is false and if p is false, then $\neg p$ is true.

Example 1

- a) Let p : “Kamana is a student”, then $\neg p$: “Kamana is not a student”.
- b) p : “The earth is round”, $\neg p$: “The earth is not round”

Example 2

Let p be a proposition. Construct the truth table of $\neg p$

Solution

p	$\neg p$
T	F
F	T

Note: The following propositions all have the same meaning:

- p : All people are intelligent.
 q : Every person is intelligent.
 r : Each person is intelligent.
 s : Any person is intelligent.

(a) The negation of the proposition p : “All students are intelligent” can be stated in any of the following ways:

- $\neg p$: Some students are not intelligent.
 $\neg p$: There exists a student who is not intelligent.
 $\neg p$: At least one student is not intelligent

Note that “No student is intelligent” is not a negation of p .

(b) The negation of the proposition q : “No student is intelligent” may be stated as:

$\neg q$: At least one student is intelligent.

Note that “All students are intelligent” is not a negation of q .



Application activity 1.4

1) Answer by True or False

- The negation of a true statement is false.
- The negation of a false statement is true.
- $\neg p$ is the same as q .
- $\neg$ (12 is divisible by 3) is false.
- Negation can be applied to mathematical statements and daily-life statements.

2) Which of the following is a negation of “I am at school”?

- I am at school
- I am not at school
- I go to school
- I study at school

3) Explain in your own words why negation is important in logical reasoning?

4) Complete the following truth table

p	q	r	$\neg p$	$\neg q$	$\neg r$
T		T		F	
	T	F	F		
T	F	T			
T	F				T
	T	T	T		
F	T	F			
F				T	F
F	F	F			

1.5. Conjunction “and”



Activity 1.5

Given two propositions: p : I am at school and q : it is raining. Discuss the truth value of the compound propositions:

- “I am at school and it is raining”.
- “I am not at school and it is raining”.
- “I am not at school and it is not raining”.



Content Summary

If two simple statements p and q are connected by the word “**and**”, then the resulting compound statement p **and** q is called a conjunction of p and q and is written in symbolic form $p \wedge q$. It has the truth value **true** whenever both p and q have the truth value **true**; otherwise it has the truth value **false**. In language, the conjunction “and” joins two similar ideas together.

Example 1

Let p be “It is raining today” and q be “There are fifteen chairs in this class room”. Assuming that p and q are true statements, construct simple sentences which describe each of those statements and construct the truth table of $p \wedge q$.

Solution

From the given two simple statements, one of the resulting compound statements is

- (i) “It is raining today and there are fifteen chairs in this class room” which is True.
- (ii) “It is not raining and there are not fifteen chairs in this classroom” which is False.

You can give the remaining two sentences and the truth table of related compound statements as follows:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example 2

Consider the two statements:

p : Kigali is the capital city of Rwanda. (True)

q : Rwamagana is the largest District in Rwanda. (False)

The compound proposition (p and q) is “Kigali is the capital city of Rwanda and Rwamagana is the largest District in Rwanda. Since, the statement q is false, the compound statement is false even though p is true.

Example 3

Let p and q be propositions. Construct the truth table for

- a) $\neg p \wedge q$
- b) $(\neg p \wedge q) \wedge \neg p$

Solution

p	q	$\neg p$	$\neg p \wedge q$	$(\neg p \wedge q) \wedge \neg p$

T	T	F	F	F
T	F	F	F	F
F	T	T	T	T
F	F	T	F	F



Application activity 1.5

- 1) If p stands for the statement “It is cold” and q stands for the statement “It is raining”, then what does $\neg q \wedge \neg q$ stands for? Construct its truth table.
- 2) Let p and q be two propositions. Construct the truth table of
 - a) $p \wedge q$
 - b) $\neg p \wedge q$
 - c) $p \wedge \neg q$
 - d) $\neg(p \wedge q)$
 - e) $\neg q \wedge (\neg p \wedge q)$
- 3) Determine the truth value of each of the following statements
 - a) Paris is in France **and** it is a Capital city
 - b) $4 + 4 = 9$ **and** $5 + 8 = 11$
 - c) Paris is in England **and** $4 + 3 = 7$
 - d) Kigali is the Capital city of Burundi **and** $1 + 1 = 2$
 - e) Alphabets are the basic of any languages **and** the digits is not the basic in counting
 - f) m^2 is “a unit of area” **and** kg is “one of the units of weight.”

1.6. Disjunction “or”



Activity 1.6

Read the following sentences:

- 1) I will study Mathematics or I will attend Physics class.
- 2) 7 is less than 10.
- 3) A student is at home or at school.
- 4) 12 is divisible by 3 and 4.

Answer the following questions:

- a) Which sentences express one idea only?
- b) Which sentences combine two ideas?

- c) What words connect the ideas in sentences 1 and 3?
 d) For sentences 1 and 3, what do you think is the meaning of “or” in this context?



Content Summary

If two simple statements p and q are connected by the word “or”, then the resulting compound statement p or q is called a **disjunction** of p and q and it is written in symbolic form by $p \vee q$.

It has the truth value **false** only when p and q both have truth value **false**; otherwise it has **true** as the truth value.

Example 1

Let p be “Paris is in France” and q be “London is in England”. Construct simple verbal sentences which describe different related compound statements each of the following statements and construct the truth table of $p \vee q$.

Solution

From the given two simple statements, the resulting compound statement is

- (i) “Paris is in France or London is in England” (True)
- (ii) “Paris is in France or London is not in England” (True)
- (iii) “Paris is not in France or London is in England” (True)
- (iv) “Paris is not in France or London is not in England” (False).

The Truth table of related compound statement is as follows:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example 2

Construct the truth tables for $\neg(p \vee q)$ and $\neg p \wedge \neg q$ then compare their results.

Solution

First, we make a table that displays all the possible combinations of truth-values for p and q , then we add two columns for the truth values of $p \vee q$ and $\neg(p \vee q)$ into its top cell and then start completion of your truth table with appropriate truth values.

p	q	$p \vee q$	$\neg(p \vee q)$
T	T	T	F
T	F	T	F

F	T	T	F
F	F	F	T

Note: Ambiguity of “or” in English language:

In natural language “or” has two meanings:

The English word “or” can be used in two different ways; as an inclusive (“and/or”) or exclusive (“either/or”) disjunction. The correct meaning is usually inferred from the context in which the word is used. However, when it is important to be precise (as it often is in mathematics, business, science, etc.), we must carefully distinguish between the two meanings of “or.”

i. **Inclusive (or):** Where $p \vee q$ is true if either p or q or both are true. This is the inclusive we explored here above.

Example

“Numbers or measurements may be taken as prerequisites for geometry”. It means that take either one but you may also take both.

We define $p \vee q$ to be true if at least one of the propositions p , q is true. That is, $p \vee q$ is true if both p and q are true, if p is true and q is false, or if p is false and q is true. It is false only if both p and q are false.

ii. **Exclusive (or):** denoted as $\oplus$: where $p \oplus q$ is true if either p or q but not both are true. It can also be noted as $p \underline{\vee} q$

Example

“You will be paid money or a computer”. It means that do not expect to get both.

We define $p \underline{\vee} q$ to be true if exactly one of the propositions p , q is true. That is, $p \underline{\vee} q$ is true if p is true and q is false, or if p is false and q is true. It is false if both p and q are false or if both p and q are true.



Application activity 1.6

- 1) Write **True** if the statement is correct and **False** if it is not.
 - a) The statement “5 is even OR 7 is prime” is true.
 - b) The statement “10 is odd OR 8 is odd” is true.
 - c) In logic, the word “or” means at least one of the statements is true.
- 2) Choose the correct answer.
 - (i) The symbol used for disjunction (OR) is:

A. $\wedge$ B. $\vee$ C. $\rightarrow$ D. $\neg$
 - (ii) The statement $p \vee q$ is false when:

A. p is true and q is false

B. p is false and q is true

- C. p is true and q is true
 - D. p is false and p is false
- (ii) Which of the following represents a disjunction?
- A. It is raining and it is cold.
 - B. I will study or watch TV.
 - C. If it rains, the ground is wet.
 - D. The book is not interesting.
- 3) Open Questions
- a) Give two examples of statements that use the word “OR”.
 - b) Give a real-life example where a decision involves “OR”.
 - c) Let p and q be two propositions provided on (3.a). Construct the truth table of $p \vee q$

1.7. Conditional statement “if ... then...”



Activity 1.7

- 1) Given the following conditional statement: “If Amanda’s health is good, then she will go to the party”. Discuss and identify the hypothesis (premise) and conclusion (consequent) of the conditional statement.
- 2) Complete these sentences referring to the conditional statement given in question one.
 - a) If Amanda’s health is not good, ...
 - b) Amanda will go to the party, ...
 - c) Amanda will not go to the party, ...



Content Summary

Consider the following statement: If you eat too much meat, then you will get fat. This is a conditional statement because there is a condition to get fat. A conditional statement is a logic statement used when the statement is in the form “if ...then ...” It can be written as x “if p then q ” or $p \rightarrow q$, (or $p \Rightarrow q$) read as p implies q . In this case the proposition p is called **antecedent, hypothesis or premise** while the proposition q is the **conclusion or the consequent**.

In language, p can be called the first clause and q is the second clause.

The conditional $p \Rightarrow q$ can also be read: If p , then q , q follows from p , q if p , p only if q , p is sufficient for q or q is necessary for p .

Notation: The conditional uses double arrow: $p \Rightarrow q$ but it can also be noted using the right arrow: $p \rightarrow q$.

Let us consider the logical conditional as an obligation or contract, for example:

“If you get 100% on the final exam, then you will earn an A”. For this compound statement we have the following simple statements:

p : If you get 100% on the final exam,

q : you will earn an A.

Symbolically, it is written as: $p \Rightarrow q$.

	q : earn an A	$\neg q$: don't get A
p : get 100%	$p \Rightarrow q$ is true	$p \Rightarrow q$ is false
$\neg p$: don't get 100%	$p \Rightarrow q$ is true	$p \Rightarrow q$ is true

This shows that $F \Rightarrow T$ does not violate the obligation, the only time the obligation is broken is when $T \Rightarrow F$. Therefore, the statement $p \Rightarrow q$ has the truth value True in all cases except when p is true while q is false. The related truth table is as follows:

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Example 1

Rephrase the sentence “If it is Sunday, you go to church”, then construct the related truth values.

Solution

Here are some various ways of rephrasing the sentence:

- (i) “You go to church if it is Sunday.”
- (ii) “It is Sunday only if you go to church.”
- (iii) “It can't be Sunday unless you go to church.”

Let denote the given statements as: p : It is Sunday and q : you go to church. The statement “If it is Sunday, you go to church” is symbolized as follow: $p \Rightarrow q$.

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Example 2

Construct the truth value of the following statement: “If either John takes Calculus or Betty takes Sociology then Peter will take English.”

Solution

Let us denote the statements as: p : John takes Calculus, q : Betty takes Sociology, and r : Peter takes English. The given statement can be symbolised as follows: $(p \vee q) \Rightarrow r$ and the truth values are in the following truth table

p	q	r	$p \vee q$	$(p \vee q) \Rightarrow r$
T	T	T	T	T
T	T	F	T	F
T	F	T	T	T
T	F	F	T	F
F	T	T	T	T
F	T	F	T	F
F	F	T	F	T
F	F	F	F	T

Note: As in the previous section, we express a compound proposition symbolically by replacing each component statement and connective by an appropriate symbol. For example, denoting “I study” and “I will pass” by a and b , respectively, the proposition “If I study, then I will pass” is written as $a \Rightarrow b$. This proposition can also be read as “A sufficient condition for passing is to study”.

To understand “**implication**” better, look at an implication as a conditional promise. If the promise is broken, the implication is false; otherwise, it is true. For this reason the only circumstance under which the implication $(p \rightarrow q)$ is false is when p is true and q is false. The word “then” in an implication merely serves to separate the conclusion from the hypothesis.



Application activity 1.7

1) Write True or False for each statement.

- A conditional statement has two parts: a hypothesis and a conclusion.
- In the statement “If it rains, then the ground becomes wet”, “it rains” is the conclusion.
- The word “if” introduces the hypothesis in a conditional statement.
- In a conditional statement, the part after “then” is called the hypothesis.

2) Open Question

Write the following statements in symbolic form and their truth values

- If n is prime, then n is odd or n is 2.
- If x is non-negative, then x is positive or x is 0.
- If Manzi is Keza’s father, then Stephen is her uncle and Micheline is her aunt.

1.8. Converse and contra-positive of a conditional statement



Activity 1.8

Read the following statements carefully and answer to questions:

p : You are a guitar player; q : You are a musician.

- 1) If you are a guitar player, then you are a musician.
- 2) If you are a musician, then you are a guitar player.
- 3) If you are not a musician, then you are not a guitar player.

Questions:

- a) Write each statement using symbols p and q .
- b) Which statement has the form If p , then q ?
- c) Which statement switches the order of p and q ?
- d) Which statement negates both p and q ?
- e) Decide whether each statement is True or False.



Content Summary

a) Converse

The statement formed by changing the order of the hypothesis and the conclusion in a conditional statement is called the **converse**. So, the converse of $p \Rightarrow q$ is $q \Rightarrow p$.

This means that the statement “if q , then p ” is called the **converse** of the statement “if p , then q ”. In symbols: the converse of $p \Rightarrow q$ is $q \Rightarrow p$.

p	q	$p \Rightarrow q$	$q \Rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Example 1

If you are a student, then you should study. The converse is “if you study, then you are a student”. This is not always true.

Example 2

Consider the statements; p : You are a thief and q : You go to jail. The implication $p \Rightarrow q$ states that “If you are a thief, you go to jail”. The converse of this implication, namely $q \Rightarrow p$, states that “If you go to jail, you are a thief”. To say that thieves go to jail is not the same as saying that everyone who goes to jail is a thief. This example illustrates that the truth of an implication does not imply the truth of its converse. Many common fallacies in thinking arise from confusing an

implication with its converse. So, $p \Rightarrow q$ and $q \Rightarrow p$ are not equivalent; they do not give the same truth values.

b) Contra-positive

The contrapositive of a conditional statement is obtained by first writing the converse, then negating both the antecedent (hypothesis) and the consequent (conclusion). Given the conditional $p \Rightarrow q$, then the converse of $p \Rightarrow q$ is $q \Rightarrow p$, $(\neg q \Rightarrow \neg p)$ is called the **contra-positive** of $p \Rightarrow q$.

Example 1

Consider the statements; p : You are a thief and q : You go to jail. The implication $p \Rightarrow q$ states “If you are a thief, then you go to jail.” The contra-positive, namely $\neg q \Rightarrow \neg p$ is “If you do not go to jail, then you are not a thief”.

p	$\neg p$	q	$\neg q$	$p \Rightarrow q$	$q \Rightarrow p$	$\neg q \Rightarrow \neg p$
T	F	T	F	T	T	T
T	F	F	T	F	T	F
F	T	T	F	T	F	T
F	T	F	T	T	T	T

Example 2

If it is a PC, then it is a computer. The contrapositive is “if it is not a computer, then it is not a PC”. This is true. From the two examples above, we learn that: A conditional statement has the same meaning as its contrapositive. This means a conditional statement can be replaced by its contrapositive and the truth value will stay the same.



Application activity 1.8

1) Short Answer Questions

- Write the contrapositive of the statement:
If a number is greater than 10, then it is greater than 5.
- Write the contrapositive of the statement:
If a shape is a triangle, then it has three sides.
- Write the contrapositive of the statement:
If a person is a teacher, then he or she works at a school.

2) Open Questions

- Explain whether the following statement and its contrapositive are true:
If a number is divisible by 2, then it is even.
- Write the converse and contrapositive of the false conditional statement below and determine whether each of the statements found is true or false.
“If x is an even number, then the last digit of x is 2”.

1.9. Inverse of a conditional statement



Activity 1.9

- 1) Consider the conditional proposition: "If this year is a leap year, then February has 29 days".
 - a) What is the truth value of that proposition?
 - b) Deduce the inverse of the proposition and the related truth value.
- 2) Write the inverse of the false conditional statement below and determine whether the statement found is true or false. "If a number is even, then its last digit is 4".



Content Summary

Given the conditional proposition $p \Rightarrow q$, the implication $\neg p \Rightarrow \neg q$ is called its **inverse**. It is obtained by negating the antecedent and negating the consequent.

Example 1

If it is a PC, then it is a computer, the inverse is "If it is not a PC, then it is not a computer". This is false.

Example 2

Consider the statements: p : You are a thief. and q : You go to jail. The implication $p \Rightarrow q$ states "If you are a thief, then you go to jail". The inverse of $p \Rightarrow q$ namely $\neg p \Rightarrow \neg q$, is "If you are not a thief, then you do not go to jail."

p	$\neg p$	q	$\neg q$	$p \Rightarrow q$	$\neg p \Rightarrow \neg q$
T	F	T	F	T	T
T	F	F	T	F	T
F	T	T	F	T	F
F	T	F	T	T	T

We have shown that a conditional statement and its contrapositive always have the same truth value.

Also, the converse and the inverse of a conditional statement have the same truth as each other, but neither the converse nor the inverse always has the same truth as the original statement.



Application activity 1.9

- 1) Answer by True or False
 - a) The statement "If a number is a multiple of 10, then the number is a multiple of 5" is true.
 - b) The inverse of the statement "If a number is even, then it is divisible by 2" is always true.
- 2) Open Questions
 - a) Write the inverse of: "If a number is a multiple of 10, then the number is a multiple of 5."

- b) Write the inverse of: “If a number is even, then it is divisible by 2.”

1.10. Bi-conditional statement “if and only if”



Activity 1.10

Read the sentences and answer to questions:

- 1) A shape is a square if and only if it has four equal sides and four right angles.
- 2) A number is divisible by 2 if and only if it is an even number.
- 3) A person is a teacher if and only if he/she works at a school.

Questions:

- (a) Which sentences show both directions?
- (b) Which sentences show only one direction?
- (c) In sentence 1, can a shape have four equal sides but not be a square? Explain.
- (d) In sentence 2, can a number be even but not divisible by 2? Explain.



Content Summary

Let us consider the proposition p : Two lines are perpendicular and q : Two lines form a right angle. We have, $p \Rightarrow q$: If two lines are perpendicular, then the two lines form a right angle (True). The converse is $q \Rightarrow p$ means: if two lines form a right angle, then the two lines are perpendicular (True). We see that a statement and its converse are both true statements; that is $p \Rightarrow q$ and $q \Rightarrow p$ are both true.

Generally, $p \Rightarrow q$ is not the same as $q \Rightarrow p$. It may happen, however, that both $p \Rightarrow q$ and $q \Rightarrow p$ are true. The statement $p \Leftrightarrow q$ defined by $(p \Rightarrow q) \wedge (q \Rightarrow p)$, is a compound statement with the biconditional connective. For this reason, the double headed arrow $p \Leftrightarrow q$ is called the **biconditional**.

Example

Let p and q denote the statements; p : Cauchy is happy. and q : Cauchy is studying.

Then “A necessary condition for Cauchy to be happy is that Cauchy is studying” means “If Cauchy is happy, he is studying”. Moreover, “A sufficient condition for Cauchy to be happy is that Cauchy is studying” means “If Cauchy is studying, he is happy”.

If both implications $p \Rightarrow q$, and $q \Rightarrow p$ are true statements, then the bi-conditional is true.

In other words, Cauchy is happy if and only if he is studying.

We can restate the definition of the bi-conditional connective by using its truth table:

p	q	$p \Rightarrow q$	$q \Rightarrow p$	$(p \Rightarrow q) \wedge (q \Rightarrow p)$	$p \Leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

The bi-conditional ($p \Leftrightarrow q$), which we read “ **p if and only if q** ” or “ **p is equivalent to q** ” is **true** if both p and q have the same truth values and **false** if p and q have opposite truth values.

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Equivalent statements

Let us compare the truth values of $\neg(p \vee q)$ and $\neg p \wedge \neg q$.

- a) Considering the truth values of $\neg p \wedge \neg q$, we can determine the case in which this statement $\neg p \wedge \neg q$ is true.

p	q	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

The first two columns of the truth table give all possible combinations of the truth values of p and q , and the second two columns of the truth table are merely negations of the first two columns. The statement $\neg p \wedge \neg q$ is the conjunction of two statements $\neg p$ and $\neg q$. Therefore, the only case in which $\neg p \wedge \neg q$ is true is when both A and B are false.

- b) The truth values of $\neg(p \vee q)$ can be summarized in the truth table:

Like the previous question, the truth table consists of four rows. The third column of the truth table gives the truth values for the disjunction $p \vee q$. The fourth column gives the truth values for the negation of disjunction. Therefore, $\neg(p \vee q)$ is true only when both p and q are false.

p	q	$p \vee q$	$\neg(p \vee q)$
T	T	T	F
T	F	T	F
F	T	T	F
F	F	F	T

Comparing the truth values of $\neg(p \vee q)$ and $\neg p \wedge \neg q$ in the last columns of the two tables, we find that their truth values are identical. Whenever two statements have the same truth values, the statements are said to be logically equivalent. The symbol for equivalent statements is $\equiv$. Thus, we can write: $\neg(p \vee q) \equiv \neg p \wedge \neg q$. In spoken language, the equivalence of two propositions means they are similar, when you express one you have the same idea with which expressed the other.

Example 2

Use a truth table to show that the conditional $p \Rightarrow q$ and its contrapositive $\neg q \Rightarrow \neg p$ are logically equivalent.

Solution

Let us construct the truth table containing both $p \Rightarrow q$ and its contrapositive $\neg q \Rightarrow \neg p$.

p	q	$p \Rightarrow q$	$\neg q$	$\neg p$	$\neg q \Rightarrow \neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

The order of $\neg q$ and $\neg p$ when organizing the columns of the table must be respected. It is clear that $p \Rightarrow q$ and $\neg q \Rightarrow \neg p$ have the same truth values in the third and the sixth columns.

Note: Two propositions differ if you can find at least one row of the truth table where values differ.



Application activity 1.10

1) Construct a truth table for each of these statements:

a) $p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$

b) $p \leftrightarrow q$ and $(\neg p \vee q) \wedge (\neg q \vee p)$

- 2) Explain in your own words what a bi-conditional statement (“if and only if”) means. Give one example from everyday life.

1.11. Tautology and contradiction



Activity 1.11

Mama Diane has a basket of fruits. She says:

- 1) “Either I sold all the apples today or I didn’t sell all the apples today.”
- 2) “I sold all the apples today and I did not sell all the apples today.”

Answer the following questions:

- a) Which of Mama Diane’s statements is always true?
- b) Which statement is never true?
- c) Can you think of other examples in real life that are always true or always false?



Content Summary

1. Tautology

This is a statement that is always true, no matter the truth of its components.

For instance,

- “It is raining or it is not raining.” Always true
- “A number is divisible by 2 or it is not divisible by 2.” Always true

Example

The statement “I will either get paid or not get paid” is a tautology since it is always true.

We can use p to represent the statement “I will get paid” and not p (written $\neg p$) to represent “I will not get paid.” p : I will get paid; $\neg p$: I will not get paid . So, $[p \vee (\neg q)]$: I will either get paid or not get paid. A truth table for the statement would look like:

p	$\neg p$	$p \vee (\neg q)$
T	F	T
F	T	T

Looking at the final column in the truth table, one can see that all the truth values are T (for true). Whenever all of the truth values in the final column are true, the statement is a tautology. So, our statement 'I will either get paid or not get paid' is always a true statement, a tautology.

2. Contradiction

This is a statement that is always false, no matter the truth of its components.

For instance,

- “It is sunny and it is not sunny.” Always false
- “I am a student and I am not a student.” Always false

Example

- 1) The statement “I don’t believe in reincarnation, but I did in my past life”, is always false.
- 2) The statement $p \wedge (\neg p)$ is always false, because p and $\neg p$ cannot both be true.

P	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

...



Application activity 1.11

- 1) Read each statement carefully and Decide whether it is a tautology (always true) or a contradiction (always false).
 - a) “It is raining or it is not raining.”
 - b) “It is sunny and it is not sunny.”
 - c) “ $2 + 2 = 4$ or $2 + 2 \neq 4$.”
 - d) “I will eat lunch and I will not eat lunch at the same time.”
 - e) “A number is divisible by 2 or it is not divisible by 2.”
- 2) Show that the statement $(p \vee q) \wedge [(\neg p) \wedge (\neg q)]$ is a contradiction.
- 3) Show that the statement $[(p \vee q) \Rightarrow r] \vee [-(p \vee q) \Rightarrow r]$ is a tautology.

1.12. Universal quantifier “for all”



Activity 1.12

- 1) Let $A = \{1, 2, 3, 4, 5\}$. Determine the truth value of each of the following statements:
 - a) For all $x \in A$, $x + 3 < 9$
 - b) For all $x \in A$, $x + 3 \leq 7$
- 2) Determine the truth value of the following statement where $U = \{1, 2, 3\}$ is the universal set:
 There exist $x \in U$ such that $x + 3 \leq 5$ What is the truth value of p , q if p : “All men eat banana”, q : All men are mortal.



Content Summary

Let $p(x)$ be a propositional function defined on a set A . Consider the expression $(\forall x \in A)p(x)$ or $\forall x p(x)$ which is read by “**For every x in A , $p(x)$ is a true statement**” or, simply, “**For all x , $p(x)$.**” The symbol universal quantifier “For all” symbolized by $\forall$ is read by “**for all**” or “**for every**” is called the **universal quantifier**. The statement $(\forall x \in A)p(x)$ means that the truth set of $p(x)$ is the entire set A .

Example 1

All men are mortal. This is true if for every man, dying is applicable. The expression $p(x)$ by itself is an open sentence or condition and therefore has no truth value. However, $\forall x \in p(x)$, that is $p(x)$ preceded by the quantifier $\forall$, does have a truth value which follows from the equivalence. Specifically: Q_1 : if $\{x / x \in A, p(x)\} = A$ then $\forall x p(x)$ is **True**; otherwise $\forall x p(x)$ is **False**.

Example 2

- (a) The proposition $(\forall n \in \mathbb{N})(n+4 > 3)$ is true since $(n | n+4 > 3) = \{1, 2, 3, \dots\} = \mathbb{N}$
- (b) The proposition $(\forall n \in \mathbb{N})(n+2 > 8)$ is false since $n+2 > 8$ is possible for $n \in \{7, 8, 9, \dots\} \neq \mathbb{N}$.



Application activity 1.12

- 1) Given the statement p : “for every real number x , $x + y > 10$.” discuss the values of y which make p true.
- 2) Give the truth value for the following statements:
 - a) Some rectangles are squares
 - b) All squares are rectangles
 - c) Every language student-teacher must take a logic mathematics course.
 - d) Given $\forall x \in \mathbb{N}$, $x + 4 > 4$.

1.13. Existence quantifier “there exists”



Activity 1.13

- 1) Let $A = \{1, 2, 3, 4, 5\}$. Determine the truth value of each of the following statements:
 - a) There exist $x \in A$, $x + 3 = 10$
 - b) For some $x \in A$, $x + 3 < 5$
- 2) Let $A = \{1, 2, 3, \dots, 8, 9, 10\}$. Consider each of the following sentences; determine the set of y for which the following statement is true. Such set is called a truth set.

- a) $(\forall x \in A)(\exists y \in A)(x + y < 14)$
- b) $(\exists y \in A)(x + y < 14)$.



Content Summary

Let $p(x)$ be a propositional function defined on a set A . Consider the expression: there exists some $x \in A$ that satisfy $p(x)$. This can be written as $(\exists x \in A)p(x)$ or “There exists an x in A such that $p(x)$ is a true statement” or, simply, “For some x , $p(x)$.” The quantifier “there exists” or “for some”, or “for at least one”, symbolized by $\exists$ is an **existential quantifier**. The statement $(\exists x)p(x)$ is read as “there exists an x such that $p(x)$ is true”. Specifically: if $\{x / p(x)\} \neq \emptyset$, then $\exists x p(x)$ is True; otherwise $\exists x p(x)$ is False.

Example

Let us consider the set $\mathbb{N}$ as the set of all counting numbers: $\mathbb{N} = \{1, 2, 3, \dots\}$,

- (a) The proposition $\{\exists n \in \mathbb{N} : n + 4 < 7\}$ is true since $\{n : n + 4 < 7\} = \{1, 2\} \neq \emptyset$
- (b) The proposition $\{\exists n \in \mathbb{N} : n + 6 < 4\}$ is false since $\{n : n + 6 < 4\} = \emptyset$.
- c) Some men do not have wives, is true since some men such as Priests remain single.
- d) There exists a natural number between 3 and 6. This is statement is true because $\{4, 5\}$ exists.
- e) Some student teachers teach at university. This is false because the set of such students is empty.



Application activity 1.13

- 1) Translate the following into symbolic form:
 - a) Somebody cried out for help and called the police.
 - b) Nobody can ignore her.
- 2) Consider the predicates: $r(x) : x - 7 = 2$ and $s(x) : x > 9$. If the universe of discourse is the real numbers, give the truth value of propositions: $(\exists x)s(x) \wedge \neg(\forall x)r(x)$
- 3) Formulate different statements with existential quantifiers and give their truth values.

1.14. Negation of quantifiers



Activity 1. 14

Negate the following statements

- 1) All grapefruits have a red color.
- 2) Some celebrities are beautiful.
- 3) No one weighs more than five hundred Kilograms.

- 4) Some people are more than 2m tall.
5) All birds can fly.



Content Summary

Consider the statement: “All dogs have tails”. Its negation is “Not all dogs have tails” which means that: “There is at least one dog that does not have a tail.” Let us express these ideas symbolically: Let $p(x)$ mean “dog x has a tail”. Then all dogs have tails is represented by $\forall x, p(x)$ and dog x doesn’t have a tail is $\neg p(x)$. So, not all dogs have tails is $\neg[\forall x, p(x)]$.

Abbreviating there is at least one dog that doesn’t have a tail or there exists a tailed dog, we can say that there exists a tailless dog is $\exists x, \neg p(x)$. Since the last two statements have, in words, the same meaning; we can declare the predicate formulae representing them to be logically equivalent: $\neg[\forall x, p(x)] \equiv \exists x[\neg p(x)]$.

By generalizing the argument that we have used to produce two results, it can be shown that the logical equivalence holds true no matter what predicate is represented by $p(x)$. In fact, this is one of the generalized of the De Morgan’s laws. For predicate logic, the other is similar:

$\neg[\exists x, p(x)] \equiv \forall x[\neg p(x)]$ that can be interpreted as “there are no dogs with tails” has the same meaning as “every dog is tailless.”

Example 1

In statements that involve the words “all, every, some, none or no”, forming the negation is not as easy. The following table shows examples

Statement	Negation
All men are mortal	Some men are not mortal
Some of the numbers are not positive	All of the numbers are positive
No birds are fish	Some birds are fish
Some women can jump	No women can jump
None of the small children worked	Some of the small children worked

This shows that: “Negating a universally quantified statement changes it into an existentially quantified statement and vice-versa with the part of the statement after the quantifier becoming negated”.

Example 2

Let $p(x)$ mean “country x has a president.” Interpret the following statements. Establish the equivalence of their negation.

- a) $\forall x, p(x)$ b) $\exists x, \neg p(x)$

Solution

a) Since $p(x)$ mean “country x has a president, thus $\forall x, p(x)$ means “all countries have president” The negation is $\neg[\forall x, p(x)]$ that is interpreted as “it is not true that all counties have

presidents.” This sentence can be express as “there exist a country without a president” and symbolically it is written as $\exists x[\neg p(x)]$. Thus, $\neg[\forall x, p(x)] \equiv \exists x[\neg p(x)]$

b) $\exists x, p(x)$ means “there is a country with a president.” Negating the given statements, we get $\neg[\exists x, p(x)]$ which means “it is not true that there is a country with a president.” This sentence is equivalent to” all countries have not a president” symbolically $\forall x[\neg p(x)]$. Hence,
 $\neg[\exists x, p(x)] \equiv \forall x[\neg p(x)]$.

The basic forms for negative statements that involve “all, every, some, none or no” can be summarized as follows:

Statement	Negation
All/every	Some not.
Some ... not	All/ every
None/no	Some
Some	None/No



Application activity 1.14

Negate each of the following statements and write the answer in symbolic form:

- 1) Some students are mathematics majors.
- 2) Every real number is positive, negative or zero.
- 3) Every good boy does fine.
- 4) There is a broken desk in our classroom.
- 5) Lockers must be turned in by the last day of class.

1.15. Hypothetical syllogism



Activity 1.15

It is known that you must have the correct form and true premises to reason deductively toward a true conclusion.

The following are examples of arguments that need true conclusions. Try to conclude:

- a) If you live in Nyarugenge, then you live in Kigali. If you live in Kigali, then you live in Rwanda. Therefore, if you live in Nyarugenge, ...
- b) If you study the whole student book, then you will pass the exam. If you pass the exam, you will be eligible to sponsorship. Therefore, if you study the whole student book, ...



Content Summary

A **hypothetical syllogism** is a valid form of reasoning in logic. It is a type of syllogism where one or both statements are conditional statements. These statements usually start with the word “if” The form of hypothetical syllogism is: “If p , then q . If q , then r . Therefore, if p , then r .” It may also be written as:

If a , then b	a implies b	$a \Rightarrow b$
If b , then c	b implies c	$b \Rightarrow c$
Therefore, if a , then c	Therefore, a implies c	Therefore, $a \Rightarrow c$

In a hypothetical syllogism, the argument can still be logically correct even if one or both of the premises are false. The truth or falsity of the premises does not change the logical form of the argument. Logic focuses on the relationship between the premises and the conclusion, not on whether the premises are true or false. When we say a deductive argument is correct, we mean that the premises are connected to the conclusion in such a way that if the premises are true, the conclusion must also be true. If the logical form is correct and the premises are true, then the conclusion cannot be false. The following are examples of the hypothetical syllogism argument form:

Example 1

If it rains, we will not have a picnic. If we don't have a picnic, we won't need a picnic basket. Therefore, if it rains, we won't need a picnic basket.

Example 2

Is the following argument a hypothetical syllogism? Why or why not?

“If you have a party, you should invite your friends. If you are graduating from college, you should invite your friends. Therefore, if you are having a party, you are graduating from college.”

Solution

The argument is not a hypothetical syllogism. The premises do not link properly. The conclusion of the first premise should be the hypothesis of the second premise, and no logical rearrangement can accomplish the proper linking of the statements



Application activity 1.15

1) Deduce the conclusion of the following arguments

- “If the people elect their own government, then the political system is democratic.”
“If the political system is democratic, then the economy is prosperous.”
- “If it snows, then I will study discrete math.”
“If I study discrete math, I will get an A”
- “If I do not wake up, then I cannot go to work.”
“If I cannot go to work, then I will not get paid.”

- 2) Determine whether or not the following arguments are correct. For those which are not correct,
- Explain what is wrong with the argument;
 - Change the minor premise and make a correct argument.
 - All good chess players wear glasses. Sylvia is a good chess player.
Therefore, Sylvia wears glasses.
 - If ABCD is a square, it has four sides. If it has four sides, then it is a quadrilateral. Therefore, if ABCD is a square, it is a quadrilateral.

1.16. Affirming the Antecedent



Activity 1.16

Rephrase the following arguments, by completing the gaps

- “If you were standing out in the rain, then you would be wet now. You must have been standing out in the rain; so, you
- “If you have a driver’s license, then you must have taken the driver’s test. You do have a driver’s license. So, you



Content Summary

Affirming the antecedent is a valid form of reasoning. It starts by saying that the first part of a conditional statement (the “**if**” part, called the **antecedent**) is true. Then it concludes that the second part (the “**then**” part, called the **consequent**) is also true. This type of argument is called “**modus ponens**”. It is written as: If p , then q . Or p . Therefore, q

Example 1

If Martha loves sugar, she will enjoy her cake. Martha loves sugar, thus she will enjoy her cake.

Example 2

If I study for 6 hours, I will pass the exam. I studied for 6 hours.

Therefore, I will pass the exam.

Example 3

This classical argument is another example of affirming the antecedent. All men are mortal. Makuza is a man. Therefore, Makuza is mortal. This can be written so that the correct form is apparent. If one is a man, then one is mortal. Makuza is a man. Therefore, Makuza is mortal.

Notice:

If an argument has the correct logical form, it is a logical argument. However, for the conclusion to be true, the premises must also be true. You can use affirming the antecedent to make a deductive argument when the argument has the correct form and the premises are true.

Example 4

Is the following argument a good one? Explain. “If you want to run a marathon, then you should train for the race”. Mukamurenzi wants to run a marathon. Therefore, Mukamurenzi should train for the race.

Solution

The argument has the correct form for affirming the antecedent. If we take its first premise as true because of commonly accepted notions about the physical stamina needed to run a marathon, the argument is a good one. Even though the argument has the correct form of an argument using the technique of affirming the antecedent, the major premise is not true. Thus, it is a correct argument but it does not arrive at a true conclusion. You need both the correct form and true premises to ensure true conclusions.



Application activity 1.16

- 1) From two propositional statements, draw the corresponding truth table for affirming the antecedent.
- 2) Give the conclusion, by affirming the antecedent.
 - a) If anyone is born of God, then he loves his brothers. Adolph is born of God.
 - b) If the national debt is paid off with inflated money, then borrowers will benefit and lenders will be harmed. The national debt is being paid off with inflated money.

1.17. Denying the Consequent



Activity 1.17

Rephrase the following arguments, by completing the gaps

- 1) “If you have a driver’s license, then you must have taken the driver’s test.

You have not taken the driver’s test. So, you _____”

- 2) “If you were standing out in the rain, then you would be wet now.

You must not be wet now. So, you _____.”



Content Summary

Denying the consequent is a type of logical argument. It starts with a conditional statement that says if one thing happens, then another thing happens. Then it states that the second thing does not happen, and concludes that the first thing also does not happen. This type of argument is called “**modus tollens**.” If **A** and **B** represent statements, the argument that denies the consequent has the following form: Major premise: $A \Rightarrow B$; Minor premise: $\sim B$;

Conclusion: $\therefore \sim A$

Example 1

If Keza is at school, then she has notebooks. Keza does not have notebooks. Therefore, Keza is not at the school.

Example 2

If you pay the bill on time, then you are not charged a penalty. You are charged a penalty. Therefore, you did not pay the bill on time.

Example 3

Is the following argument a good one? Explain.

If a number is not positive, then the number is negative. Five is not negative. Therefore, five is positive.

Solution

The argument has the form of an argument using denying the consequent, so it is a correct argument. The conclusion is correct but the premise is not true. It should be “a number is not positive or zero”.



Application activity 1.17

- 1) Give the conclusion, by denying the consequence
 - a) If Bwenge is competent, he will get a promotion. Bwenge is not getting a promotion, Therefore....
 - b) If I have cash, I will loan your money. I am not loaning your money. So....
 - c) If a world government is established, wars will cease. Wars are not ceasing. Therefore, ...
 - 2) Determine whether or not the following arguments are correct. For those which are not correct,
 - a) Explain what is wrong with the argument.
 - b) Change the minor premise and make a correct argument.
- When it is midnight, I am asleep. I was asleep. Therefore, it was midnight.
 - All Rhode Island Red hens lay brown eggs. My hen, Motopa is a Rhode Island Red. Therefore, Motopa lays brown eggs
 - If a triangle is equilateral, then it has three equal sides. ABC does not have three equal sides. Therefore, ABC is not equilateral.



End of unit assessment

Assessing Knowledge and Understanding

- 1) Answer by either **True** or **False** for each statement.
 - a) A simple statement expresses a single idea that can be either true or false.

- b) The negation of a true statement is false.
- c) If $p \Rightarrow q$ is true, then its contrapositive ($\neg q \Rightarrow \neg p$) is also true.
- d) A bi-conditional statement “ p if and only if q ” is true only when p and q have the same truth values.
- e) The universal quantifier “for all” ($\forall$) applies to every element in a set.

2) Multiple Choice Questions

- (i) Which of the following is a compound statement?
 - A. Kigali is the capital of Rwanda
 - B. 5 is greater than 9
 - C. Bwenge reads a Mathematics book and a newspaper
 - D. 7 is a prime number
- (ii) The negation of “All students are in class” is:
 - A. No students are in class
 - B. Some students are not in class
 - C. Every student is absent
 - D. Every student is in class
- (iii) Which of the following is an example of a conditional statement?
 - A. If it rains, then we will stay home
 - B. I like Mathematics or I like Science
 - C. Rwanda is in Africa
 - D. $3 + 4 = 7$

3) Open-Ended Questions

Assessing Skills

- 1) Show that $[p \wedge (p \rightarrow q)] \rightarrow q$ is a tautology.
- 2) Find the truth value of the bi-conditional “The moon is made of cheese iff $1=2$ ”.

Assessing attitudes and values

- 1) A student says: “If I don’t understand a topic, I will not ask questions. Do you agree? Explain.
- 2) Why is it important to check all possible cases in a truth table?



UNIT 2: POINTS, LINES AND GEOMETRIC SHAPES IN 2D

Key unit competence: Apply 2D shapes properties in calculating the area of land.

Learning objectives: By the end of this unit, you should be able to:

Knowledge and Understanding:

- Describe coordinate of a point in 2D.
- State properties of a straight line.
- Provide conditions of parallelism and perpendicularity of two straight lines.
- Classify the geometric shapes in 2D

Skills:

- Represent a point and a vector in 2D.
- Calculate the distance between two points in 2D and the mid - point of a segment in 2D.
- Calculate the angle between two straight lines.
- Determine the equations of a straight line (parametric and cartesian equation).
- Determine the intersection of lines.
- Calculate the perimeter and area of geometric shapes in 2D.

Attitudes and Values:

- Appreciate that a point is a fixed position in a plane.
- Be accurate in representations and calculations.
- Manifest a satisfaction that an angle is made by the intersection of two straight lines.
- Appreciate the importance of determining perimeter and area in real life.

Introduction

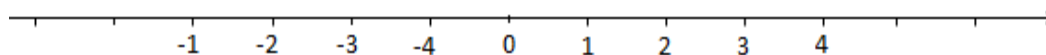
Geometry is all around us in the buildings we see, the plots of land we measure, and the objects we use every day. In this unit, you will explore points, lines, line segments, and 2D shapes, and learn how to use their properties to calculate areas and distances. Understanding these concepts will help you solve real-life problems, such as measuring land, planning constructions, or

designing layouts. You will also learn how to use vectors and coordinates to describe points and lines accurately, making your calculations more precise.



Introductory activity 2

Rukundo uses a number line to locate the points with x-coordinates -4, -2, 3, and 4. Her classmate Isimbi notices that -4 is closer to zero than -2 as shown in the figure below.



Isimbi told Rukundo that he was mistaken. Rukundo replied that he was not sure about his diagram because he did not have time to repeat his course yesterday evening.

- Use what you know about a vertical number line to determine if Rukundo made a mistake or not. Support your explanation with a number line diagram.
- What is number line?
- What is Cartesian plane?

2.1. Cartesian coordinates of a point



Activity 2.1

Consider the points $A(1,2)$ and $B(5,4)$.

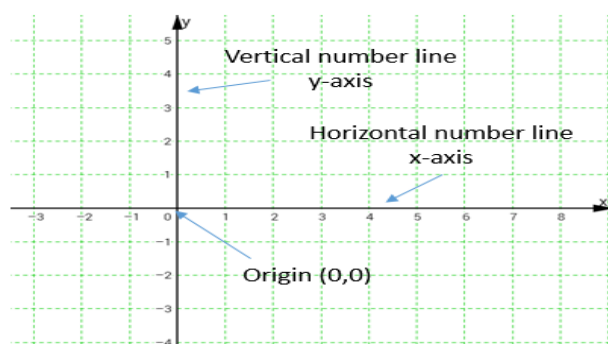
- Represent these points in xy -plane
- Draw a line segment from A to B



Content Summary

In two dimensions, a point is represented by Cartesian coordinates (also called rectangular coordinates). Cartesian coordinates are a pair of numbers that specify signed distances from the coordinate axes. They are specified in terms of the x - coordinates and the y - coordinates. The origin is the intersection of the two axes. The position of a point on the Cartesian plane is represented by a pair of numbers. The pair is called an ordered pair or coordinates (x, y) . The first number, x , called the x -coordinate and the second number, y , is called the y -coordinate.

Cartesian Plane



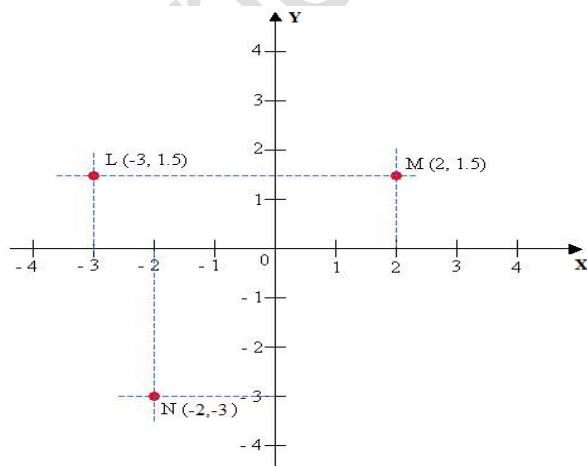
The origin is indicated by the ordered pair or coordinates $(0, 0)$. To get to the point (x, y) in the Cartesian plane, we start from the origin. If x is positive, then we move x units right from the origin otherwise if x is negative then we move x units left from the origin. Then, if y is positive, we move y units up otherwise if y is negative, we move y units down

Example

Represent the points $M(2, 1.5)$, $L(-3, 1.5)$ and $N(-2, -3)$ in the Cartesian plane.

Solution

- Point M has coordinates $M(2, 1.5)$. To get to point M, we move 2 units to the right of x -axis from the origin (positive side) and 1.5 units up on y -axis from the origin (positive side).
- Point L is represented by the coordinates $L(-3, 1.5)$. To get to point L, we move 3 units to the left of x -axis from the origin (negative side) and 1.5 units up on y -axis from the origin (positive side).
- Point N has coordinates $N(-2, -3)$. To get to point N, we move 2 units to the left of the origin on x -axis (negative) and 3 units down on y -axis from the origin (negative side).



Points on Coordinate Plane



Application activity 2.1

1) Choose the correct answer.

A point P is located 3 units to the right of the origin and 4 units above the origin. What are its coordinates?

- A) (4, 3) B) (3, 4) C) (-3, 4) D) (4, -3)

2) Plot the following points on a 2D grid and write their coordinates:

A (2, 3), B (0, -4) and C (-3, 1)

3) A treasure is hidden at point (4, -2) on a map. If you move 3 units left and 1 unit up, what are the new coordinates of the treasure? Show your work.

2.2. Distances between two points



Activity 2.2

Kalisa and Mugisha live in the same Sector but at two different hills. Kalisa's house is located at $A(1, 2)$ from the Sector and Mugisha's house is at $B(4, 6)$.

- If the office of the sector is considered as the origin, present this situation on Cartesian plan.
- Use the ruler to measure the distance between Kalisa's house and Mugisha's house.



Content Summary

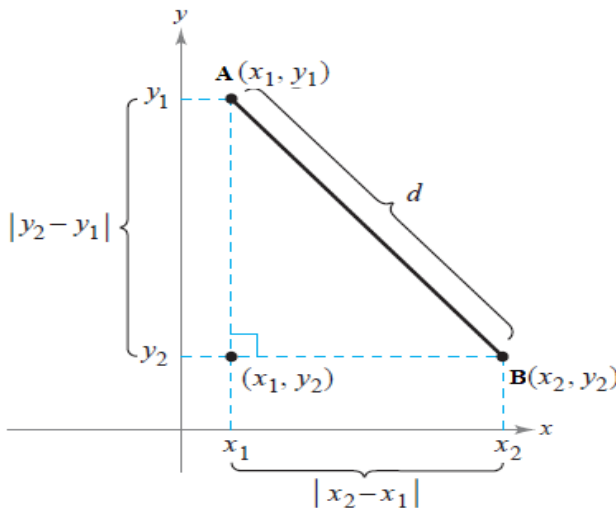
In a two-dimensional (2D) plane, the position of a point is shown by its coordinates. A point is written as (x, y) , where x shows the horizontal position and y shows the vertical position. The distance between two points is the length of the straight line joining them. If the coordinates of the two points are:

$A(x_1, y_1)$ and $B(x_2, y_2)$; the distance between them is given by the formula:

$$d(A, B) = \|AB\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This formula helps us calculate the length of the line segment between two points on a coordinate plane. If A and B are two points on xy -coordinates, we can form a vector $\overrightarrow{AB}$ and the distance between these two points denoted $d(A, B)$ is given by $\|\overrightarrow{AB}\|$.

Suppose you want to determine the distance “ d ” between the two points (x_1, y_1) and (x_2, y_2) in the plane. If the points lie on a horizontal line, then $y_1 = y_2$ and the distance between the points is $d = |x_2 - x_1|$. If the points lie on a vertical line, then $x_1 = x_2$ and the distance between the points is $d = |y_2 - y_1|$. If the two points do not lie on a horizontal or on a vertical line, they can be used to form a right triangle, as shown in the figure below.



The length of the vertical side of the triangle is $|y_2 - y_1|$, and the length of the horizontal side is $|x_2 - x_1|$. By Pythagorean theorem, it follows that: $d^2(A, B) = (x_2 - x_1)^2 + (y_2 - y_1)^2$ and $d(A, B) = \|\overrightarrow{AB}\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Thus, if $A(x_1, y_1)$ and $B(x_2, y_2)$ are points of the Cartesian plane; then, The distance “ d ” between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ in the Cartesian plane, is given by: $d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Example 1

Consider the points $A(1, 4)$, $B(-2, -3)$ in the Cartesian plane. Find the distance between the point A and B.

Solution

The distance is $d(A, B) = \|\overrightarrow{AB}\| = \sqrt{(-2-1)^2 + (-3-4)^2} = \sqrt{9+49} = \sqrt{58}$ units

Example 2

Consider the points $C(k, -2)$ and $D(0, 1)$ in Cartesian plane. Find the number k if the distance between point A and B is 5 units.

Solution

$$d(C, D) = \sqrt{k^2 + 9}$$

$$\sqrt{k^2 + 9} = 5 \Leftrightarrow k^2 + 9 = 25 \Leftrightarrow k^2 = 16 \Rightarrow k = \pm 4$$

Thus the value of k is -4 or 4



Application activity 2.2

1) Calculate the distance between the points given below:

a) $S(-2, -5)$ and $Q(7, -2)$

b) $A(2, 7)$ and $B(-3, 5)$

c) $A(x, y)$ and $B(x+4, y-1)$

2) The length of $CD = 5$. Find the missing coordinate if:

a) $C(6, -2)$ and $D(x, 2)$. b) $C(4, y)$ and $D(1, -1)$.

2.3. Midpoint of a line segment



Activity 2.3

Three friends: Pascal, Steve, and Benjamin live on the same side of a street from Nyabugogo to Ruyenzi. Steve's house is halfway between Pascal's and Benjamin's houses. If the locations of Pascal can be given by the coordinates $(2, 5)$, Benjamin's house at $(4, 7)$.

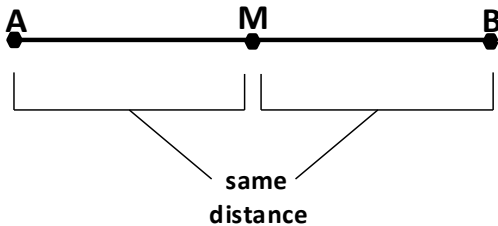
- Draw a Cartesian plane and locate Pascal's and Benjamin's houses
- Show the line segment joining the locations of Pascal's and Benjamin's houses
- Given that Steve's house is in the half way, locate his house, estimate the coordinate of that location and explain how to find it.



Content Summary

A line segment from point A to point B , denoted $[AB]$ is the set of all points on the part of the line that joins A and B , including A and B . The midpoint of this segment is point M such that the

distance $[AM] = [MB]$ and it is given by $M = \frac{1}{2}(A + B)$. The midpoint of a segment is the point in the middle of the line segment. It has the same distance from both endpoints of the segment. For example, consider the line segment on the left.



It has endpoints named A and B. The midpoint of the segment is labelled M. It is at the same distance from each of the endpoints. Suppose that the coordinates of the line segment $\overline{AB}$, are $A(x_1, y_1)$ and $B(x_2, y_2)$. The coordinates of the midpoint, are obtained using the midpoint formula, such that; **The midpoint formula:**

$$\text{Coordinates of midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Example 1

Find the midpoint of the segment joining points $A(3, 0)$ and $B(1, 8)$.

Solution

$$\text{The midpoint is } M = \frac{1}{2}(A + B) = \frac{1}{2}(4, 8) = (2, 4).$$

Example 2

If $(-3, 5)$ is the midpoint of $(2, 6)$ and (a, b) , find the value of a and b.

Solution

$$\begin{aligned} (-3, 5) &= \frac{1}{2}(2 + a, 6 + b) \Leftrightarrow (-3, 5) = \left(\frac{2 + a}{2}, \frac{6 + b}{2} \right) \\ \Rightarrow \begin{cases} \frac{2 + a}{2} = -3 \\ \frac{6 + b}{2} = 5 \end{cases} &\Leftrightarrow \begin{cases} 2 + a = -6 \\ 6 + b = 10 \end{cases} \Rightarrow \begin{cases} a = -8 \\ b = 4 \end{cases} \end{aligned}$$



Application activity 2.3

1) Choose the correct answer

- a) The midpoint of the segment joining $A(2, 4)$ and $B(6, 8)$ is:
 A. $(4, 6)$ B. $(3, 5)$ C. $(5, 7)$ D. $(2, 6)$
- b) The midpoint of the segment joining $A(1, 3)$ and $B(5, 7)$ is:
 A. $(2, 4)$ B. $(3, 5)$ C. $(4, 6)$ D. $(5, 7)$

2) Answer by Yes or No.

- a) The midpoint of a line segment is always at the same distance from both endpoints.
 b) The midpoint of the segment joining $(2, 2)$ and $(6, 6)$ is $(4, 4)$.
 c) The midpoint of a segment lies outside the line segment.

2.4. Vector in 2D



Activity 2.4

Read and answer the questions:

A student is standing at the school gate. The teacher asks the student to move 4 steps to the right and then 3 steps forward to reach the library. Another student starts at the same place but only walks 5 steps forward.

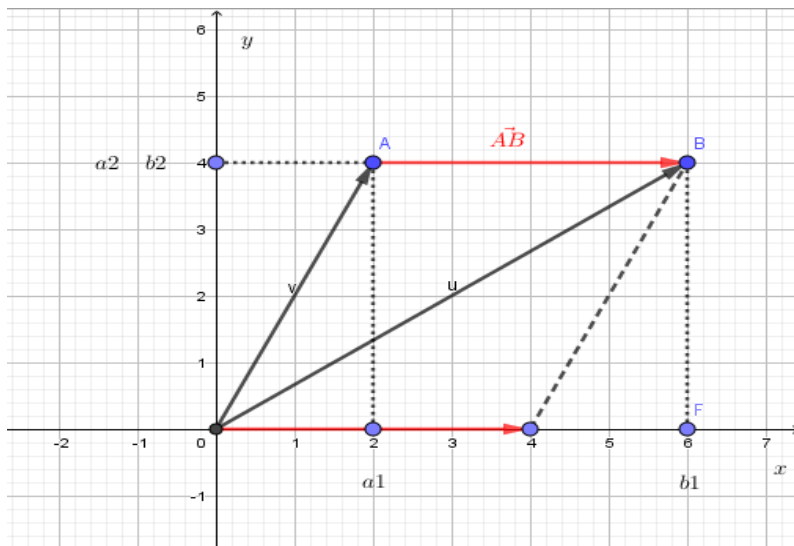
Questions

- (a) Which student moved far from the starting point?
 (b) Why is it important to know both the distance and the direction of movement?
 (c) If we want to describe the movement clearly, what information should we give?
 (d) How can we represent the movement using arrows on a grid?



Content summary 2.4

A vector is a line segment that shows direction. It is a quantity that has both size (magnitude) and direction. A vector is usually shown by an arrow. The length of the arrow shows the size of the vector, and the arrowhead shows its direction. This means a vector has a certain length and a certain direction. The vector joining the point A and the point B is denoted by $\overrightarrow{AB}$ and to find its position, we subtract the coordinates of point A from the coordinates of point B . For example the vector $\overrightarrow{AB}$ defined by two points $A(a_1, a_2)$ and $B(b_1, b_2)$



On the figure, it is clear that the vector

$$\vec{AB} = \vec{OB} - \vec{OA} = (b_1, b_2) - (a_1, a_2) = (b_1 - a_1, b_2 - a_2) = \begin{pmatrix} b_1 - a_1 \\ b_2 - a_2 \end{pmatrix}$$

Therefore, $\overrightarrow{AB} = (b_1 - a_1, b_2 - a_1)$, which can also be written in the form of a column vector

$\vec{AB} = \begin{pmatrix} b_1 - a_1 \\ b_2 - a_2 \end{pmatrix}$. The point A is called the **initial point** or tail of $\overrightarrow{AB}$ and B is called the **terminal point** or **tip**. The **zero vector** is $(0,0)$ is denoted by $\vec{0}$.

Example 1

The vector defined by the point $A(1,2)$ and $B(4,3)$ is $\overrightarrow{AB} = (3,1)$

Example 2

The vector $\overrightarrow{CD} = (-4,2)$ is defined by the point $C(-3,4)$ and D .

Find the coordinates of the point D .

Solution

Let the point D be $D(x, y)$, then $\overrightarrow{CD} = (x + 3, y - 4) = (-4, 2)$

$$x + 3 = -4 \Rightarrow x = -7$$

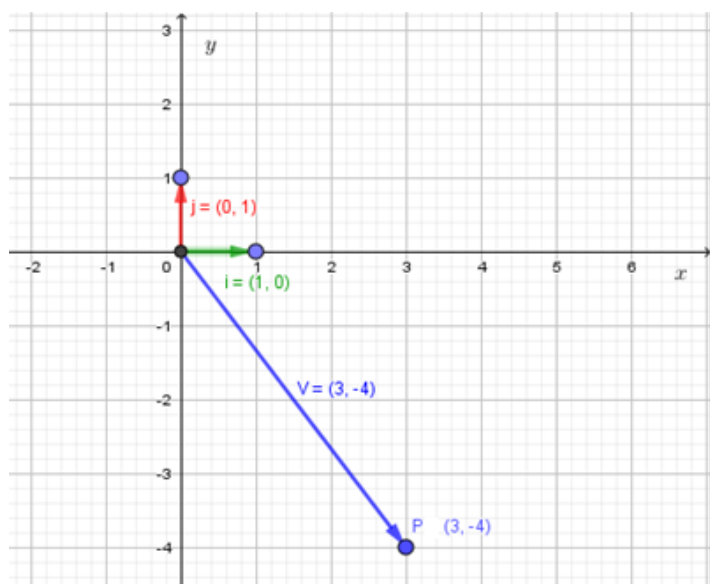
$$y - 4 = 2 \Rightarrow y = 6$$

Thus, the point D has the coordinates $(-7, 6)$.

Components of a vector

We now present an alternative representation of a vector in the plane that is common in the physical sciences. Let $\vec{i}$ denote the unit vector whose direction is along the positive x -axis; let $\vec{j}$ denote the unit vector whose direction is along the positive y -axis. Then, $\vec{i} = (1, 0)$ and $\vec{j} = (0, 1)$. Any position vector $\vec{v} = (a, b)$ can be written using the unit vectors $\vec{i}$ and $\vec{j}$ as follows: $\vec{v} = (a, b) = a(1, 0) + b(0, 1) = a\vec{i} + b\vec{j}$

We call a and b the **horizontal** and **vertical components** of the position vector $\vec{v} = (a, b)$, respectively. For example, if $\vec{v} = (3, -4) = 3\vec{i} - 4\vec{j}$, then 3 is the **horizontal component** and -4 is the **vertical component**.



Note: It is now easy to show that if we have two points $A(a_1, a_2)$ and $B(b_1, b_2)$, the vector

$$\vec{AB} = \vec{OB} - \vec{OA} = (b_1, b_2) - (a_1, a_2) = (b_1 - a_1, b_2 - a_2)$$

Since,

$$\vec{OA} = a_1\vec{i} + a_2\vec{j}, \vec{OB} = b_1\vec{i} + b_2\vec{j}$$

$$\vec{AB} = \vec{OB} - \vec{OA} = (b_1\vec{i} + b_2\vec{j}) - (a_1\vec{i} + a_2\vec{j})$$

$$= b_1\vec{i} + b_2\vec{j} - a_1\vec{i} - a_2\vec{j} = b_1\vec{i} - a_1\vec{i} + b_2\vec{j} - a_2\vec{j} = (b_1 - a_1)\vec{i} + (b_2 - a_2)\vec{j}$$

$$\vec{AB} = (b_1 - a_1)\vec{i} + (b_2 - a_2)\vec{j} = (b_1 - a_1, b_2 - a_2)$$

Therefore, $\overrightarrow{AB} = (b_1 - a_1, b_2 - a_2)$ or $\overrightarrow{AB} = \begin{pmatrix} b_1 - a_1 \\ b_2 - a_2 \end{pmatrix}$



Application activity 2.4

- 1) In the xy – plane, which of the following represents a vector with initial point $A(1,2)$ and terminal point $B(4,6)$? A. $\overrightarrow{AB} = (3,4)$ B. $\overrightarrow{AB} = (5,6)$ C. $\overrightarrow{AB} = (4,2)$ D. $\overrightarrow{AB} = (1,2)$
- 2) Answer by either **Yes** or **No** for the following questions
 - (i) A vector has both magnitude and direction.
 - (ii) The vector from $P(1,1)$ to $Q(4,5)$ is $\overrightarrow{PQ} = (3,4)$.
 - (iii) A vector can be drawn without direction.

2.5. Dot product of vectors in 2 D



Activity 2.5

Imagine two students are walking on a flat field: Student A walks 3 steps east and 2 steps north. Student B walks 4 steps east and 1 step north.

- 1) Who is moving more in the same direction as the other?
- 2) Can we measure how much their movements match in direction?
- 3) How can we find a number that shows their movements together?



Content Summary 2.5

The dot product of two vectors is a way to multiply them to get a single number (**scalar**). It is not a vector, just a number. The dot product is also called scalar product (or sometimes inner product). If we have two vectors: $u = (a_1, a_2)$ and $\vec{v} = (b_1, b_2)$, then the **dot product** is written as $\vec{u} \cdot \vec{v} = a_1 b_1 + a_2 b_2$

Notice: Two vectors are perpendicular if their scalar product is zero.

Example 1

Show that $(2,4) \cdot (10,4) = \left(2\vec{i} + 4\vec{j} \right) \left(10\vec{i} + 4\vec{j} \right)$

Solution

$$(2,4) \cdot (10,4) = 2 \cdot 10 + 4 \cdot 4 = 20 + 16 = 36$$

$$\left(2\vec{i} + 4\vec{j} \right) \left(10\vec{i} + 4\vec{j} \right) = 2 \cdot 10 \vec{i} \cdot \vec{i} + 2 \cdot 4 \vec{i} \cdot \vec{j} + 4 \cdot 10 \vec{i} \cdot \vec{j} + 4 \cdot 4 \vec{j} \cdot \vec{j} = 20 + 0 + 0 + 16 = 36$$

With $ii = 1, jj = 1$ and $ij = 0, ji = 0$

Example 2

Given vectors

$$\vec{u} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$$

Verify if vector $\vec{u}$ and $\vec{v}$ are perpendicular.

$$\vec{u} \cdot \vec{v} = (3 \times 4) + (4 \times -3) = 12 - 12 = 0 \Leftrightarrow \vec{u} \cdot \vec{v} = 0$$

Hence, the given vectors are perpendicular.



Application activity 2.5

1) Find the dot product of $\vec{u} = (3, 2)$ and $\vec{v} = (4, 5)$

A. 23 B. 14 C. 22 D. 15

2) Yes or No Questions

- a) The dot product of two vectors is always a number.
- b) If the dot product of two vectors is 0, the vectors are perpendicular.

3) Practical questions

In xy -plane :

- a) Represent the points $A(1, 2)$ and $B(-3, 1)$
- b) Find the distance between point A to point B

2.6. Properties of dot product in 2D



Activity 2.6

Imagine two friends are pushing a box on the floor:

Friend A pushes the box to the right while Friend B pushes the box in the same direction.

- 1) What happens if both push together in the same direction?
- 2) What happens if they push in opposite directions?
- 3) What if one pushes to the side while the other pushes forward?



Content Summary

Properties of scalar product

- a) If $\vec{u} = \vec{0}$ or $\vec{v} = \vec{0}$, then $\vec{u} \cdot \vec{v} = 0$
- b) If $\vec{u} \parallel \vec{v}$ and $\vec{u}, \vec{v}$ have same direction, then $\vec{u} \cdot \vec{v} > 0$
- c) If $\vec{u} \parallel \vec{v}$ and $\vec{u}, \vec{v}$ have opposite direction, then $\vec{u} \cdot \vec{v} < 0$
- d) Orthogonality: If $\vec{u} \perp \vec{v}$, then $\vec{u} \cdot \vec{v} = 0$
- e) Commutativity: $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
- f) $\vec{u} \cdot (a\vec{u} + b\vec{w}) = a\vec{u} \cdot \vec{u} + b\vec{w} \cdot \vec{u}$, $(a\vec{u} + b\vec{v}) \cdot \vec{w} = a\vec{u} \cdot \vec{w} + b\vec{v} \cdot \vec{w}$
- g) $\vec{u} \cdot \vec{u} > 0, \vec{u} \neq \vec{0}$. We define the square of $\vec{u}$ to be $\vec{u} \cdot \vec{u} = (\vec{u})^2$

Example 1

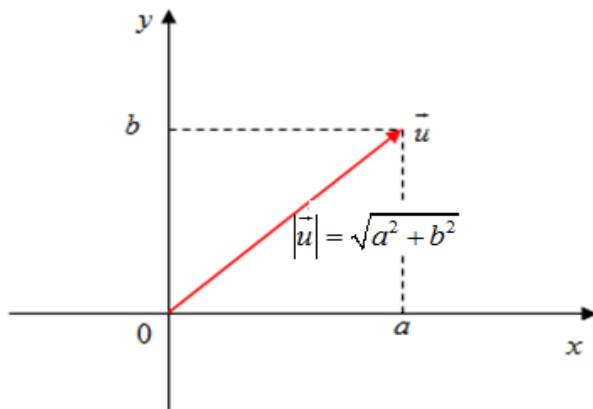
The scalar product of the vector $\vec{u} = (2, 4)$ and vector $\vec{v} = (-5, 0)$ is $\vec{u} \cdot \vec{v} = 2(-5) + 0 = -10$

The square of the vector $\vec{u} = (10, 4)$ is $(\vec{u})^2 = 10(10) + 4(4) = 100 + 16 = 116$

Magnitude or Modulus or norm of a vector

The magnitude of the vector u noted by $\|\vec{u}\|$ is defined as its length and is given by the square root of the sum of the squares of its components.

The magnitude of a vector u is also noted by $|\vec{u}|$.



That is $\|\vec{u}\| = \sqrt{(\vec{u})^2}$ or $\|\vec{u}\|^2 = (\vec{u})^2$ or $\|\vec{u}\|^2 = (\vec{u})^2$.

Thus, if $\vec{u} = (a, b)$

$$\vec{u} \cdot \vec{u} = (\vec{u})^2$$

$$\vec{u} = (a, b), \vec{u} \cdot \vec{u} = (a, b)(a, b) = a^2 + b^2$$

$$(\vec{u})^2 = a^2 + b^2$$

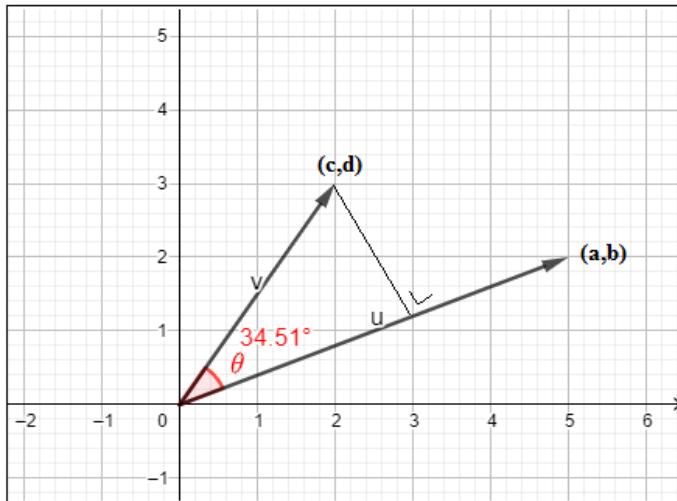
$$\|\vec{u}\|^2 = a^2 + b^2; \quad \|\vec{u}\| = \sqrt{a^2 + b^2}$$

Consequences

a) If $\vec{u} = \vec{0}$ then $\|\vec{u}\| = 0$

b) If k is a real number, $\|k\vec{u}\| = |k|\|\vec{u}\|$.

c) Geometrically, the scalar product of two vectors $\vec{u} = (a, b)$ and $\vec{v} = (c, d)$ of a plane is given by $\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\|\cos\theta$. Where θ is the angle between vectors $\vec{u}$ and $\vec{v}$.



From this relation we have $\cos\theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\|\|\vec{v}\|}$ or $\cos\theta = \frac{a \cdot c + b \cdot d}{\sqrt{a^2 + b^2} \sqrt{c^2 + d^2}}$.

Example 2

Find the norm of the vector $\vec{v} = (3, 4)$.

Solution

The norm is $\|\vec{v}\| = \sqrt{9 + 16} = \sqrt{25} = 5$

Example 3

Find the norm of the vector $\vec{u} = (-1, 4)$

Solution

The norm is $\|\vec{u}\| = \sqrt{1 + 16} = \sqrt{17}$



Application activity 2.6

1) Find the norm(magnitude) of the following vectors:

a) $\vec{u} = (-3, 4)$

b) $\vec{v} = (3, 1)$

2) Consider the following points $A(3, 4), B(-2, 3)$ and the vectors $\vec{u} = (4, 5), \vec{v} = (-3, 1)$ in Cartesian plane. Find

a) vector $\overrightarrow{AB}$

b) vector $\vec{w} = 2\overrightarrow{AB} - 3\vec{u} - \vec{v}$

c) the norm of vector $\overrightarrow{AB}$ and norm of vector $\vec{w}$

d) the scalar product of vector $\vec{u}$ and vector $\vec{v}$

the scalar product of vector $\vec{u}$ and vector $\vec{w}$

2.7. Equation of a straight line passing through a point and parallel to a direction vector



Activity 2.7

In a city map, a road goes from point $A(1, 2)$ in the direction of a vector $(3, 2)$. A city engineer wants to design another road that passes through a water tank located at point $B(4, 5)$ and is parallel to the first road so that both roads have the same direction. To construct this new road, the engineer needs to determine the equation of the straight line passing through a point and parallel to a given direction vector.

Answer the following questions:

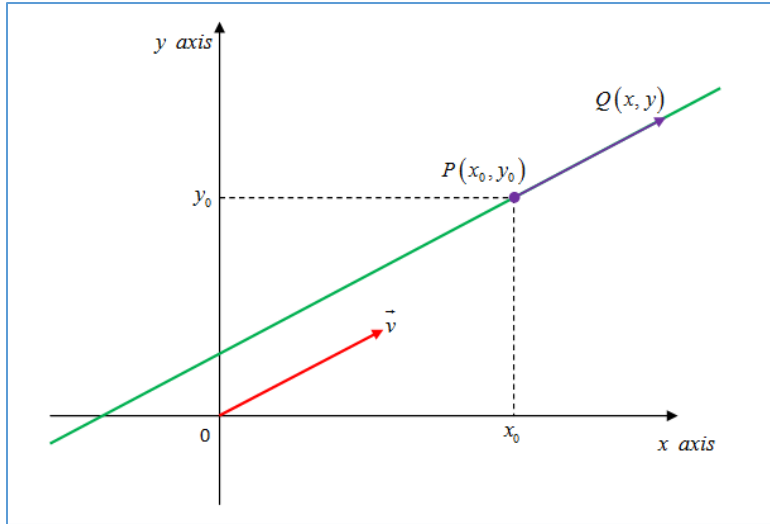
- 1) What do we mean when we say two lines are parallel?
- 2) If two lines are parallel, what can we say about their direction vectors?
- 3) How can we find the equation of a line if we know a point on the line and its direction vector?



Content Summary

To determine the equation of the line passing through the point $P(x_0, y_0)$ and parallel to the direction vector $v = (a, b)$, we will use our knowledge that parallel vectors are scalar multiples.

The vector $\overrightarrow{OP}$ is called the **position vector** of P .



Thus, the vector through $P(x_0, y_0)$ and any other point on the line is the product of a scalar and the direction vector $\vec{v} = (a, b)$.

$\overrightarrow{PQ} = r\vec{v}$ with r a parameter and $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$. Hence, the **vector equation** of the line that is parallel to the vector $\vec{v} = (a, b)$ and which passes through the point P with position vector $\overrightarrow{OP} = (x_0, y_0)$ is given as $\overrightarrow{OQ} = \overrightarrow{OP} + r\vec{v}$; where $Q(x, y)$ is any point of the line.

$$(x, y) = (x_0, y_0) + r(a, b) \quad \text{or} \quad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + r \begin{pmatrix} a \\ b \end{pmatrix}$$

The **parametric equations** of the line that is parallel to the vector $\vec{v} = (a, b)$ and which passes through the point P with position vector $\overrightarrow{OP} = (x_0, y_0)$ are given by:
$$\begin{cases} x = x_0 + r a \\ y = y_0 + r b \end{cases}$$

The **symmetric equation** or **Cartesian equation** is found after eliminating the parameter r .

$$\frac{x - x_0}{a} = \frac{y - y_0}{b}$$

This can be expanded to the standard form $bx - ay = bx_0 - ay_0$.

Example

Find the vector equation of the straight line that is parallel to the vector $\vec{u} = (2, -1)$ and which passes through the point with position vector $\vec{v} = (3, 2)$.

Solution

The vector equation is, r is a parameter. The parametric equations: $\begin{cases} x = 3 + 2r \\ y = 2 - r \end{cases}$

The Cartesian equation: $\frac{x-3}{2} = \frac{y-2}{-1}$ or $2y - 4 = -x + 3$ or $x + 2y = 7$

In general, The **vector equation** of the line can be rewritten as

$(x, y) = (x_0, y_0) + r(a, b)$ or $x\vec{i} + y\vec{j} = x_0\vec{i} + y_0\vec{j} + r(a\vec{i} + b\vec{j})$ where $\{\vec{i} = (1, 0), \vec{j} = (0, 1)\}$ form the standard basis of $\mathbb{R}^2$.



Application activity 2.7

1) Answer by True or False

- Two parallel lines have the same direction vector.
- A line can be determined by a point and a direction vector.
- If a line passes through $A(1, 2)$ with direction vector $\vec{u}(2, 3)$, its parametric equation can be $x = 1 + 2t, y = 2 + 3t$.

2) Multiple Choice Questions

i) A line passes through point $A(3, 1)$ and has direction vector $\vec{u}(2, 5)$. The parametric equation is:

- $x = 3 + 2t, y = 1 + 5t$
- $x = 3 + 5t, y = 1 + 2t$
- $x = 1 + 2t, y = 3 + 5t$

ii) If a line is parallel to vector $(4, -1)$, its direction vector can also be:

- A. $(1, 1)$ B. $(4, -1)$ C. $(2, 3)$ D. $(3, 2)$

2.8. Equation of a straight line given 2 points



Activity 2.8

On a map of a small town, a water pipeline must be installed to connect a water tank at point $A(2, 1)$ to a school at point $B(6, 5)$. The engineer wants to represent this pipeline by an equation of a straight line on the coordinate plane. To design the pipeline correctly, the engineer needs to determine the equation of the straight line passing through the two points.

Answer the following questions:

- 1) How many points are needed to determine the equation of straight line?
- 2) If two points of a line are known, how can we determine its slope?
- 3) Once the slope is known, how can we find the equation of the line?

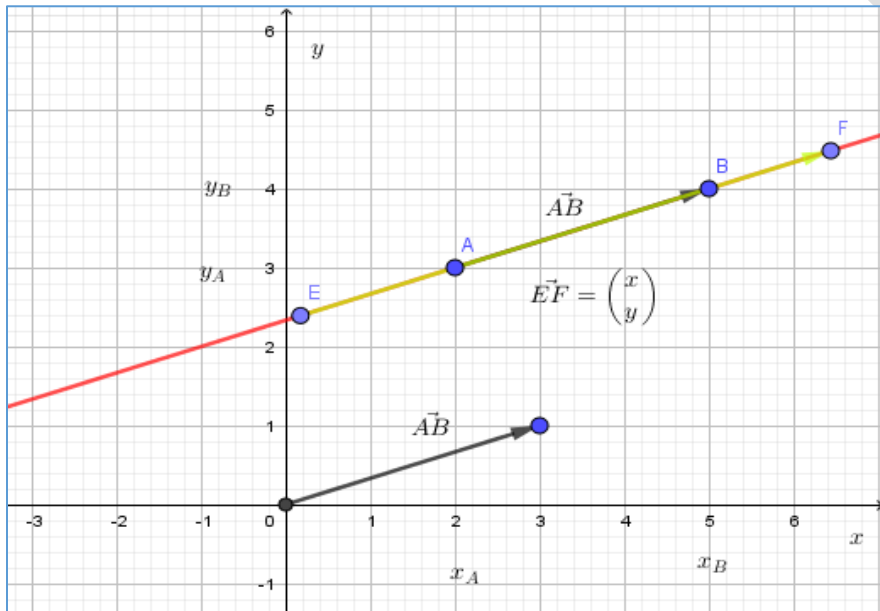


Content Summary

Two points $A(x_A, y_A)$ and $B(x_B, y_B)$ form a vector $\overrightarrow{AB} = \begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}$.

A vector parallel to $\overrightarrow{AB} = \begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}$ and passing to the point $B(x_B, y_B)$ is of the form

$$\vec{w} = (x, y) = r\overrightarrow{AB} + A$$



Therefore, The vector equation of the line is: $(x, y) = \overrightarrow{OA} + r\overrightarrow{AB}$ or $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_A \\ y_A \end{pmatrix} + r \begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}$,

where $\overrightarrow{AB}$ is the direction vector.

The parametric equations of the line are:

$$\begin{cases} x = x_A + r(x_B - x_A) \\ y = y_A + r(y_B - y_A) \end{cases}$$

The Cartesian equation is obtained in this way:

$$x = x_A + r(x_B - x_A), \quad r = \frac{x - x_A}{x_B - x_A}$$

$$y = y_A + r(y_B - y_A), \quad r = \frac{y - y_A}{y_B - y_A}$$

Equalizing the value of r , we find the Cartesian equation: $\frac{x - x_A}{x_B - x_A} = \frac{y - y_A}{y_B - y_A}$

This can be expanded to the standard form: $x(y_B - y_A) - y(x_B - x_A) = x_A(y_B - y_A) - y_A(x_B - x_A)$

Example

Find the vector equation of the straight line passing through $A(3, 2)$ and $B(4, 1)$.

Solution

The direction vector is $\overrightarrow{AB} = (1, -1)$, r is a parameter. The vector equation is $(x, y) = \overrightarrow{OA} + r\overrightarrow{AB}$ or $(x, y) = (3, 2) + r(1, -1)$.

The parametric equations: is $\begin{cases} x = 3 + r \\ y = 2 - r \end{cases}$

The symmetric equation, is $\frac{x-3}{1} = \frac{y-2}{-1}$ or $y-2 = -x+3$ or $x+y=5$



Application activity 2.8

1) Answer by True or False

- A straight line can be determined by two distinct points.
- The slope of a line passing through $A(1, 2)$ and $B(4, 8)$ is 2.
- The equation of a line passing through two points can be found using the point-slope form.

2) Determine the equation of a straight line passing through $P(2, 4)$ and $B(3, -2)$.

2.9. Equation of a straight line passing through given point and its gradient



Activity 2.9

A road engineer is designing a straight road that starts at a bus station located at point $A(2, 3)$ on a map. The road must rise steadily with a gradient (slope) of 2. To represent this road on the

coordinate map, the engineer needs to determine the equation of the straight line passing through the given point with the given gradient.

Answer the following questions:

- 1) What does the gradient (slope) of a line represent?
- 2) If we know the gradient and one point on the line, can we determine the equation of the line?
- 3) Which formula can we use to find the equation of such a line?



Content Summary 2.9

Consider a line having gradient m and passing through the point $P_1(x_1, y_1)$. Suppose that the point $P(x, y)$ is another point on the line. Then the gradient of the line is the rate at which the line rises (or falls) vertically for every unit across to the right. It is defined by the change in y to the change in x .

$m = \frac{y - y_1}{x - x_1}$. Thus, the equation of the line in point-slope form is defined by $y - y_1 = m(x - x_1)$

From the equation above, if we take any other point $P_2(x_2, y_2)$ that lies on this line, then:

Vector equation: $\overrightarrow{OP} = \overrightarrow{OP_1} + r\overrightarrow{P_1P_2}$ or $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + r \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$, $\overrightarrow{P_1P_2}$ is the direction vector.

Parametric equation: $\begin{cases} x = x_1 + r(x_2 - x_1) \\ y = y_1 + r(y_2 - y_1) \end{cases}$

Symmetric (to Cartesian) equation: $\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1}$

Example 1

Write the equation of the line that has slope -3 and passes through the point $(-1, 7)$

Solution

$y - y_1 = m(x - x_1)$ use point-slope form

$y - 7 = -3(x - (-1))$ substitute $(-1, 7)$ for (x_1, y_1) and -3 for m .

$y - 7 = -3(x + 1)$.

Note: The general equation of the line is $Ax + By + C = 0$, where A, B, C are constants, and, $A \neq 0, B \neq 0$.

a) If $A = 0$ the line is horizontal.

- b) If $B = 0$ the line is vertical.
 c) If $C = 0$ the line passes through the origin.

On the other hand, the line has slope $m = \frac{-A}{B}$ and the y-intercept is $b = \frac{-C}{B}$.

- d) If the two lines are parallel, then their slopes/gradients are equal.

Therefore, $m_1 = m_2$.

Thus, the equations $Ax + By + C = 0$ and $Ax + By + D = 0$ are parallel.

- e) If two lines are perpendicular, then the product of their slopes/gradients is equal to -1 .

Therefore, $m_1 \times m_2 = -1$.

Thus, the equations $Ax + By + C = 0$ and $Bx - Ay + D = 0$ are perpendicular.

Example 2

Find the equation for the line that contains the point $(5, 1)$ and is parallel to $y = \frac{3}{5}x + 3$.

Solution

Identify the slope of the given line

$$y = \frac{3}{5}x + 3 \Rightarrow m = \frac{3}{5}$$

Write the equation of the line through point $(5, 1)$ and $m = \frac{3}{5}$

$$y - y_1 = m(x - x_1) \Rightarrow y - 1 = \frac{3}{5}(x - 5) \Rightarrow y = \frac{3}{5}x - 3 + 1 \Leftrightarrow y = \frac{3}{5}x - 2$$

Example 3

Find the equation for the line that contains the point $(0, -2)$ and is perpendicular to $y = 5x + 3$?

Solution

Identify the slope of the given line and write its negative reciprocal.

$$y = 5x + 3 \Rightarrow m = 5 \text{ and it has negative reciprocal } \frac{-1}{5}$$

Write the equation of the line through point $(0, -2)$ and $m = \frac{-1}{5}$

$$y - y_1 = m(x - x_1) \Rightarrow y - (-2) = \frac{-1}{5}(x - 0) \Leftrightarrow y = \frac{-1}{5}x - 2.$$



Application activity 2.9

1) Answer by True or False

- A straight line can be determined if we know one point and its gradient.
- The equation $y - 4 = 3(x - 1)$ represents a line with gradient 3 passing through point $(1, 4)$.
- The gradient of the line $y = 5x + 2$ is 2.

2) Write the equation in point-slope form for the line through the given point that has the given slope.

a) $(3, -4); m = 6$

d) $(4, 2); m = \frac{-5}{3}$

b) $(-2, -7); m = \frac{-3}{2}$

e) $(4, 0); m = 1$

c) $(-5, 2); m = 0$

f) $(1, -8); m = \frac{-1}{5}$

3) Write an equation of the line that contains the point $(-3, -5)$ and the same slope as $y + 2 = 7(x + 3)$

2.10. Position of a point to the line

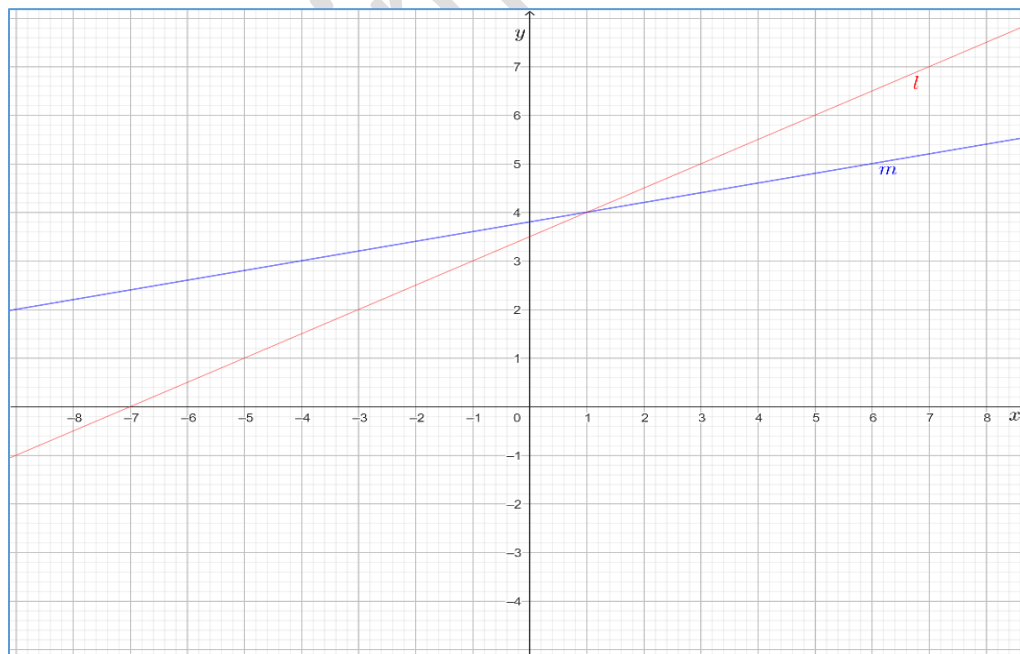


Activity 2.10

1) From the following points: $A(4, 5)$, $B(6, 2)$ and $C(8, 7)$; Select the one that is lying on the line with equation $y = \frac{1}{2}x + 3$

2) The x and y values shown in the table represent points on one of the lines shown in the graph below. Which line is it?

x	-3	1	5
y	2	4	6





Content Summary

A point represents position of an object only. It has zero size (that is, zero length, zero width, and zero height). A line (straight line) can be thought of as a set of many points infinitely connected. A line has infinite length, zero width, and zero height. To verify whether a point P with coordinates (x_1, y_1) lies on the line L with equation $ax + by + c = 0$, verify if it satisfies the given equation by replacing x by x_1 and y by y_1 to get whether $ax_1 + by_1 + c = 0$ (the line L passes through the point P). Otherwise, the point P is not on the line L (the line L does not pass through the point P).

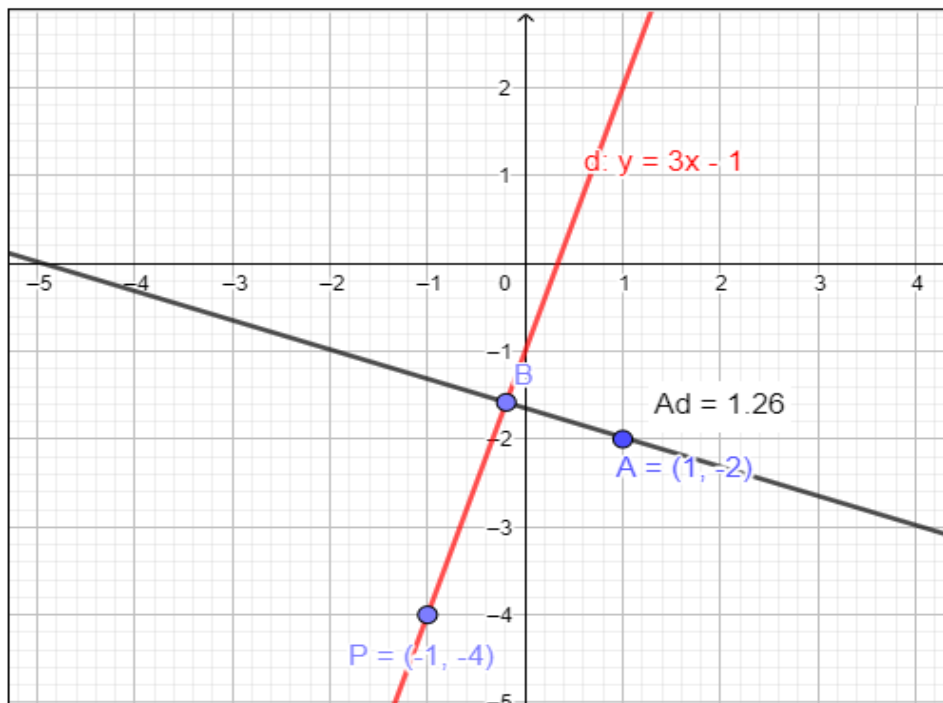
Example

Given a line d with equation $y = 3x - 1$. Show that the point $P(-1, -4)$ lies on that line and show that the point $A(1, -2)$ is outside of that line.

Solution

Replacing y by -4 and x by -1 in the equation $y = 3x - 1$ yields $-4 = 3(-1) - 1 \Rightarrow -4 = -4$

The left side is equal to the right side, thus the point $(-1, -4)$ lies on the line with equation $y = 3x - 1$.



Replacing y by -2 and x by 1 in the equation $y = 3x - 1$ yields $-2 \neq (3 \times 1) - 1$ or $-2 \neq 2$.

This means that the point A is out of the line d .

Note: We can verify the position of a point and a line by calculating the distance from that line and the point. The perpendicular distance from a point $A(x_1, y_1)$ to the line d of equation

$$ax + by + c = 0 \text{ is given by } D(d, A) = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|.$$

If this distance is zero, the given point is on the line. If the distance is different from zero, the point is outside the line.

In our example, the distance between the line d and the point A is 1.2 units which is different from zero. So, A is outside the line.



Application activity 2.10

- 1) From the following points, indicate those which do not lie on the line with equation $2x - 3y + 1 = 0$

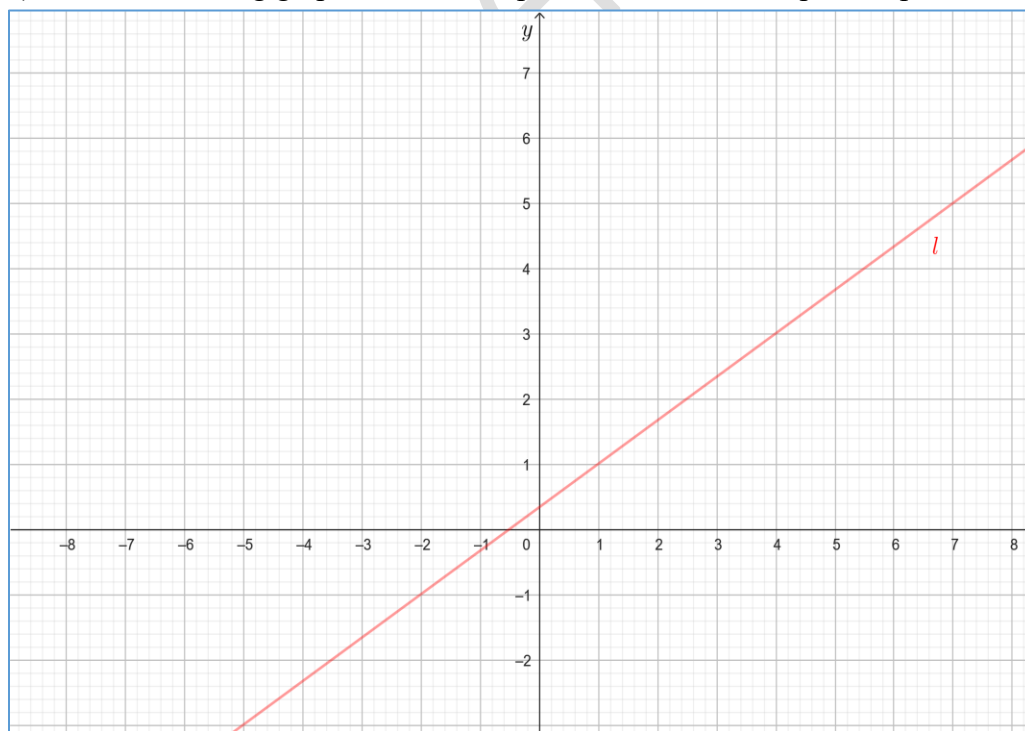
$A(-1, 2)$, $B(1, 0)$, $C(1, 5)$, $D(-2, -1)$, $E(3, 4)$, $F(4, 3)$.

- 2) Select the point that is lying on the line with equation : $y = \frac{3}{4}x + 3$

$A(4, 8)$ $B(6, 1)$ $C(7, 3)$

- 3) Given that the line with equation $y = 3x + b$ passes through the point $(1, 4)$.
find the value of b

- 4) In the following graph, the ordered pairs from the table represent points on line l



Complete the missing values in the table.

x	-4	1	—	—
y	—	—	3	5

2.11. Position of two lines



Activity 2.11

Draw two straight lines on the board that cross each other.

Answer the following questions

- 1) What do you observe about the two lines?
- 2) Do the lines meet at some point?
- 3) What do we call the point where the two lines meet?



Content Summary

Let the equations of two intersecting straight lines be $l_1 \equiv a_1x + b_1y + c_1 = 0$ (i) and $l_2 \equiv a_2x + b_2y + c_2 = 0$(ii) . Suppose the above equations of two intersecting lines meet at $P(x_1, y_1)$. Then, (x_1, y_1) will satisfy both the equations (i) and (ii). Therefore, $a_1x_1 + b_1y_1 + c_1 = 0$ and $a_2x_1 + b_2y_1 + c_2 = 0$

Solving the above two equations by using the method of cross-multiplication, we get,

$$\frac{x_1}{b_1c_2 - b_2c_1} = \frac{y_1}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

$$\text{Therefore, } x_1 = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1} \text{ and } y_1 = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}, a_1b_2 - a_2b_1 \neq 0$$

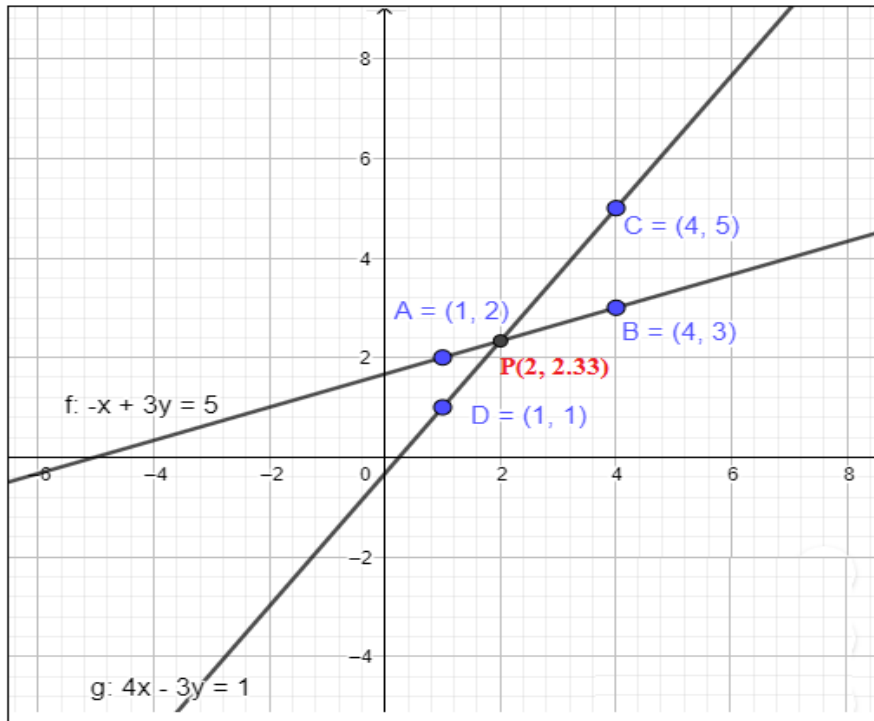
The two intersecting lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$

meet at the point $P(x_1, y_1)$ if $(x_1, y_1) = \left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right), a_1b_2 - a_2b_1 \neq 0$

Notes:

To find the coordinates of the point of intersection of two non-parallel lines, we solve the given equations simultaneously and the values of x and y obtained determine the coordinates of the point of intersection.

For example, the line $L_1 \equiv -x + 3y + 5$ passing through the point $A(1, 2)$ and the point $B(4, 3)$ and the line $L_2 \equiv 4x - 3y = 1$ passing through the point $C(4, 5)$ and the point $D(1, 1)$ meet at the point $P(2, 2.33)$:



Example

Find the coordinates of the point A where the line $L_1 \equiv x + y = 5$, and $L_2 \equiv x - y = 1$ meet.

Solution

Add the equations
$$\begin{cases} x + y = 5 \\ x - y = 1 \end{cases} \Rightarrow 2x = 6 \Leftrightarrow x = 3$$

Then replace the value of x in the equation $x + y = 5$ to get:

$3 + y = 5 \Leftrightarrow y = 2$. Thus, the L_1 and L_2 lines intersect at $A(3, 2)$



Application activity 2.11

- 1) Choose the correct answer; Two lines intersect when:
 - a) They have the same slope
 - b) They meet at one point
 - c) They never meet
 - d) They have no solution
- 2) Answer by Yes or No.
 - a) Two intersecting lines meet at exactly one point.
 - b) To find the intersection point of two lines, we solve their equations together.
 - c) Parallel lines have an intersection point.
- 3) Find the position of the given pair of straight lines and their point of intersection if any.

a) $x + y + 9 = 0$ and $3x - 2y + 2 = 0$

b) $x - y + 3 = 0$ and $-x + y = 1$

2.12. Condition of parallelism



Activity 2.12

Draw two lines on the board that never meet.

Answer the following questions:

- 1) Do the two lines meet?
- 2) How do we call lines that never meet?
- 3) What do you notice about their direction?



Content Summary

Two lines are **parallel** if they **never meet**, even if they are extended.

$$\text{If } a_1b_2 - a_2b_1 = 0, a_1b_2 = a_2b_1 \Rightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2} \Leftrightarrow -\frac{a_1}{b_1} = -\frac{a_2}{b_2}$$

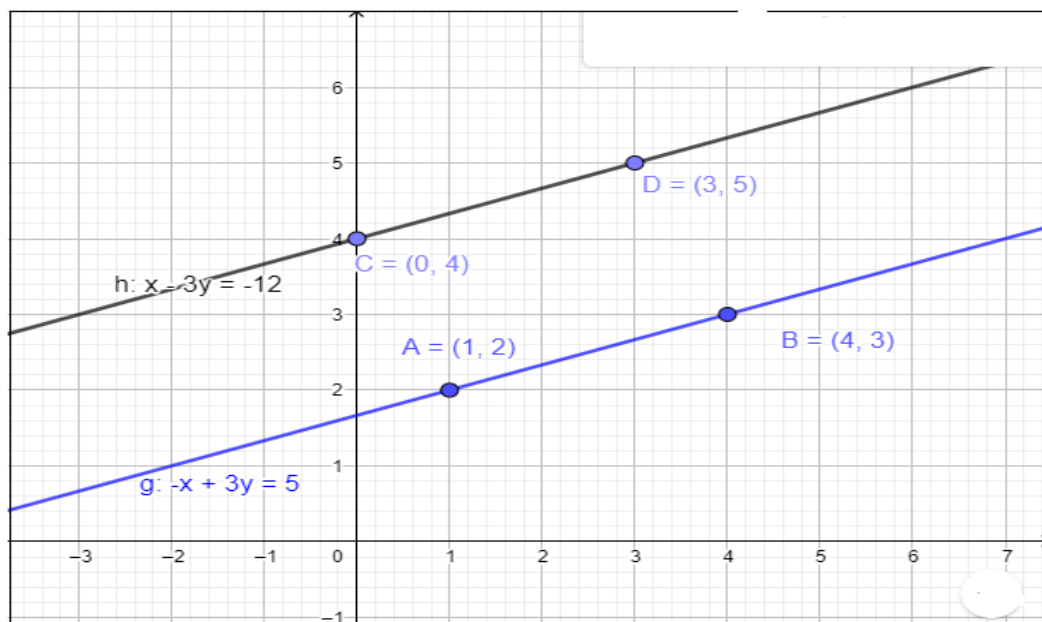
i.e., the slope of line (i) is equal to the slope of line (ii). Therefore, in this case the straight lines (i) and (ii) are parallel.

Two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are parallel if they have the same slope:

$$-\frac{a_1}{b_1} = -\frac{a_2}{b_2}.$$

For example, the line h with equation $x - 3y = -12$ and the line g with equation $x - 3y = -5$

represented below are parallel. Their slope $m = \frac{1}{3}$ is the same.



Example

Find the coordinates of the point of intersection of the lines $2x - y + 3 = 0$ and $x + 2y - 4 = 0$

Solution

We know that the co-ordinates of the point of intersection of the lines $a_1x + b_1y + c_1 = 0$ and

$$a_2x + b_2y + c_2 = 0 \text{ are } \left(\frac{b_1c_1 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right), a_1b_2 - a_2b_1 \neq 0$$

Given equations are

$$2x - y + 3 = 0 \dots (i)$$

$$x + 2y - 4 = 0 \dots (ii)$$

Here $a_1 = 2$, $b_1 = -1$, $c_1 = 3$, $a_2 = 1$, $b_2 = 2$ and $c_2 = -4$

$$\left(\frac{(-1)(-4) - (2)(3)}{(2)(2) - (1)(-1)}, \frac{(3)(1) - (-4)(2)}{(2)(2) - (1)(-1)} \right) \Rightarrow \left(\frac{4 - 6}{4 + 1}, \frac{3 + 8}{4 + 1} \right) \Rightarrow \left(\frac{-2}{5}, \frac{11}{5} \right)$$

Therefore, the co-ordinates of the point of intersection of the lines $2x - y + 3 = 0$ and

$x + 2y - 4 = 0$ are $\left(\frac{-2}{5}, \frac{11}{5} \right)$. Alternatively, by solving simultaneous equations

$2x - y + 3 = 0 \dots (i)$ and $x + 2y - 4 = 0 \dots (ii)$ using different methods you get the same answer.

$$\left(\frac{-2}{5}, \frac{11}{5} \right).$$



Application activity 2.12

- 1) Choose the correct answer; two lines are parallel if:
 - a) They meet at one point
 - b) They have the same slope
 - c) They have different slopes
 - d) They cross each other
- 2) Answer **Yes** or **No**.
 - a) Two parallel lines never meet.
 - b) Parallel lines have the same slope.
 - c) If two lines have different slopes, they are parallel.
- 3) Determine whether the following lines are either parallel or not.
 - a) $x = 3 - 2y$ and $2x - y = 3$
 - b) $3x + 6y = 9$ and $x + 2y = 3$

2.13. Angles between two lines



Activity 2.13

You are at a crossroads in your town where two streets meet. You notice that some cars turn sharply while others turn gently. The angle formed by the streets determines how sharp or gentle the turn is. How can we find the size of the angle formed between these two streets using mathematics?



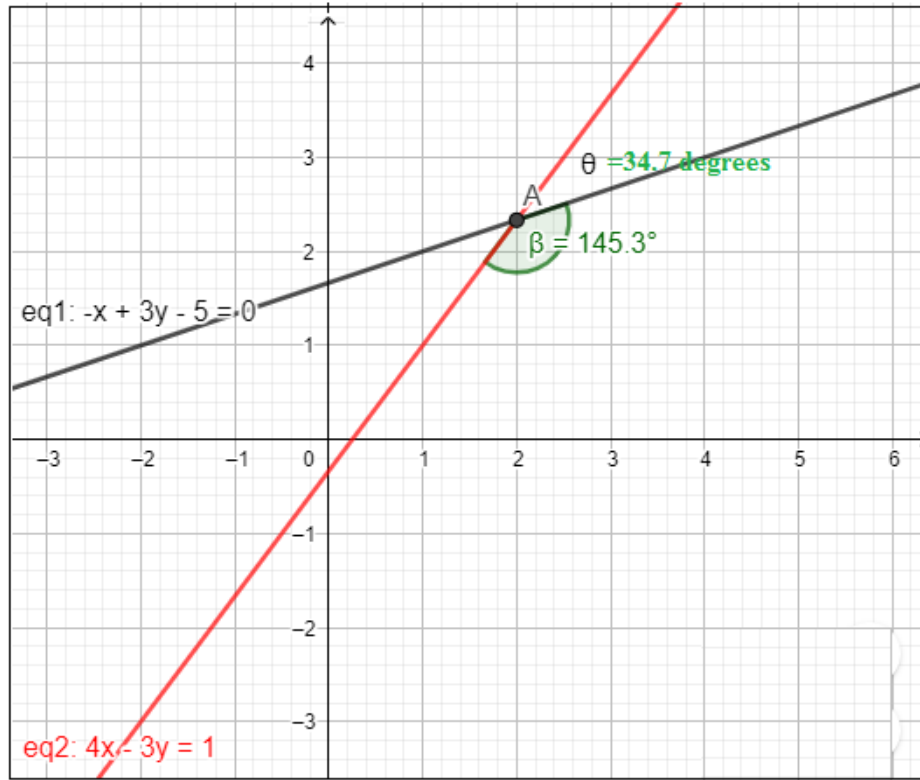
Content Summary

The angle between two lines is the measure of the inclination between them. For two intersecting lines, there are two types of angles between them; the **acute angle** and the **obtuse angle**. Here we consider the acute angle between the lines to be the angle between two lines. Instead of drawing and measuring, the angle between two lines can be calculated from the slopes of the lines or from the equation of the two lines. The simplest formula to find the angle between the two lines is from the slopes of the two lines; The angle between two lines with slopes m_1 and m_2 respectively can be computed by using the trigonometric tangent function.

The **acute angle** θ between two lines is such that $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$.

For example, the line f with equation $-x + 3y = 5$ has the slope $m_1 = \frac{1}{3}$; The line g with equation

$4x - 3y = 1$ has the slope $m_2 = \frac{4}{3}$. The angle θ between them is $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$



The second formula to find the angle between the two lines $l_1 \equiv a_1x + b_1y + c_1 = 0$, and

$l_2 \equiv a_2x + b_2y + c_2 = 0$ is given by $\tan \theta = \left| \frac{a_2b_1 - a_1b_2}{a_1a_2 + b_1b_2} \right|$. Also, the angle between the two lines can

be computed using the following formula $\cos \theta = \left| \frac{a_1b_1 + a_2b_2}{\sqrt{a_1^2 + b_1^2} \times \sqrt{a_2^2 + b_2^2}} \right|$.

Example 1

Find the angle between two lines having slopes of $m_1 = 1$ and $m_2 = \frac{1}{2}$ respectively.

Solution

The given slopes of the two lines are $m_1 = 1$ and $m_2 = \frac{1}{2}$.

The formula to find the angle between the two lines is $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$, or

$$\tan \theta = \left| \frac{1 - \frac{1}{2}}{1 + 1 \times \frac{1}{2}} \right| = \left| \frac{\frac{1}{2}}{\frac{3}{2}} \right| = \frac{1}{3}. \text{ Therefore the angle between the lines is } \theta = \tan^{-1} \left(\frac{1}{3} \right) \approx 18^\circ$$



Application activity 2.13

1) Write True or False for each statement.

- The angle between two lines is the smallest angle formed when the two lines intersect.
- If two lines are parallel, the angle between them is 0°

2) Choose the right answer.

If the slopes of two lines are $m_1 = 1$ and $m_2 = -1$, what is the angle between the two lines?

- A. 0° B. 45° C. 90° D. 180°

3) Two lines have equations : $l_1 \equiv y = 3x + 2$, $l_2 \equiv y = -\frac{1}{3}x + 1$. Find the angle θ between these lines.

2.14. Condition of perpendicularity



Activity 2.14

A student is drawing the map of a football field on the board. The field has side lines and goal lines that meet at the corners of the field. Line $L_1 : x = y$ and Line $L_2 : y = -x$

Answer the following questions:

- What type of angle is formed when the two lines intersect?
- Do the lines appear to meet at a right angle 90° ? What are the slopes of the two lines?
- Multiply the slopes. What do you observe?
- Based on your observation, what condition can you suggest for two lines to be perpendicular?



Content Summary

Two lines are **perpendicular** if they intersect and form a right angle 90° . In coordinate geometry, the condition for two lines to be perpendicular depends on their slopes. Let two lines have slopes m_1 and m_2 . The two lines are **perpendicular** if the product of their slopes is equal to -1 . $m_1 m_2 = -1$. This means that the slope of one line is the negative reciprocal of the slope of the other line.

Example 1

Derive the condition for two lines with slopes m_1 and m_2 to be parallel and perpendicular using the angle between two lines formula.

Solution

The angle between two lines with slopes m_1 and m_2 is: $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$.

Note: When the lines are parallel, the angle between two parallel lines is 0° and $\tan 0^\circ = 0$.

Substituting this in the above formula leads to $\frac{m_1 - m_2}{1 + m_1 m_2} = 0 \Rightarrow m_1 = m_2$.

Example 2

Find the angle between two straight lines having the equations $l_1 \equiv 3x + 4y - 10 = 0$, and $l_2 \equiv 4x - 5y + 2 = 0$.

Solution

The given two equations of the lines are $l_1 \equiv 3x + 4y - 10 = 0$, and $l_2 \equiv 4x - 5y + 2 = 0$.

Here, we have $a_1 = 3, b_1 = 4, a_2 = 4$, and $b_2 = -5$. The angle between the two lines can be

calculated using the formula $\tan \theta = \left| \frac{a_2 b_1 - a_1 b_2}{a_1 a_2 + b_1 b_2} \right|$.

$$\text{Thus, } \tan \theta = \left| \frac{a_2 b_1 - a_1 b_2}{a_1 a_2 + b_1 b_2} \right| = \left| \frac{(4 \times 4) - 3 \times (-5)}{(3 \times 4) + 4 \times (-5)} \right| = \left| \frac{16 + 15}{12 - 20} \right| = \left| \frac{31}{-8} \right| = \frac{31}{8}$$

Or $\theta = \tan^{-1} \left(\frac{31}{8} \right) \approx 75.5^\circ$; Therefore, the angle between the lines is about 75.5° .



Application activity 2.14

1) Which pair of slopes represents perpendicular lines?

- A. 2 and $\frac{1}{2}$ B. -5 and $-\frac{1}{5}$ C. 3 and 3 D. -2 and $-\frac{1}{2}$

- 2) Two lines are perpendicular if:
 - A. Their slopes are equal
 - B. Their slopes multiply to -1
 - C. Their slopes multiply to 1
 - D. Their slopes are positive
- 3) True or False
 - a) Two lines are perpendicular if the product of their slopes is -1 .
 - b) Lines with slopes 3 and $-\frac{1}{3}$ are perpendicular.
 - c) Two perpendicular lines always have equal slopes.
- 4) Determine whether the lines with slopes $m_1 = -3$ and $m_2 = \frac{1}{3}$ are perpendicular.

Explain your answer

2.15. Identification of Geometric shapes in 2D



Activity 2.15

- 1) For the following;
 - a) Discuss the difference between regular polygons and irregular polygons and give examples.
 - b) Name the shapes with three sides and give at least 4 examples of materials in real life with three sides.
 - c) Name the shapes with four sides and give at least 4 examples of materials in real life with four sides.
- 2) a) Plot the following points on the $2D$ coordinate plane, then use a ruler to connect the points in the order they are listed to form a polygon.
 - Polygon 1: $A(8,6)$; $B(11,1)$; $C(8,6)$
 - Polygon 2: $A(2,5)$; $B(6,5)$; $C(6,3)$; $D(2,3)$
 - Polygon 3: $A(3,5)$; $B(7,2)$; $C(7,-5)$; $D(3,-2)$
 - Polygon 4: $A(-2,5)$; $B(-2,-2)$; $C(3,-2)$; $D(5,5)$
 - Polygon 5:
 $A(-2,-2)$; $B(2,-2)$; $C(2,2)$; $D(6,2)$; $E(6,-6)$; $F(-4,-6)$; $G(-4,-4)$; $H(-2,4)$
 - b) What is the name of each geometric figure formed?
 - c) Describe the similarities and differences of those figures.



Content Summary

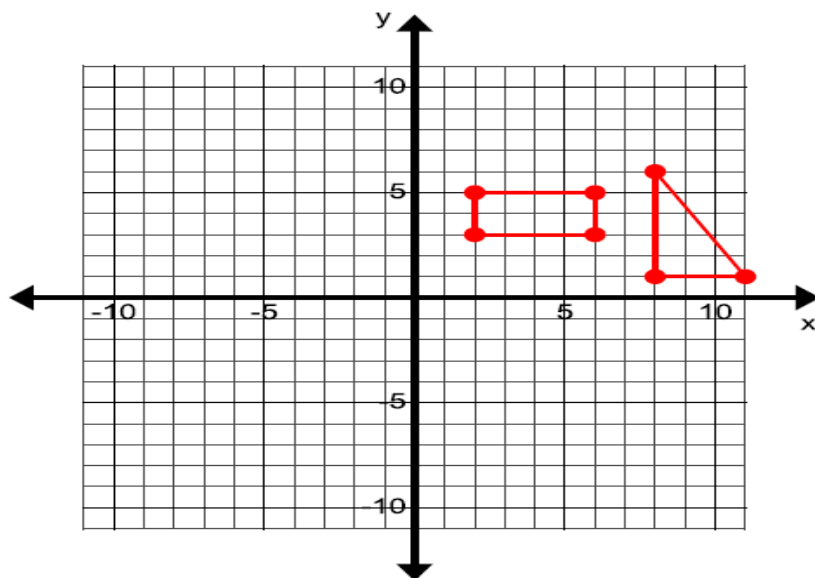
In geometry, a two dimensional shape can be defined as a flat plane figure or a shape that has two dimensions: length and width. Length on x -axis and width on y -axis or vice versa.

Two-dimensional or 2-D shapes do not have any thickness and can be measured in only two dimensions. In the two dimensional coordinate, geometric figures can be plotted:

Example 1

The following points; a) (8,6); (11,1); (8,6) and b) (2,5); (6,5); (6,3); (2,3)

Can make the following polygons in coordinates' graphs



Two dimensional shapes are for example: Triangle, square, rectangle, parallelogram, trapezium, rhombus, circle, pentagon, hexagon, etc. It is also important to take note of the hierarchy, e.g. all rectangles are parallelograms but all parallelograms are not rectangles.

Example 2

Explain why a square is a rectangle but a rectangle is not a square.

Solution

Depending on their properties, a square is a quadrilateral with four right angles and equal sides. The two parallel sides are obviously equal, which allows a square to be a rectangle. Even though the two parallel sides of a rectangle are equal, the four sides are not equal; which means that a rectangle is not a square but a square is rectangle.



Application activity 2.15

1) Answer by **yes** or **No**:

Given two points : $C(1,2)$ and $P(5,-1)$;

- Can you determine the distance between the two points using the distance formula?
- Can these two points be plotted on the Cartesian plane and used to draw a circle with center C?
- Is it possible to find other points that lie on the same circle?

- 2) When you draw a polygon on the coordinate plane by plotting given vertices and connecting them in order,
- Can you determine whether the polygon is regular or irregular?
 - Is it possible to calculate the area of the polygon after plotting it?

2.16. Perimeter and area of geometric shapes in 2D: Square, Triangle, Rectangle



Activity 2.16

A farmer wants to fence and cultivate three different pieces of land. The shapes of the land are:

- A square field
- A rectangular field
- A triangular field

The farmer wants to know:

- How much wire is needed to go around each field?
- How much land is inside each field for planting crops?

Answer the following questions

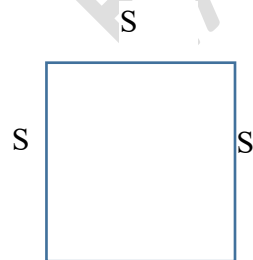
- When the farmer measures the distance around each field, what do we call this in mathematics?
- When the farmer wants to know the space inside the field, what do we call it?
- If a square field has all sides equal, how can we find the distance around it?
- How can we find the space inside a rectangle or a triangle?
- Why do you think it is important to calculate the perimeter and area of shapes in real life?



Content Summary

1) Square

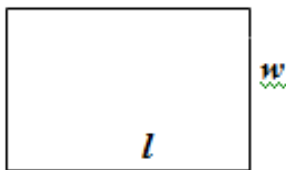
This is a quadrilateral with four equal sides and four right angles (90°).



Its perimeter is the total length around the square. Its area is the space enclosed inside the Square

Formulas: Perimeter(P) = $4S$ (where s is the length of one side) Area(A) = S^2 where, S :Side

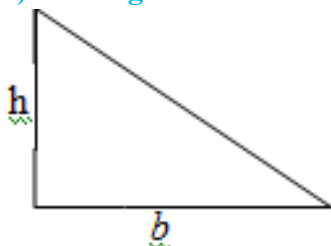
2) Rectangle



This is a quadrilateral with opposite sides equal and all angles 90° . Its perimeter is the total distance around the rectangle. Its area is the space enclosed inside the rectangle

Formulas: Perimeter(P) = $2(w + l)$, Area(A) = $(l \times w)$ l : length, w :width

3) Triangle



This is a polygon with three sides, three vertices, and three angles. Its perimeter is the total length around the triangle. Its area is the space enclosed inside the triangle.

Formulas : Perimeter(P) = $side_1 + side_2 + side_3$ Area(A) = $\frac{1}{2} \times base \times height$ or $A = \frac{1}{2}(b \times h)$

A 3-sided polygon (sum of internal angles = 180°)



Application activity 2.16

1) Answer by True or False

- The perimeter of a square with side 6 cm is 24 cm .
- The area of a rectangle with length 5 cm and width 3 cm is 15 cm^2
- The perimeter of a triangle is the sum of its three sides.

2) Multiple Choice Questions

- The side of a square is 8 m . What is the area?

A. 16 m^2 B. 32 m^2 C. 64 m^2 D. 24 m^2

- A rectangle has length 10 cm and width 4 cm . What is the perimeter?

A. 14 cm B. 28 cm C. 40 cm D. 20 cm

3) Open-ended Questions

- a) Find the perimeter and area of a square with side 12 cm .
- b) A rectangle has length 15 m and width 7 m . Calculate its perimeter and area.

2.17. Perimeter and area of geometric shapes in 2D: Parallelogram, Pentagon and Hexagon



Activity 2.17

A gardener wants to create different flower beds in the school garden:

- One flower bed is a parallelogram,
- One is a pentagon,
- One is a hexagon.

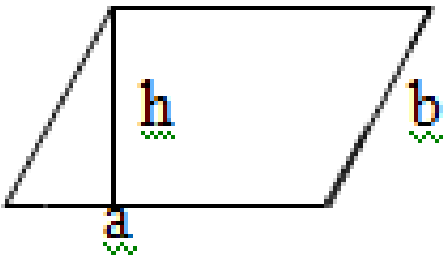
The gardener wants to know how much fencing (perimeter) is needed around each flower bed and how much soil (area) will be needed to fill each bed.

- (a) What do we call the total distance around each flower bed?
- (b) What do we call the space inside each flower bed?
- (c) How do you think the perimeter of a pentagon or hexagon is calculated?
- (d) How do you think the area of a parallelogram is different from the area of a pentagon or hexagon?
- (e) Why is it important to calculate perimeter and area before fencing and planting?



Content Summary

1) Parallelogram

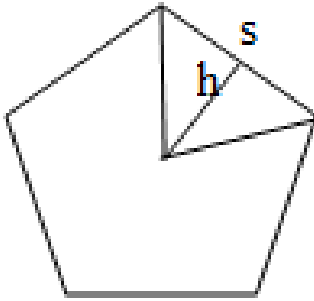


This is a quadrilateral with opposite sides parallel and equal, and opposite angles equal. Its Perimeter is Total length around the shape. Its area is the space inside it.

Formulas: $P = 2 \times (\text{base} + \text{side})$ or $P = 2(b + s)$, (where b is the base, s is the side length)

$A = \text{base} \times \text{height}$ or $A = b \times h$, (height h is perpendicular to the base).

2) Pentagon



A polygon with five sides and five angles. Its Perimeter is sum of all five sides.

Formulas: $P = side_1 + side_2 + side_3 + \dots + side_5$ or $P = 5s$, $A = 5 \left(\frac{s \times h}{2} \right)$

3) Hexagon

A polygon with six sides and six angles. Its Perimeter is sum of all six sides.

Formulas: $P = side_1 + side_2 + \dots + side_6$ or $P = 6s$ Area (regular hexagon) = $3(s \times h)$



Application activity 2.17

1) Answer by True or False

- a) The perimeter of a parallelogram is the sum of all its four sides.
- b) A regular pentagon has five sides of equal length.
- c) The perimeter of a hexagon is six times the length of one side if it is regular.
- d) A parallelogram with all sides equal is always a square.

2) Multiple Choice Questions

- a) A parallelogram has base 12 cm and height 5 cm . What is its area?
 $A. 60\text{ cm}^2$ $B. 30\text{ cm}^2$ $C. 17\text{ cm}^2$ $D. 25\text{ cm}^2$
- b) A regular pentagon has side 7 cm . What is its perimeter?
 $A. 28\text{ cm}$ $B. 35\text{ cm}$ $C. 42\text{ cm}$ $D. 30\text{ cm}$

3) Open-ended Questions

- a) Find the perimeter and area of a parallelogram with base 10 cm , side 6 cm , and height 4 cm
- b) A regular pentagon has side 8 cm and apothem 5.5 cm . Calculate Perimeter and area.



END OF UNIT ASSESSMENT

Assessing Knowledge and Understanding

Section A: True or False

- 1) The midpoint of the line segment joining points $A(2,3)$ and $B(4,5)$ is $(3,4)$.
- 2) Two lines with slopes $m_1 = 2$ and $m_2 = -\frac{1}{2}$ are perpendicular.

Section B: Multiple Choice Questions

- 1) The distance between points $P(1,2)$ and $Q(5,6)$ is:
- A. 5 units B. 4 units C. $\sqrt{32}$ D. 7 units
- 2) A parallelogram has a base of 10 m , a side of 6 m , and a height of 4 m . Which of the following is correct?
- B. Perimeter = 32 m , Area = 40 m^2
- C. Perimeter = 16 m , Area = 24 m^2
- D. Perimeter = 20 m , Area = 40 m^2
- E. A Perimeter = 32 m , Area = 24 m^2

Assessing Skills

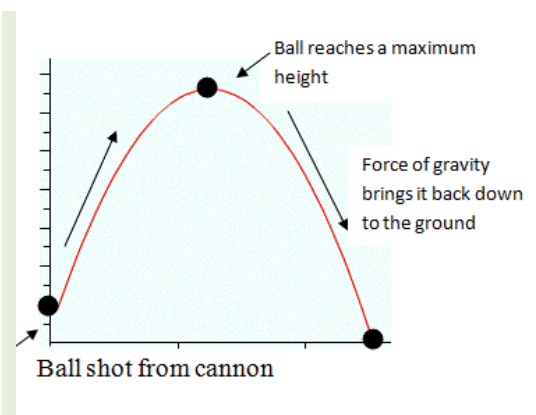
Section C: Open-ended Questions

- 1) A farmer wants to cultivate a triangular field with vertices at $A(0,0)$, $B(6,0)$ and $C(3,4)$
- a) Find the lengths of all sides of the triangle.
- b) Calculate the perimeter of the triangle.
- c) Find the area of the triangular field.
- 2) A garden has three plots:
- A square plot with side 8 m
 - A rectangular plot with length 10 m and width 6 m
 - A parallelogram plot with base 12 m and height 5 m

- c) Calculate the perimeter of each plot.
- d) Calculate the area of each plot.
- e) Which plot requires the most fencing? Which has the largest area?

Assessing attitudes and Values

- 1) Two students are measuring the distance between two points on a map. One student guesses the distance, while the other uses the distance formula.
 - a) Which student shows the correct attitude? Why?
 - b) What values are demonstrated by using correct mathematical methods?
- 2) A farmer divides land into different shapes without calculating the area and later realizes that some land is wasted.
 - a) What mistake did the farmer make?
 - b) How can mathematics help avoid such problems?
 - c) State two values that students develop by applying mathematics in such situations.



UNIT 3: GRAPHS AND FUNCTIONS

Key unit competence: Apply graphical representation of functions in solving economics and financial problems.

Learning objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Determine the domain and range of a function.
- Perform operations on functions.
- Describe constant and identity functions and recognize their properties.
- Explain monomial, polynomial, rational, and irrational functions and state their general forms.
- Explain how to determine the domain of definition for polynomial, rational, and irrational functions.
- Distinguish between even and odd functions and explain their graphical symmetry properties.
- Explain the concept of the inverse of a function and conditions for its existence.
- Describe the graphical representation and key characteristics of linear and quadratic functions.
- Explain the economic meaning of functions such as cost, revenue, profit, price, and consumption functions.

Skills:

- Draw graphs that corresponds to linear, non-linear functions;
- Find the slopes of linear functions and tangents to non-linear function by graphical analysis.
- Use the slope of a linear demand function to calculate point elasticity.
- Analyze, model and solve problems involving linear or quadratic functions and interpret the results.
- Interpret the meaning of functions with two independent variables: marginal revenue and marginal cost functions.
- Determine the domain and range of various types of functions (polynomial, rational, irrational).

- Perform operations on functions (addition, subtraction, multiplication, division, composition) correctly.
- Identify and verify the inverse of a function through algebraic and graphical methods.
- Identify whether a function is even or odd using algebraic and graphical methods.
- Calculate and interpret marginal cost, marginal revenue, and marginal profit from given data or functions.

Attitudes and Values:

- Showing interest in exploring how one quantity depends on another in real-life situations.
- Valuing precision when determining whether a rule truly defines a function.
- Appreciating logical methods used to distinguish valid functions from non-functions.
- Demonstrating patience in analyzing and testing values to find valid inputs and outputs.
- Increase self-confidence and determination to appreciate and explain the importance of functions
- Evaluate cost and profit situations using mathematical reasoning to support decision-making.
- Apply teamwork, discussion, and communication skills when exploring graphical and applied problems.
- Appreciate the usefulness of mathematical models in describing and predicting real-world phenomena.

Introduction

In many real-life situations, one quantity depends on another. For example, the cost of buying items depends on the number of items purchased, and the profit of a business depends on its cost and revenue. Mathematics uses functions to describe and analyze these relationships between quantities. In this unit, learners will study the concept of functions and their graphical representations. They will learn how to determine the domain and range of functions, perform operations on functions, and understand different types of functions such as linear, quadratic, polynomial, rational, and irrational functions. Learners will also study even and odd functions, inverse functions, and how these functions behave when represented graphically.

Furthermore, the unit connects mathematics with economics and financial situations. Students will use graphs and functions to analyze important economic concepts such as cost, revenue, profit, demand, and consumption. Through graphical interpretation and problem solving, students will develop the ability to model real-life situations and make logical decisions using mathematical reasoning.



Introductory activity

Suppose that average weekly household expenditure on food C depends on average net household weekly income y according to the relationship $C = 12 + 0.3y$.

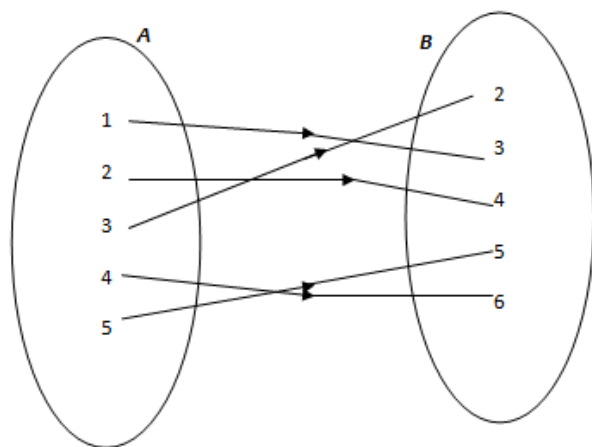
- Can you find a value of y for which c is not a real number?
- Complete a table of value of c from $y = 0$ to $y = 10$ and use it to draw the graph of $C = 12 + 0.3y$
- If $y = 90$, what is the value of c ?

3.1. Generalities on numerical functions



Activity 3.1

In the following Venn diagram, for each element of set A state which element of B is mapped to.



- What is the set of elements of A which have images in B ?
- Determine the set of elements in B which have antecedent in A .
- Is there any element of A which has more than one image?



Content Summary

Function

A function is a rule that assigns to each element in a set A one and only one element in set B . We can even define a function as any relationship which takes one element of one set and assigns to it one and only one element of second set. If x is an element in the domain of a function f , then the element that f associates with x is denoted by the symbol $f(x)$ (read f of x) and is called the image of x under f or the value of f at x .

The set of all possible values of $f(x)$ as x varies over the domain is called the range of f and it is denoted $R(f)$. The set of all values of x in A which have images in B is called the domain of f and denoted $Dom f$.

We shall write $f(x)$ to represent the image of x under the function f . The letters commonly used for this purpose are f , g and h .

Example 1

Given that $f(x) = x^2$, find the values of $f(0)$, $f(2)$, $f(3)$, $f(4)$, $f(5)$.

Solution

$$f(0) = 0^0 = 0; f(2) = 2^2 = 4; f(3) = 3^2 = 9;$$

$$f(4) = 4^2 = 16; f(5) = 5^2 = 25$$

Note: $f(x) = x^2$ can also be written as $f : x \rightarrow x^2$ which is read as “ f is a function which maps x onto x^2 ”

Example 2

Draw arrow diagrams for the functions. Use the domain $\{1, 2, 3\}$

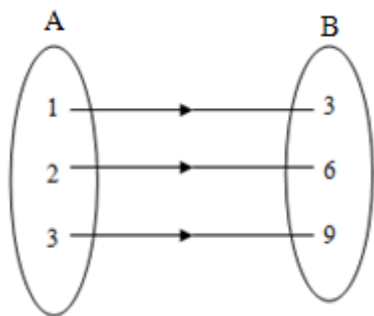
a) $f : x \rightarrow 3x$

b) $x \rightarrow x^2 + 1$

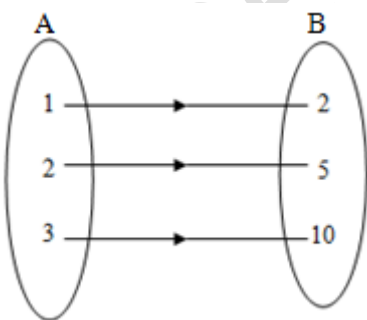
c) $g : x \rightarrow 2x + 1$

Solution

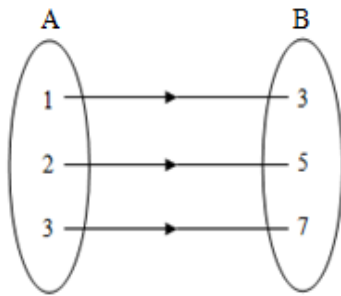
a) $f : x \rightarrow 3x$



b) $h : x \rightarrow x^2 + 1$



c) $g : x \rightarrow 2x + 1$



Example 3

The functions f and g are given as $f(x) = x + 3$ for $x \geq 0$ and $g(x) = x^2$ for $-2 \leq x \leq 3$. State the range of each of these functions.

Solution

If $x \geq 0$, then $x + 3 \geq 3$. Thus, the range of f will be $f(x) \geq 3$

If $-2 \leq x \leq 3$, then $(-2)^2 \leq x^2 \leq (3)^2$. Thus, the range of g will be $4 \leq g(x) \leq 9$

Injective, surjective and bijective functions

Given sets A and B , a function defined from A to B is a correspondence, or a rule that associates to any element of A either one image in B , or no image in B . A function such that every element of A has an image in B is called a **mapping**, thus, under a mapping any element of A has exactly one image in B (not less than one, and not more than one).

A mapping such that every element of B is image of either one element of A , or of no element of A , is called a **one- to- one mapping**, or an **injective mapping** or simply an **injection**; under a one-to-one mapping no two elements of A share the common image in B .

Mathematically, $(\forall x_1 \in A)(\forall x_2 \in A); f(x_2) = f(x_1) \Rightarrow x_2 = x_1$.

A mapping such that every element of B is image of at least one element of A (image of one element of A , or image of more than one element of A), is called an **onto mapping**, or a **surjective mapping** or simply a **surjection**.

Mathematically, $(\forall y \in B)(\exists x \in A); f(x) = y$ or simply $B = f(A)$. A mapping satisfying properties of both one-to-one and onto is said to be a **bijective mapping**, or simply a **bijection**.

In particular, **linear function** $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto f(x) = ax + b$, where $a \neq 0$, is bijective, there is no restrictions on the variables (independent or dependent).

Quadratic function

The function $f: \mathbb{R} \rightarrow \mathbb{R}$

$x \mapsto f(x) = ax^2 + bx + c$, where $a \neq 0$, is not bijective, since some real numbers share images, or some real numbers are not images under function f . But the following restrictions are bijective:

$$1) f : [\frac{-b}{2a}, +\infty[\rightarrow [-\frac{\Delta}{4a}, +\infty[$$

$$x \mapsto f(x) = ax^2 + bx + c, \quad a > 0;$$

$$2) f :]-\infty, -\frac{b}{2a}] \rightarrow [-\frac{\Delta}{4a}, +\infty[$$

$$x \mapsto f(x) = ax^2 + bx + c, \quad a > 0;$$

$$3) f : [\frac{-b}{2a}, +\infty[\rightarrow]-\infty, -\frac{\Delta}{4a}]$$

$$x \mapsto f(x) = ax^2 + bx + c, \quad a < 0 \text{ and}$$

$$4) f :]-\infty, -\frac{b}{2a}] \rightarrow]-\infty, \frac{\Delta}{4a}]$$

$$x \mapsto f(x) = ax^2 + bx + c, \quad a < 0.$$

The homographic function; $f : \mathbb{R} \rightarrow \mathbb{R}$

$x \mapsto f(x) = \frac{ax+b}{cx+d}$ is not bijective, since it is not a mapping ($x = -\frac{d}{c}$ has no image under

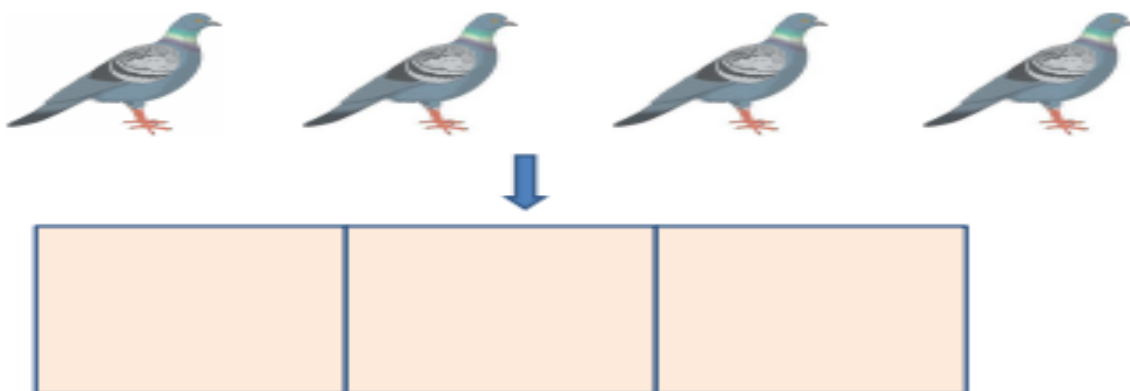
function f , or $y = \frac{a}{c}$ is not image under function f). But the restriction

$$\left. \begin{aligned} f : \mathbb{R} - \{-\frac{d}{c}\} &\rightarrow \mathbb{R} - \{\frac{a}{c}\} \\ x \mapsto f(x) &= \frac{ax+b}{cx+d} \end{aligned} \right\} \text{ is bijective}$$

Example 4

Consider the set of pigeons and the set of pigeonholes on the diagram below to answer the questions. Determine whether it can be established or not between the two sets:

- A mapping,
- A one-to-one mapping,
- An onto mapping,
- A bijective mapping:



Solution

Let the pigeons be numbered a, b, c, d and the pigeonholes be numbered $1, 2, 3$.

- It is possible to establish a mapping between the two sets. For example $\{(a, 1); (b, 2); (c, 3); (d, 3)\}$. This function is a mapping since each pigeon is accommodated in exactly one pigeonhole, though pigeons c and d are in the same pigeonhole.
- It is not possible to establish a one-to-one mapping, since sharing images is not allowed. A function from one finite set to a smaller finite set cannot be one-to-one: there must be at least two elements that have the same image.
- The example given in part (a) illustrates a mapping that is onto: no pigeonhole is empty.
- It is impossible to define a bijection, since it is already impossible to establish a one-to-one mapping.

Example 5

Determine whether function $f: \mathbb{R} \rightarrow \mathbb{R} \quad x \mapsto f(x) = 3x - 5$ is (or is not)

- One-to-one
- Onto
- Bijjective.

Solution

- Let x_1 and x_2 be real numbers such that $f(x_1) = f(x_2)$. Then, $3x_1 - 5 = 3x_2 - 5$. This is equivalent, successively to $3x_1 = 3x_2$ (by adding 5 on both sides); $x_1 = x_2$ (by dividing both sides by 3). Since the equality $f(x_1) = f(x_2)$ implies $x_1 = x_2$, the function is one-to-one.
- Suppose y a real number. Let us look for real number x , if possible, such that $f(x) = y$.

Then, $3x - 5 = y$. It follows that $x = \frac{y+5}{3}$; such x exists for any value of y ;

$$f\left(\frac{y+5}{3}\right) = 3\left(\frac{y+5}{3}\right) - 5 = y \text{ Therefore, function } f \text{ is } \mathbf{onto}.$$

- Since, from points (a) and (b), f is one-to-one and onto, function f is bijective.

Example 6

Show that function f defined by $f: \mathbb{R} \rightarrow \mathbb{R} \quad x \mapsto f(x) = x^2 + 2x - 3$ is neither one-to-one, nor onto.

Solution

$f(-2) = (-2)^2 + 2(-2) - 3 = -3$ and $f(0) = (0)^2 + 2(0) - 3 = -3$. Since, $f(-2) = f(0)$ and $-2 \neq 0$, the function is not one-to-one. On the other side, there is no x such that $f(x) = -5$;

in fact, $f(x) = -5 \Leftrightarrow x^2 + 2x - 3 = -5 \Leftrightarrow x^2 + 2x + 2 = 0$

No such x since $\Delta = 2^2 - 4(1)(2) = -4 < 0$. Therefore, the function is not onto

Example 7

Consider the function f defined by $f: \mathbb{R} \rightarrow \mathbb{R} : x \mapsto f(x) = -x^2 + 4x$.

Determine the greatest subsets A and B of $\mathbb{R}$ such that function

$f: A \rightarrow B : x \mapsto f(x) = -x^2 + 4x$ is bijective

Solution

The maximum value of the function occurs for $x = 2$ and $f(2) = 4$. Therefore,

$A = [2, +\infty[$ and $B =]-\infty, 4]$, or $A =]-\infty, 2]$ and $B =]-\infty, 4]$

Example 8

Function $f: \mathbb{R} - \{a\} \rightarrow \mathbb{R} - \{b\}$

$$x \mapsto f(x) = \frac{2x-5}{3-x} \text{ is bijective}$$

- a) Find the values of a and b
- b) Show that f is one-to-one
- c) Find the real number whose image is 2

Solution

a) f is bijective if $a = 3$ and $b = -2$

b) Let $x_1 \neq 3$ and $x_2 \neq 3$ be such that $f(x_1) = f(x_2)$, that is $\frac{2x_1-5}{3-x_1} = \frac{2x_2-5}{3-x_2}$

Then, $6x_1 - 2x_1x_2 - 15 + 5x_2 = 6x_2 - 2x_1x_2 - 15 + 5x_1$

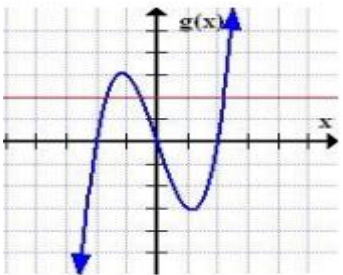
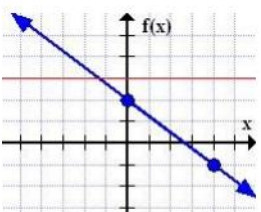
$6(x_1 - x_2) - 5(x_1 - x_2) = 0 \Leftrightarrow x_1 - x_2 = 0 \Leftrightarrow x_1 = x_2$. Therefore, f is one-to-one

c) $f(x) = 2 \Leftrightarrow \frac{2x-5}{3-x} = 2 \Leftrightarrow 2x-5 = 6-2x \Leftrightarrow 2x+2x = 6+5 \Leftrightarrow 4x = 11$

finally, $x = \frac{11}{4}$ (or 2.75). Hence, The real number whose image is 2 is **2.75**.

Horizontal line test

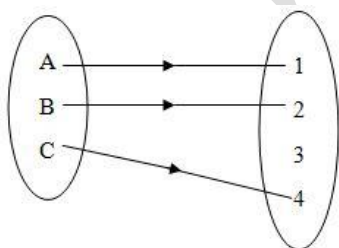
Horizontal Line Test states that a function is a one to one(injective) function if there is no horizontal line that intersects the graph of the function at more than one point.

Graph representation	Interpretation	Conclusion
	You can see that for this graph, there are horizontal lines that intersect the graph more than once.	Not injective
	You can see that for this graph any horizontal line intersects the graph only once.	Injective function

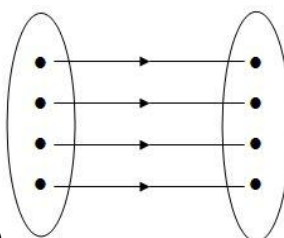


Application activity 3.1

1) State which of the following relations shows a function.



(a)



(b)

2) If $f(x) = 2x + 4$. find:

- a) $f(2)$
- b) $f(-2)$
- c) $f(d)$
- d) The value of a if $f(a) = a$

3.2. Constant function and identity functions



Activity 3.2

Give the same mark of 10 marks to every student who participated in a class activity.

Answer the following questions

- 1) If student A , B , and C all receive 10 marks, does the output change when the student changes?
- 2) What kind of relationship is this, if the output is always the same value?
- 3) Suppose another rule says every number stays the same, for example $1 \rightarrow 1$, $2 \rightarrow 2$, $3 \rightarrow 3$. What do you notice?



Content Summary

a) Constant function

This is a function that assigns the same value to every member of its domain is called a **constant function**. This is $f(x) = c$ where c is a given real number. The function f given by $f(x) = 3$ is constant.

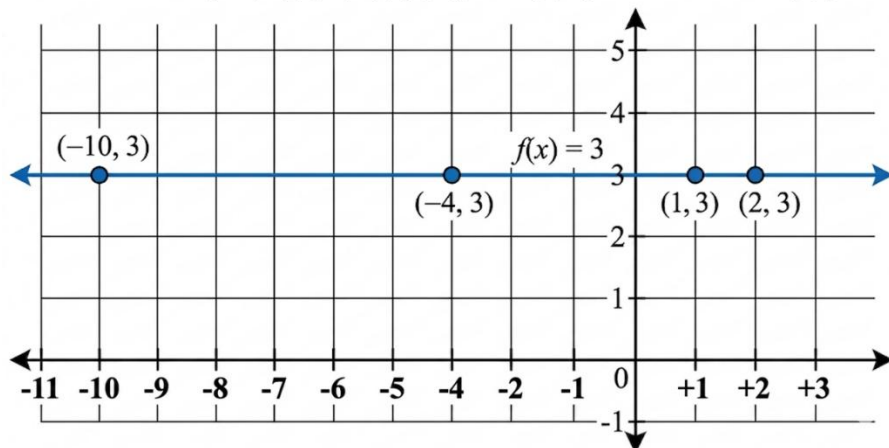
Remark: The constant function that assigns the value c to each real number is sometimes called **the constant function c** .

Example

Graph of a constant function: $f(x) = 3$

x	1	2	-4	-10
$f(x)$	3	3	3	3

Plot of Points $(-10, 3)$, $(-4, 3)$, $(1, 3)$, and $(2, 3)$ on the Function $f(x)=3$



Key properties

- The graph is a horizontal straight line.
- The output value does not change.

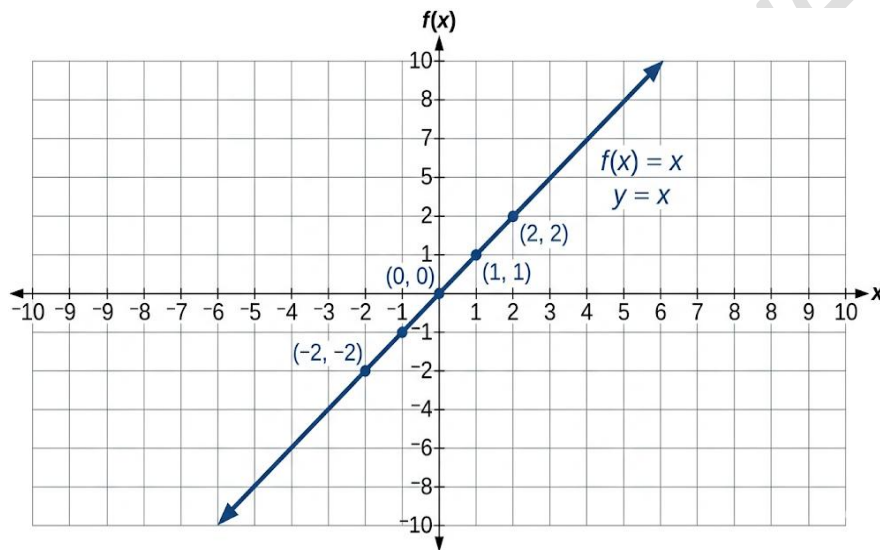
b) Identity Function

An identity function is a function where each input equals its output. General form: $f(x) = x$

Example

Graph of identity function: $f(x) = x$

x	0	1	-2	2
$f(x)$	0	1	-2	2



Key properties

- The graph is a straight line passing through the origin (0,0).
- The slope of the line is 1.



Application activity 3.2

Answer by True or False

- 1) A constant function has the same output for all values of x .
- 2) In an identity function, the value of the input is equal to the output.

3.3. Monomial and polynomial functions



Activity 3.3

State whether the following functions are **Monomials** or **Polynomials**:

- a) $f(x) = 15$
- b) $f(x) = 4x^3 - 2x + 1$
- c) $f(x) = \frac{1}{2}x$
- d) $f(x) = x^2 + 9$
- e) $f(x) = 7xy^2$



Content Summary

a) Monomial

This is a function of the form cx^n , where c is a constant and n is a non-negative integer called a **monomial** in x .

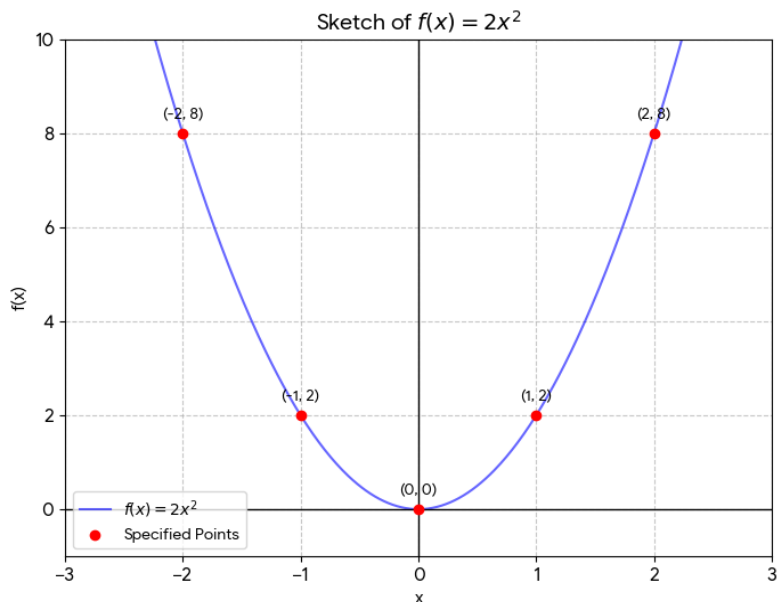
Examples

- 1) $f(x) = 0.3x$ is a monomial in x .
- 2) $f(x) = 2x^3$; $f(x) = \pi x^7$; $f(x) = 4x^0$; $f(x) = -6x$ and $f(x) = x^{17}$ are also monomials

The functions $f(x) = 4x^{\frac{1}{2}}$ and $f(x) = x^{-3}$ are not monomials because the powers of x are not non-negative integers.

Graph of $f(x) = 2x^2$ function and is parabola

x	0	1	-1	2	-2
$f(x) = y$	0	2	2	8	8



b) Polynomial Function

A **polynomial function** is a function made of two or more monomials added or subtracted together. General form: $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, Where the powers of x are whole numbers.

Examples

- 1) $f(x) = x^3 + 4x + 7$ and $f(x) = 17 - \frac{2}{3}x$ are polynomials.
- 2) $g(x) = (x - 2)^3$ is a polynomial in x because it is expressible as a sum of monomials.

In general, f is a polynomial in x if it is expressible in the form;

$f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$ where n is a non-negative integer and $a_0, a_1, \dots, a_n$ are real constants.

Types of polynomials

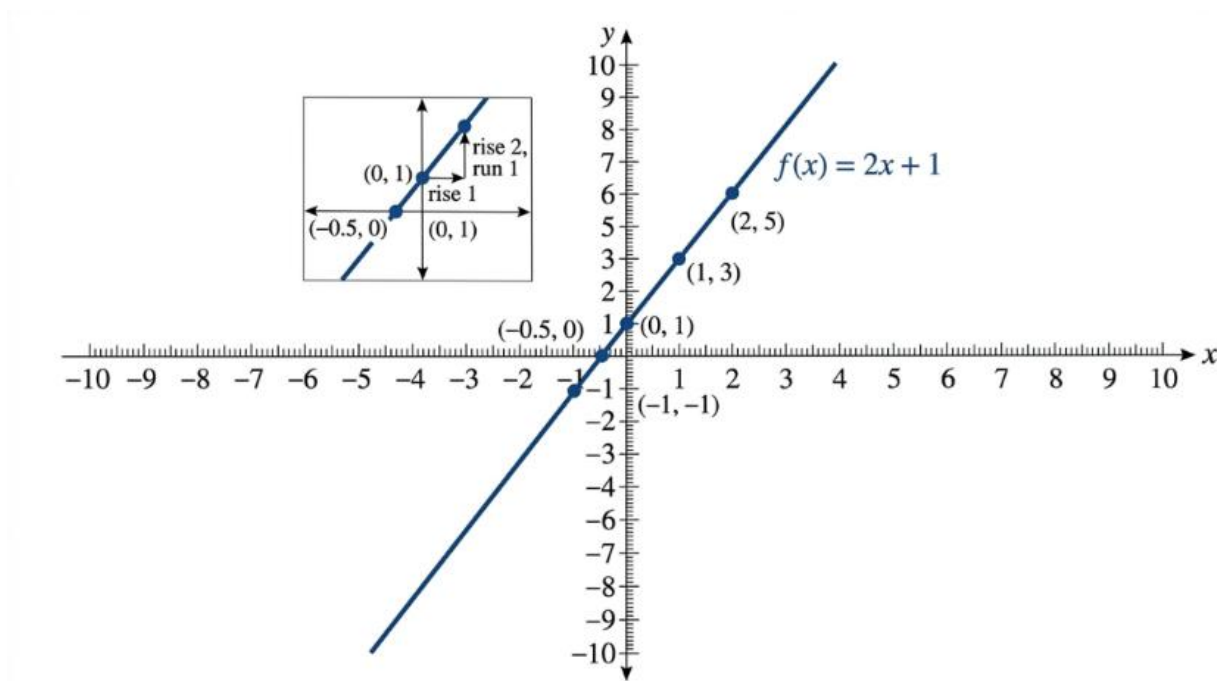
1) Linear polynomial

Is the linear equation with degree of polynomial (n) = 1, it has the form: $a_0 + a_1 x$, $a_1 \neq 0$, e.g:

$$f(x) = 2x + 1$$

Graph of $f(x) \equiv y = 2x + 1$. The following is its table values.

x	0	1	-0.5	2
$f(x)$	1	3	0	5



2) Quadratic polynomial:

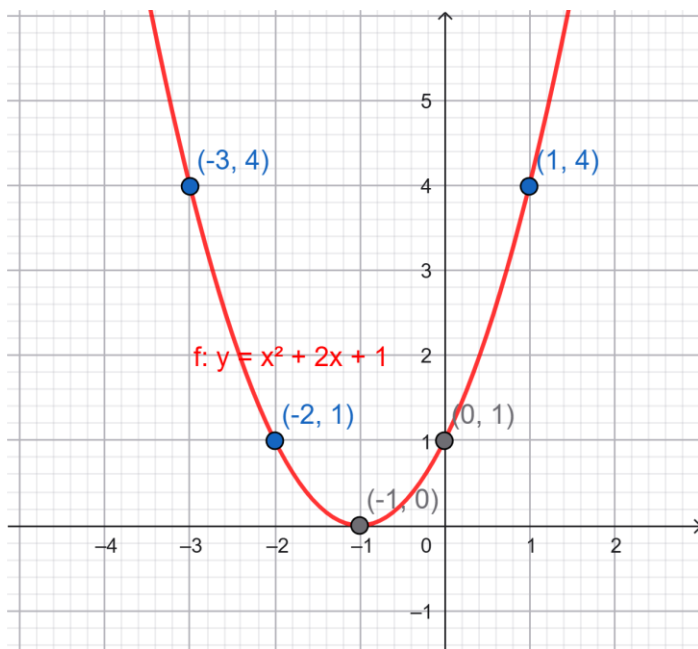
Is a polynomial with the highest power of the variable (n) equal to 2 . it has the form:

$$a_0 + a_1x + a_2x^2, a_2 \neq 0,$$

Example

Given $f(x) = x^2 + 2x + 1$.The following is its table values.

x	0	1	-1	2	-4
$y = f(x)$	1	4	0	9	9



Application activity 3.3

1. Answer by True or False

- A monomial function contains only one term.
- A polynomial function can contain two or more terms.

2. Multiple Choice Questions

(i) Which of the following is a monomial function?

- $f(x) = x^2 + 3x$
- $g(x) = 4x^3$
- $h(x) = x^2 + x + 1$
- $f(x) = 2x + 5$

(ii) Which of the following is a polynomial function?

- $g(x) = \frac{1}{x}$
- $h(x) = x^2 + x + 1$
- $f(x) = \sqrt{x}$
- $f(x) = \frac{3}{x+1}$

3.4. Rational and irrational functions



Activity 3.4

A company produces x items per day. The average cost per item is given by $C(x) = \frac{100}{x} + 5$

- 1) What happens to the value of $C(x)$ when x increases?
- 2) Can the value of x be equal to 0? Explain.
- 3) What type of function might this represent?

Discuss your answers with your classmates.



Content Summary

a) Rational Functions

A function that is expressible as a ratio of two polynomials is called **rational function**. It has the

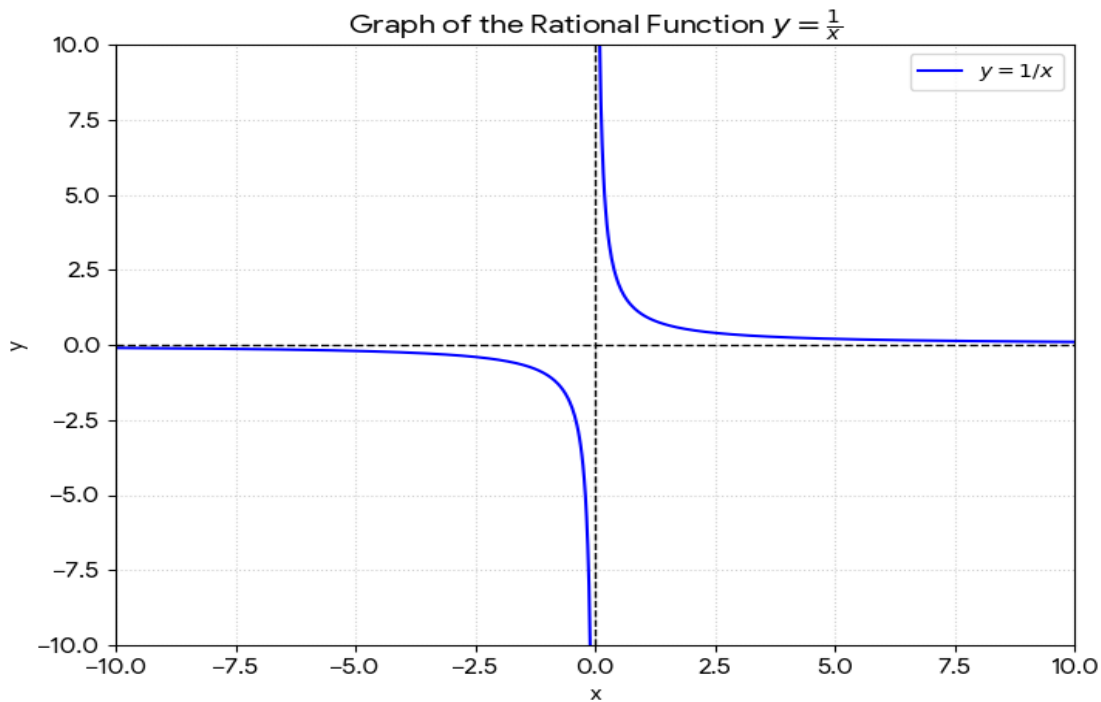
form : $f(x) = \frac{a_0 + a_1x + a_2x^2 + \dots + a_nx^n}{b_0 + b_1x + b_2x^2 + \dots + b_mx^m}$.

Example 1

Given $f(x) = \frac{x^2 + 4}{x - 1}$ and $g(x) = \frac{1}{3x - 5}$ two rational functions

Example 2

A simple rational function is: $y = \frac{1}{x}$



Properties

- The function is not defined at $x = 0$
- The graph has two branches.
- The axes $x = 0$ and $y = 0$ act as **asymptotes**.

b) Irrational functions

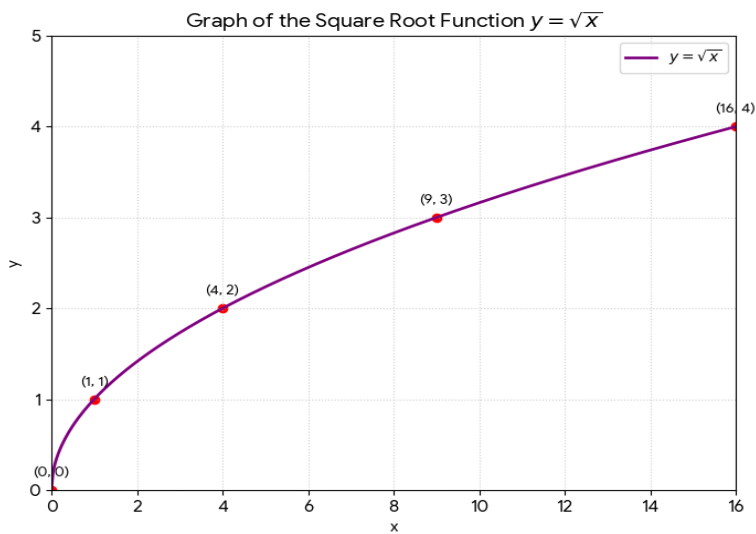
A function that is expressed as a root extractions is called an irrational function. It has the form $\sqrt[n]{f(x)}$, where $f(x)$ is a polynomial or rational function and n is positive integer greater or equal to 2.

Example 1

Given $f(x) = \frac{\sqrt{x^2 + 4}}{\sqrt[3]{x-1}}$ and $g(x) = \sqrt{\frac{x}{x-5}}$ two irrational functions

Example 2

Consider the function: $y = \sqrt{x}$



Properties

- The value inside the square root must be greater than or equal to zero.
- The graph begins at $x = 0$ and increases gradually.



Application activity 3.4

(1) Answer by True or False

- A rational function can be written as the ratio of two polynomials.
- The function $f(x) = \frac{3}{x}$ is defined for all real values of x .

(2) What is the type of the following function?

A. $f(x) = x^3 + 2x^2 - 2$

B. $g(x) = -2$

C. $h(x) = \frac{x^3 + 2x^2 - 2}{x - 5}$

D. $f(x) = \frac{x^2 - 2}{x^2 - 8x + 15}$

3) Provide other types of functions and explain your reasons with examples.

3.5. Domain of definition of polynomial functions



Activity 3.5

Consider the following functions:

a) $f(x) = x^2 + x - 2$

- b) $g(x) = \frac{1}{x}$
- c) $h(x) = \sqrt{x}$

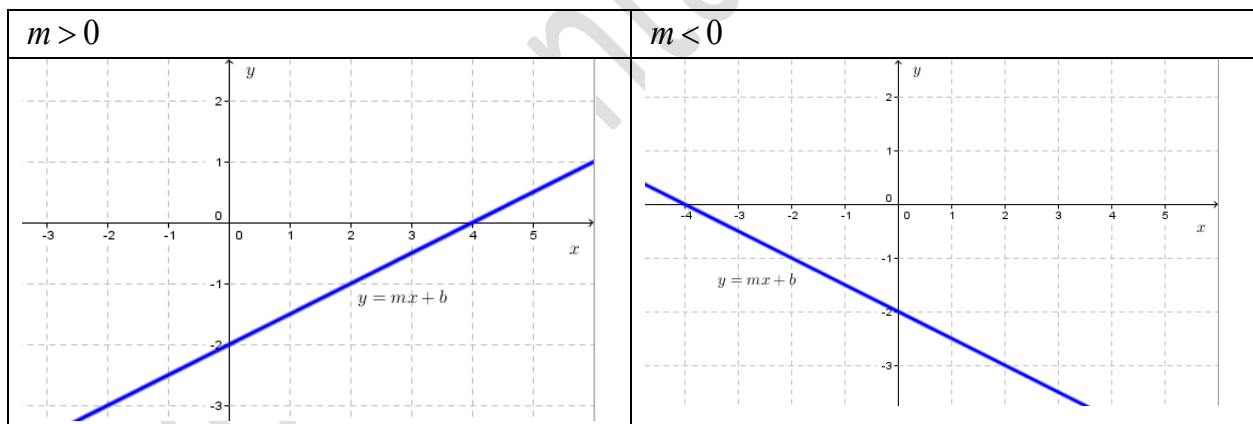
Answer the following questions

- 1) For which values of x can each function be calculated?
- 2) Which function can accept any real number as input?
- 3) What do you notice about the structure of the first function?



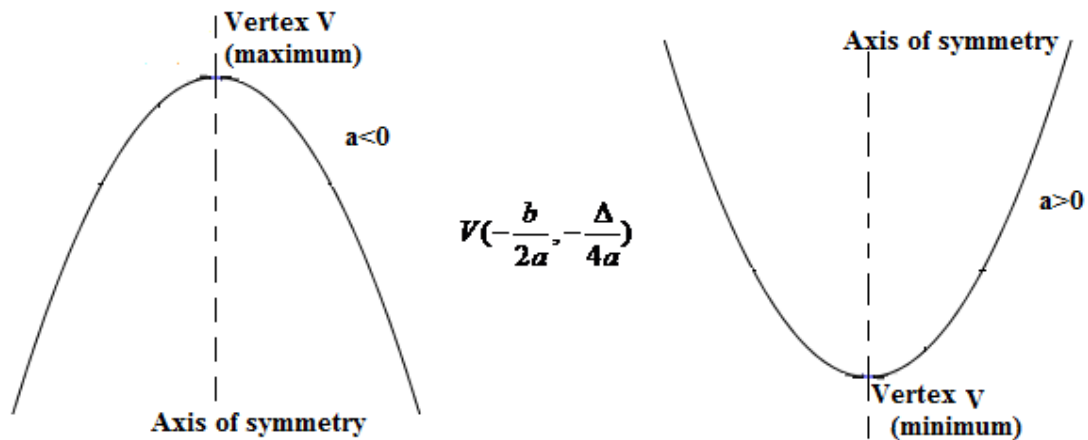
Content Summary

The domain of any **linear function** f is the set of all real numbers, that is $Domf = \mathbb{R}$ or $Dom f =]-\infty, +\infty[$. Similarly, the **range** (the set of all values $f(x)$ for all $x \in Domf$) of a linear function f , denoted $Im f$ is the set of all real numbers, that is $Im f =]-\infty, +\infty[$. Where, Im is the image. Depending on the sign of m in the equation $y = mx + b$, the trend of the graph is as follows:



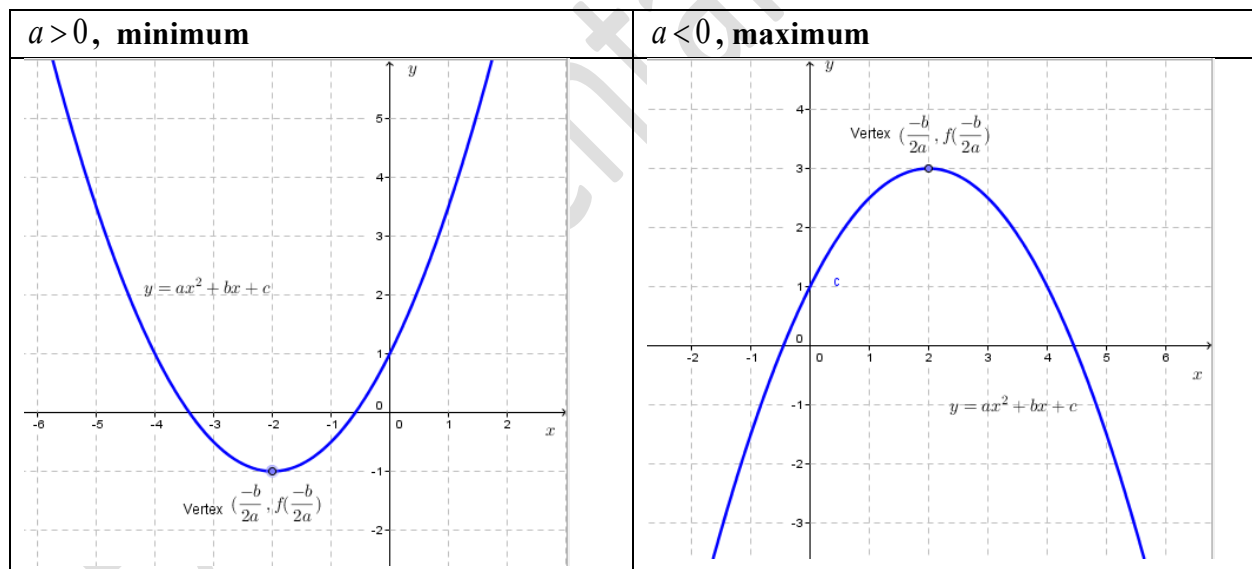
From the graphs, one can observe that each value of x has its corresponding y value.

For quadratic functions $y = ax^2 + bx + c$, the main features are summarized on the graph below:

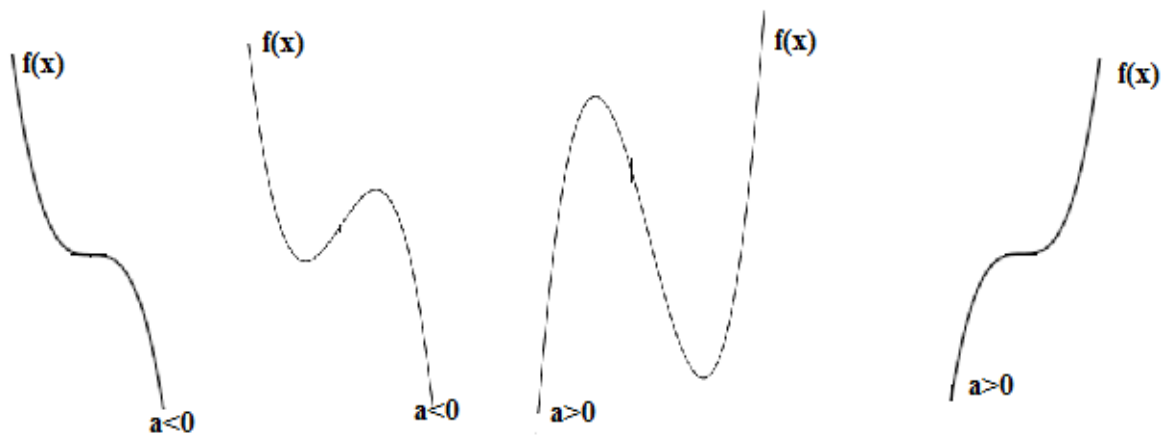


If $a > 0$, then the range of function $y = ax^2 + bx + c$ is $[-\frac{\Delta}{4a}, +\infty[$

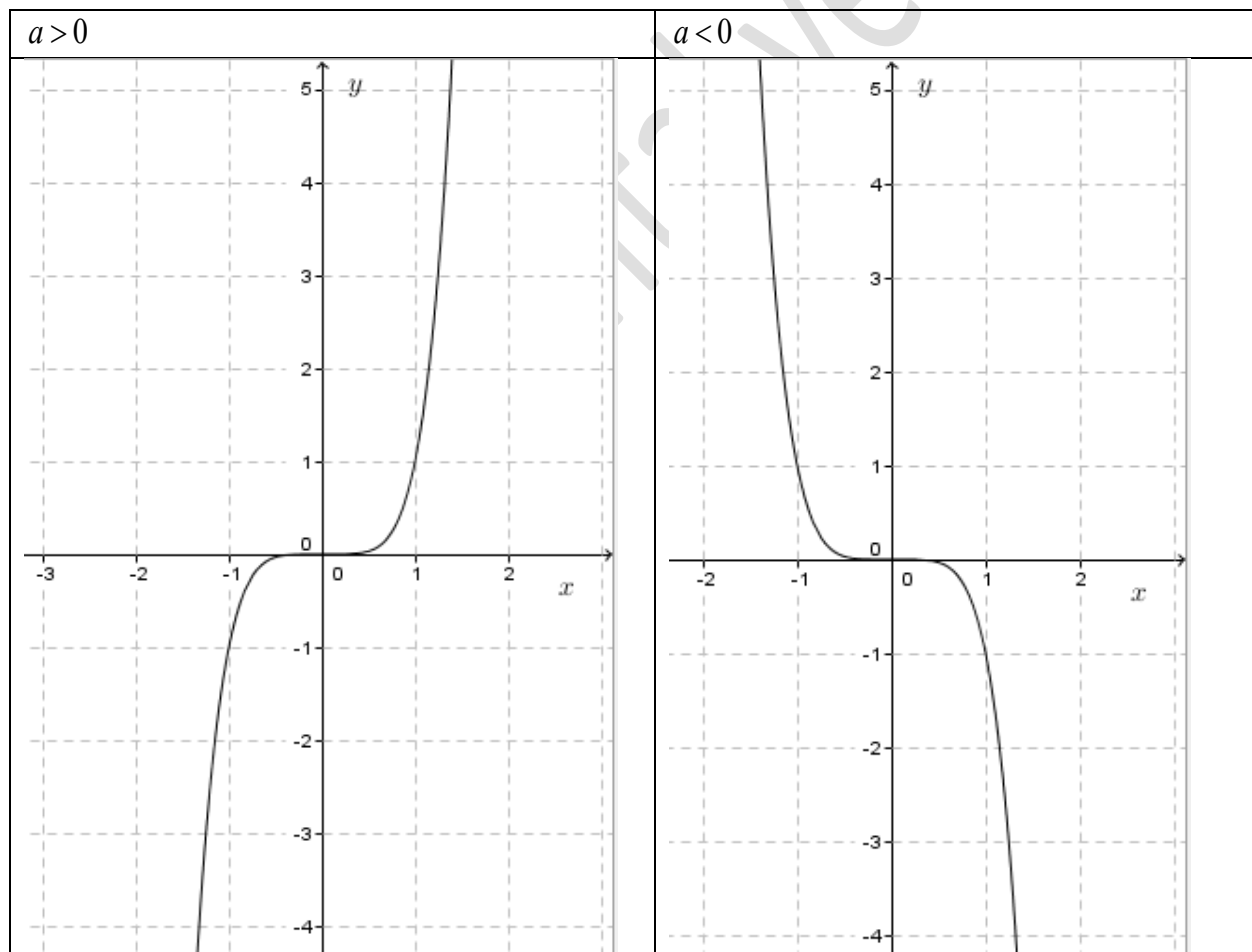
If $a < 0$, then the range of function $y = ax^2 + bx + c$ is $] -\infty, -\frac{\Delta}{4a}]$



For cubic functions $f(x) = ax^3 + bx^2 + cx + d; a \neq 0$, the trends of the graphs are as shown below:



In each case, the domain is $] -\infty, +\infty[$ and the range is $] -\infty, +\infty[$.

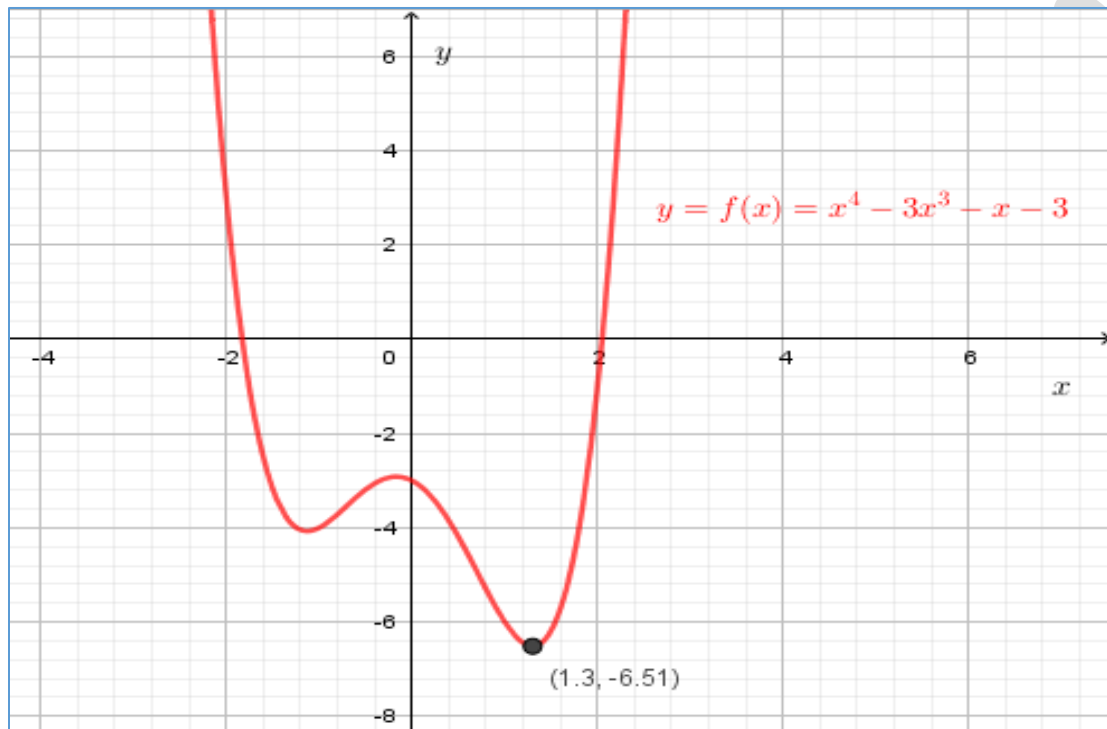


$Dom f =] -\infty, +\infty[$ and $\mathbb{R} =] -\infty, +\infty[$.

Even though the range for polynomials of odd degrees is the set of all real numbers, it is not the case for polynomials of even degree greater or equal to 4. The determination of the range is not easy unless the function is given by its graph; in this case, find by inspection, on the y -axis, the set of all points such that the horizontal lines through those points cut the graph.

Example 1

Determine the domain and range of $f(x) = x^4 - 3x^3 - x - 3$ shown on the graph below:



Solution

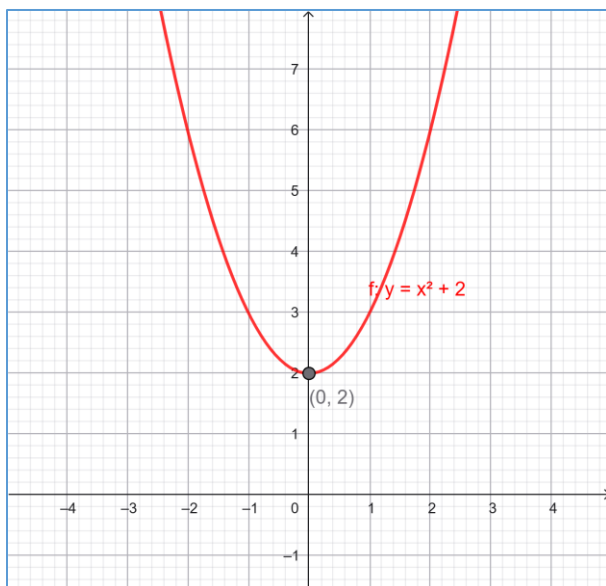
$\text{dom } f =]-\infty, +\infty[$ and $\text{Im } f = [-6.51, +\infty[$

Example 2

Find the range for the function $f(x) = x^2 + 2$

Solution

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by $f(x) = x^2 + 2$. Then the range is $[2, +\infty[$ as shown on the graph below:



Application activity 3.5

Determine the domain of definition and range of the following functions

a) $f(x) = -x^3 + 2x - 1$

b) $f(x) = x^4 - 4$

c) $f(x) = 2x^5 - 3x^2 + 2x - 1$

d) $f(x) = x^2 + 3$

3.6. Domain of definition of rational functions



Activity 3.6

Consider the following functions: $f(x) = \frac{2}{x}$ and $g(x) = \frac{3}{x-4}$

- What happens when $x = 0$ in the $f(x)$ function?
- What happens when $x = 4$ in the $g(x)$ function?
- Why are these values not allowed in the domain?



Content Summary

Given that $f(x) = \frac{g(x)}{h(x)}$ where $g(x)$ and $h(x)$ are polynomials, then the domain of definition is

the set of real numbers excluding all values where the denominator is zero. That is

$$\text{Dom} f = \{x \in \mathbb{R} : h(x) \neq 0\}$$

Example 1

Find the domain of each of the following functions:

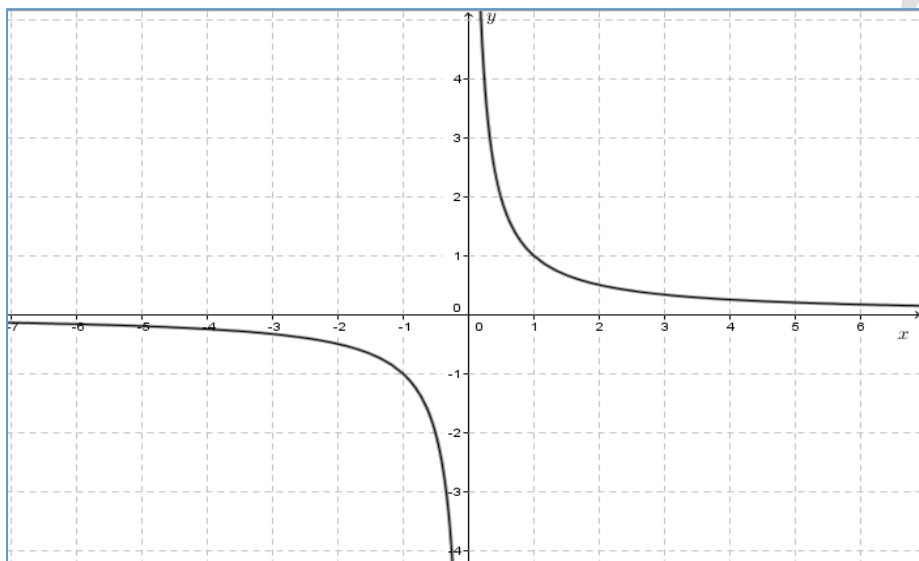
a) $f(x) = \frac{1}{x}$ b) $f(x) = \frac{x}{(x-1)(x+3)}$ c) $f(x) = \frac{x+1}{3x+6}$

Solution

a) The denominator should be different from zero ($x \neq 0$), domain of definition is

$\{x \in \mathbb{R} : x \neq 0\}$ or $\mathbb{R}^* \text{ or } \mathbb{R}^+$. The domain can be written as an interval as follows:

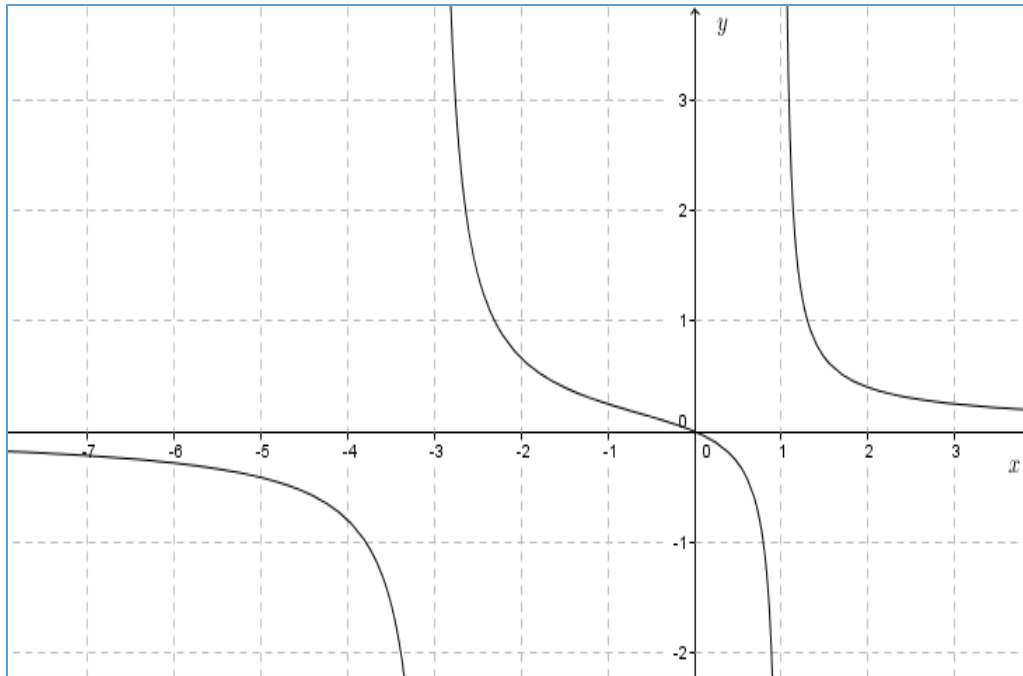
$]-\infty, 0[\cup]0, +\infty[$. Observing the graph of the function $f(x) = \frac{1}{x}$, one can easily realize that the function has no value only if $x = 0$.



b) $f(x) = \frac{x}{(x-1)(x+3)}$. The denominator should be different from zero, $(x-1)(x+3) \neq 0$.

Domain of definition is $\{x \in \mathbb{R} : x \neq 1 \text{ and } x \neq -3\}$ or simply $]-\infty, -3[\cup]-3, 1[\cup]1, +\infty[$.

Observing the graph of the function $f(x) = \frac{x}{(x-1)(x+3)}$, one can early realize that the function has no value only if $x = 1$ and $x = -3$.



c) Condition: $3x+6 \neq 0$; $3x+6=0 \Rightarrow x=-2$.

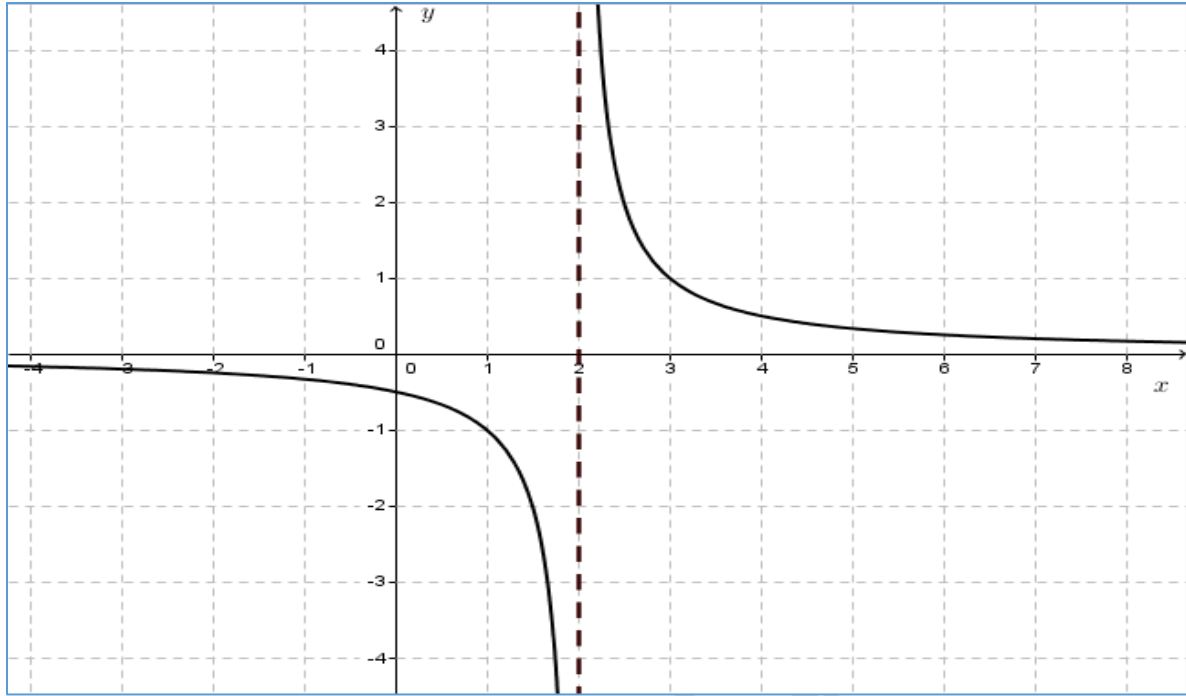
Then, $\text{Dom}f = \mathbb{R} \setminus \{-2\}$ or $\text{Dom}f =]-\infty, -2[\cup]-2, +\infty[$

Example 2

Find the range for the function $f(x) = \frac{1}{x-2}$

Solution

Put $y = f(x) = \frac{1}{x-2}$ and Solve for x , $y = \frac{1}{x-2} \Leftrightarrow x = \frac{1}{y} + 2$. Note that x can be solved if and only if $y \neq 0$. The range of $f(x)$ is $\{y \in \mathbb{R} : y \neq 0\} = \mathbb{R} \setminus \{0\}$. Alternatively, one can see on the graph that the range of $f(x)$ is $\mathbb{R} \setminus \{0\}$.



Example 3

Find the range for the function $f(x) = \frac{2x+1}{x^2+2}$

Solution

Put $y = f(x) = \frac{2x+1}{x^2+2}$ and Solve for x , $y = \frac{2x+1}{x^2+2} \Leftrightarrow yx^2 + y = 2x+1$.

$$yx^2 + y = 2x+1 \Leftrightarrow yx^2 - 2x + (y-1) = 0$$

$$x = \frac{2 \pm \sqrt{4 - 4y(y-1)}}{2y} \text{ if } y \neq 0, \quad x = -\frac{1}{2} \text{ if } y = 0$$

$$= \frac{1 \pm \sqrt{1 - y^2 + y}}{y} \text{ if } y \neq 0$$

Comparing the two, we see that x exists in the set of real numbers if and only if $1 - y^2 + y \geq 0$, that is $y^2 - y - 1 \leq 0$. The range of $f(x)$ is $\{y \in \mathbb{R} : y^2 - y - 1 \leq 0\}$. Solving the inequality

$y^2 - y - 1 \leq 0$, we get $y = \frac{1 \pm \sqrt{5}}{2}$. Then studying the sign of the quadratic expression,

$$y^2 - y - 1 = \left(y - \frac{1 - \sqrt{5}}{2}\right) \left(y - \frac{1 + \sqrt{5}}{2}\right).$$

2) Multiple Choice

(i) The function $f(x) = \frac{1}{x-5}$ is undefined at:

- A. $x=1$ B. $x=5$ C. $x=0$ D. $x=-5$

(ii) The domain $f(x) = \frac{3}{x+2}$ is:

- A. $x \neq -2$ B. $x \neq 2$ C. $x > 0$ D. All real numbers

3) Open Questions

Determine the domain of definition for the following functions

a) $f(x) = \frac{x}{1-x}$ b) $f(x) = \frac{x+1}{x^2-2}$

3.7. Domain of definition of irrational functions



Activity 3.7:

Consider the function: $f(x) = \sqrt{x-3}$

- 1) What happens if $x = 2$?
- 2) Can the expression inside the square root be negative?
- 3) For which values of x is the function defined?



Content summary

Given that $f(x) = \sqrt[n]{g(x)}$ where $g(x)$ is a polynomial, there are two cases

- a) If n is odd number, then the domain is the set of real numbers. That is $\text{dom} f = \mathbb{R}$
- b) If n is even number, then the domain is the set of all values of x such $g(x)$ is positive or zero. That is $\text{dom} f = \{x \in \mathbb{R} : g(x) \geq 0\}$

Example 1

Given the function $g(x) = \sqrt{x^2-1}$, determine the domain.

Solution

Condition of existence: $x^2-1 \geq 0$ this implies that we need to determine the interval where x^2-1 is positive. The corresponding equation is $x^2-1=0$, solving for x , we obtain:

$$x^2-1=0 \Leftrightarrow (x-1)(x+1)=0$$

$$x-1=0 \Rightarrow x=1$$

$$x+1=0 \Rightarrow x=-1$$

x	$-\infty$	-1				1				$+\infty$			
$x+1$	-	-	-	-	-0	+	+	+	+	+	+	+	+
$x-1$	-	-	-	-	-	-	-	-	-	0	+	+	+
x^2-1	+	+	+	+	+0	-	-	-	-	-0	+	+	+

Therefore $Domf =]-\infty, -1] \cup [1, +\infty[$

Example 2

Find the domain of the function $h(x) = \frac{\sqrt{1-x^2}}{x}$

Solution

Conditions of existence: $1-x^2 \geq 0$ and $x \neq 0$

- For the corresponding equation is $1-x^2 = 0$ and solving for the variable, we get:

$$(1-x)(1+x) = 0 \Rightarrow \begin{cases} 1-x=0 \text{ or } x=1 \\ 1+x=0 \text{ or } x=-1 \end{cases}$$

- For $x \neq 0$ all real numbers are accepted except from zero. Combining the two conditions we get:

x	$-\infty$	-1				0				1				$+\infty$
$1+x$	-	-	-	-	-0	+	+	+	+	+	+	+	+	+
$1-x$	+	+	+	+	+	+	+	+	+	+	+	+	0	-
$1-x^2$	-	-	-	-	-0	+	+	+	+	+	+	+	0	-
x	-	-	-	-	-	-	-	-	0	+	+	+	+	+
$f(x) = \frac{\sqrt{1-x^2}}{x}$	undefined				0	-	-	-		+	+	+	0	undefined

$-1 \leq x \leq 1$ and $x \neq 0$. Therefore, $domf = [-1, 0[\cup]0, 1]$

Example 3

Find domain of definition of $f(x) = \sqrt[3]{x+1}$

Solution

Since the index in radical sign is odd number, then $Domf = IR$

Example 4

Find the domain of definition of $g(x)$ if $g(x) = \sqrt[4]{x^2 + 1}$

Solution

Condition: $x^2 + 1 \geq 0$; Clearly $x^2 + 1$ is always positive. Thus $Dom g = \mathbb{R}$

Example 5

Find domain of $f(x) = \frac{x}{\sqrt{x^3 - 4x^2 + x + 6}}$

Solution

Conditions: $x^2 - 4x^2 + x + 6 \geq 0$ and $x^3 - 4x^2 + x + 6 \neq 0$.

The two conditions are combined in one: $x^3 - 4x^2 + x + 6 \geq 0$ and $x^3 - 4x^2 + x + 6 \neq 0$

$$x^3 - 4x^2 + x + 6 = (x+1)(x-2)(x-3)$$

x	$-\infty$	-1	2	3	$+\infty$															
$x+1$	-	-	-	-	-0	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
$x-2$	-	-	-	-	-	-	-	-	-	-0	+	+	+	+	+	+	+	+	+	+
$x-3$	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-0	+	+	+	+	+
x^3-4x^2+x+6	-	-	-	-	-0	+	+	+	+	+0	-	-	-	-	-0	+	+	+	+	+

Then, $Dom f =]-1, 2[\cup]3, +\infty[$

Example 6

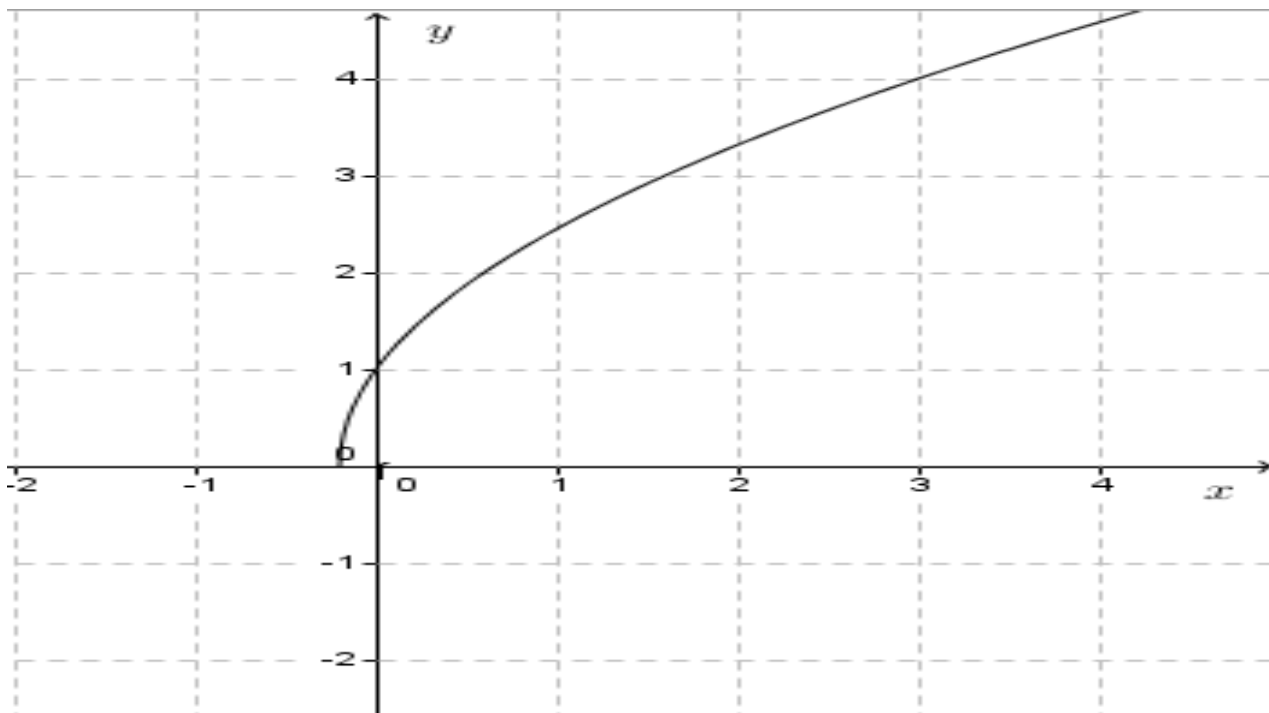
Find the range for the function $f(x) = \sqrt{1+5x}$

Solution

$1+5x \geq 0$ (Restrictions on x); $\sqrt{1+5x} \geq 0$ (Taking the square roots); But $f(x) = \sqrt{1+5x}$.

Therefore, $f(x) \geq 0$. The range of $f(x)$ is $\text{Im } f = [0, +\infty[$.

The graph below illustrates the range:



Application activity 3.7

Find the domain of definition for each of the following functions:

$$1. f(x) = \sqrt{4x-8} \quad 2. g(x) = \sqrt{x^2+5x-6} \quad 3. h(x) = \frac{x^3+2x^2-2}{\sqrt[3]{x+4}}$$

$$4. f(x) = \frac{x-2}{\sqrt[4]{x^2-25}} \quad 5. f(x) = \sqrt{\frac{(x-1)^2}{x+4}}$$

3.8. Parity of a function (odd or even).



Activity 3.8

1. Consider the following function: $f(x) = x^2$

- Calculate $f(2)$ and $f(-2)$.
- Compare the two values.
- What do you notice about the outputs?

2. Now consider the function: $g(x) = x^3$

- Calculate $g(2)$ and $g(-2)$

b) What relationship do you observe?. Discuss your observations.



Content Summary 3.8

Even function

Let $f(x)$ be a numerical function whose domain is $Domf$. $f(x)$ is said to be an **even function** if and only if:

- (i) $(\forall x \in Domf), -x \in Domf$
- (ii) $f(-x) = f(x)$, that is, any two opposite values of the independent variable have the same image under the function.

The graph of an even function is symmetrical about the y -axis.

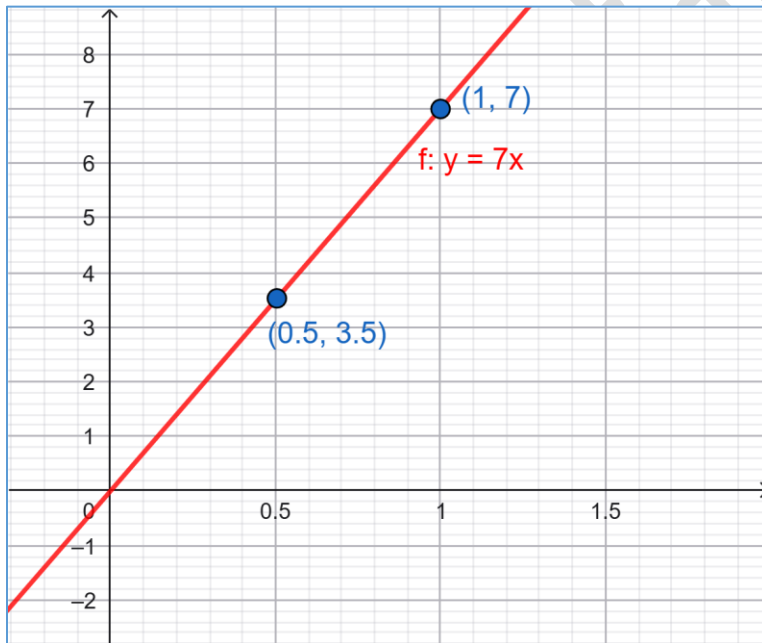
Example 1

Determine whether the function $f(x) = 7x$ is even or not.

Solution

The domain of function f is the set of all real numbers. For any real number x , the opposite $-x$ is also a real number and $f(-x) = -7x \neq 7x = f(x)$. Since, $f(-x) \neq f(x)$, function f is not even.

Graphically,



The graph is not symmetrical about the y -axis

Example 2

Determine whether the function $f(x) = 3x^2 - 4$ is even or not.

Solution

$$f(-x) = 3(-x)^2 - 4 \leftrightarrow 3x^2 - 4 = f(x).$$

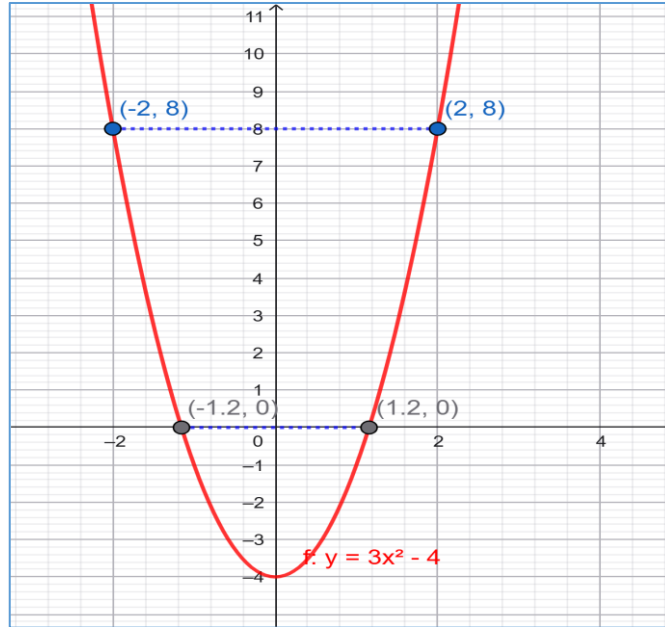
Since, $f(-x) = f(x)$, function f is even.

Remember that,

$$(-x)^n = \begin{cases} x^n, & \text{if } n \text{ is even} \\ -x^n, & \text{if } n \text{ is odd} \end{cases}$$

The graph of the function is symmetrical about the y -axis as shown on the diagram:

$$f(-2) = f(2) \text{ and } f(-1) = f(1).$$

**Example 3**

Determine whether the function $g(x) = x^6 - x^4 + x^2 + 9$ is even or not.

Solution

$$g(-x) = (-x)^6 - (-x)^4 + (-x)^2 + 9 = x^6 - x^4 + x^2 + 9 = g(x)$$

Therefore, the function is even.

Example 4

Determine whether the function $f(x) = \frac{3x+1}{x^2-25}$ is even or not.

Solution

$$f(-x) = \frac{3(-x)+1}{(-x)^2-25} \leftrightarrow f(-x) = \frac{-3x+1}{x^2-25}. \text{ Therefore the function is not even since } f(-x) \neq f(x)$$

Example 5

Given functions $f(x) = \sqrt{x}$ and $g(x) = \sqrt{x-4}$, find the $f(x) \cdot g(x)$ and determine if the result is an even function or not.

Solution

$$\text{Provided } x \geq 0 \text{ and } x-4 \geq 0, f(x) \cdot g(x) = \sqrt{x} \cdot \sqrt{x-4} \leftrightarrow f(x) \cdot g(x) = \sqrt{x^2-4x}$$

$$f \cdot g(-x) = \sqrt{(-x)^2-4(-x)} \leftrightarrow f \cdot g(-x) = \sqrt{x^2+4x}. \text{ Therefore the function is not even since } f(-x) \neq f(x)$$

Notice that the conclusion could have been drawn from the fact that $x \geq 4$ does not imply $-x \geq 4$, thus function f is not even

Odd function

Let $f(x)$ be a numerical function whose domain is $Domf$. $f(x)$ is said to be an **odd function** if and only if:

- (i) $(\forall x \in Domf), -x \in Domf$
- (ii) $f(-x) = -f(x)$, that is, any two opposite values of the independent variable have opposite images under the function.

The graph of an even function is symmetrical about the origin.

Note: Some functions are neither even nor odd

Example 6

Determine whether the function $f(x) = x^3$ is odd or not

Solution

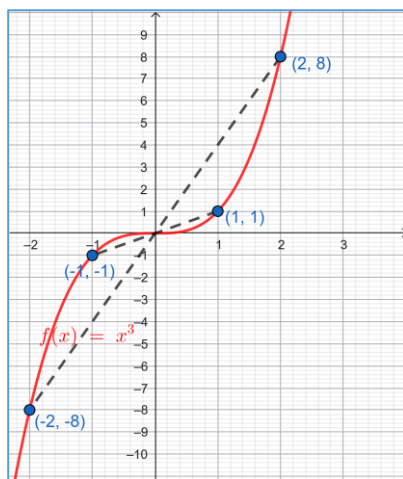
$$f(-x) = (-x)^3 \Leftrightarrow f(-x) = -x^3$$

and $-f(x) = -x^3$.

Therefore, $f(-x) = -f(x)$ and

the function $f(x) = x^3$ is odd.

Graphically, the point $(0,0)$ is the centre of symmetry for the graph of the function $f(x) = x^3$.



Example 7

Determine whether the function $f(x) = x^2 + 2x + 1$ is odd, even or neither

Solution

$$f(-x) = (-x)^2 + 2(-x) + 1.$$

$$\Leftrightarrow f(-x) = x^2 - 2x + 1$$

$$\text{and } -f(x) = -x^2 - 2x - 1$$

$$f(-x) \neq -f(x) \text{ and}$$

$$f(-x) \neq f(x).$$

Therefore, the function $f(x) = x^2 + 2x + 1$ is not

odd neither even. Graphically,

point $(0,0)$ is not the centre of

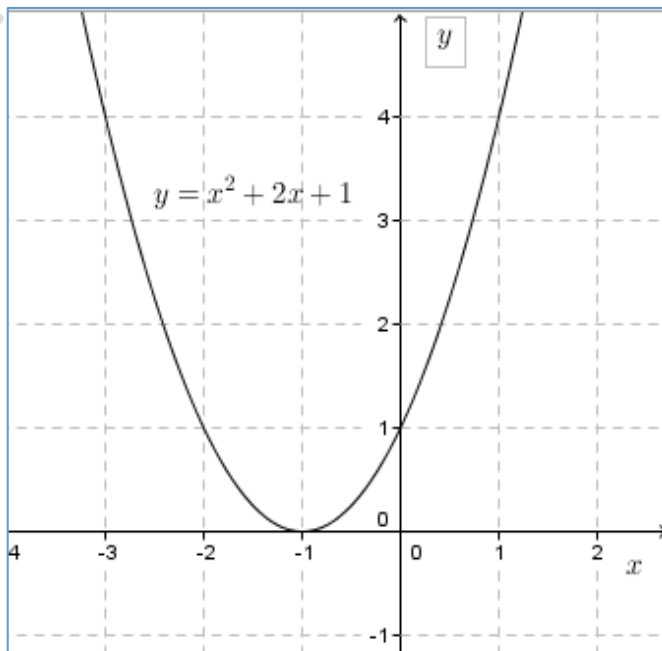
symmetry for the graph of the

function $f(x)$, and the line $x = 0$ is

not the axis of symmetry for the

graph of function

$$f(x) = x^2 + 2x + 1.$$



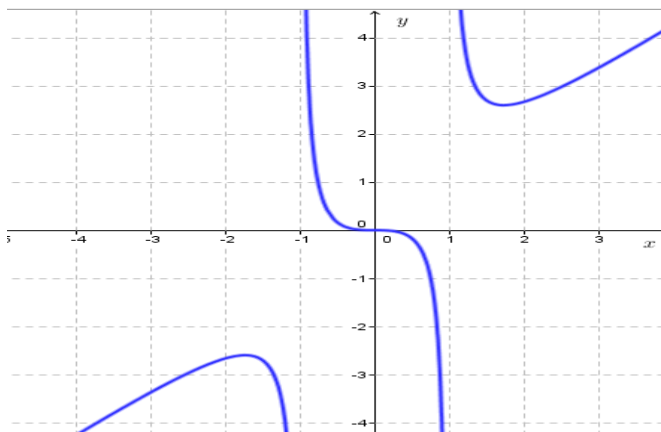
Example 8: Determine if the function $f(x) = \frac{x^3}{x^2 - 1}$ is even, odd, or neither and deduce the symmetry of its graph.

Solution:

$$f(-x) = \frac{(-x)^3}{(-x)^2 - 1} = \frac{-x^3}{x^2 - 1} \neq f(x).$$

Therefore, the function is not even
But, $f(-x) = -f(x)$; it follows that f is an odd function. The graph of

$f(x) = \frac{x^3}{x^2 - 1}$ is shown below. It can be seen that point $(0, 0)$ is the centre of symmetry for the graph.



...



Application activity 3.8

1. Answer by True or False

- An even function satisfies $f(-x) = f(x)$
- The graph of an odd function is symmetric about the origin.

2. Which of the following functions is even?

A. $f(x) = x^3$ B. $f(x) = x^2$ C. $f(x) = x^3 + x$ D. $f(x) = x^2 + x$

3. Study the parity of the following functions

a) $f(x) = 2x^2 + 2x - 3$

b) $f(x) = \frac{3x^3 + 2x^2 + 8}{x - 5}$

c) $g(x) = x^3 - x$

3.9. Addition and subtraction of functions



Activity 3.9

Consider the following functions representing costs:

$C_1(x) = 2x + 5$ (cost of material 1)

$C_2(x) = 3x + 2$ (cost of material 2)

- How can you find the total cost if both materials are used?
- How can you find the difference in cost between the two materials?
- Can these situations be represented using function operations?



Content Summary 3.9

1. Addition of Functions

If $f(x)$ and $g(x)$ are two functions, their **sum** is defined as: $(f + g)(x) = f(x) + g(x)$

This new function assigns to each x the sum of $f(x)$ and $g(x)$

Example 1

Let: $f(x) = 2x + 3$, $g(x) = x^2 - 1$ Then,

$$\begin{aligned}
 (f + g)(x) &= f(x) + g(x) \\
 &= (0x^2 + 2x + 3) + (x^2 + 0x - 1) = x^2 + 2x + 2
 \end{aligned}$$

Example 2

Let f and g be the functions $f(x) = 3x^4 - 5x^3 + x - 4$ and $g(x) = 4x^3 - 3x^2 + 4x + 3$. Find $(f + g)(x)$.

Solution

$$\begin{array}{r}
 f(x) = 3x^4 - 5x^3 + 0x^2 + x - 4 \\
 + \quad g(x) = 0x^4 + 4x^3 - 3x^2 + 4x + 3 \\
 \hline
 (f + g)(x) = 3x^4 - x^3 - 3x^2 + 5x - 1
 \end{array}$$

3) Subtraction of Functions

If $f(x)$ and $g(x)$ are two functions, their difference is defined as: $(f - g)(x) = f(x) - g(x)$

Example 1: Let: $f(x) = x^2 + 4$, $g(x) = 2x - 1$

$$\begin{aligned}
 (f - g)(x) &= f(x) - g(x) \\
 &= [x^2 + 4 - (2x - 1)] = x^2 - 2x + 5
 \end{aligned}$$

Example 2

Let f and g be the functions $f(x) = 3x^4 - 5x^3 + x - 4$ and $g(x) = 4x^3 - 9x^2 + 4x + 3$.

Find $(f - g)(x)$

Solution

$f(x) - g(x) = [3x^4 - 5x^3 + 0x^2 + x - 4 - (0x^4 - 9x^3 - 3x^2 + 4x + 3)]$ After multiplying minus for $g(x)$, then make sum. Finally, $(f - g)(x) = 3x^4 + 4x^3 + 3x^2 - 3x - 7$

Notes on Domain:

Domain of $f + g$ or $f - g$ is the intersection of the domains of f and g . For example, if $f(x) = \frac{1}{x}$ and $g(x) = \sqrt{x}$ then: $(f + g)(x) = \frac{1}{x} + \sqrt{x}$. The $Dom f$ is $x > 0$ because it must satisfy both functions domains.



Application activity 3.9

Answer by True or False

- 1) The sum of two functions is always defined for all real numbers.
- 2) The domain of $f + g$ is the intersection of the domains of f and g .
- 3) Given the functions: $f(x) = 2x^3 + 5x - 1$ and $g(x) = 3x - 4$. Find $(f + g)(x)$

3.10. Multiplication and division of functions



Activity 3.10

Let: $f(x) = x + 2$ and $g(x) = x - 1$ be the functions.

Answer the following questions

- a) Find $(f \cdot g)(x)$
- b) Determine the domain of $(f \cdot g)(x)$
- c) Calculate $(f \cdot g)(2)$



Content Summary 3.10

1. Multiplication of Functions

The product of two functions $f(x)$ and $g(x)$ is a new function defined by

$$(f \cdot g)(x) = f(x) \cdot g(x)$$

- The value of $(f \cdot g)(x)$ at each x is obtained by Multiplying $f(x)$ and $g(x)$.
- Domain :All values of x that belong to both domains of f and g

Example 1

Given $f(x) = x + 2$ and $g(x) = x - 1$. Then, $(f \cdot g)(x) = (x + 2)(x - 1) = x^2 + x - 2$

Domain: All real numbers ($\mathbb{R}$)

Example 2

If $f(x) = \frac{9}{x+2}$ and $g(x) = x^3$. Find $f(x) \cdot g(x)$

Solution: let $t(x)$ be a function of the product of $f(x)$ and $g(x)$; $[t(x) = f(x) \cdot g(x)]$. then,

$$t(x) = \frac{9}{x+2} \cdot x^3 = \frac{9x^3}{x+2}$$

2. Division of Functions

The quotient of two functions $f(x)$ and $g(x)$ (with $g(x) \neq 0$) is defined by : $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$,
 $g(x) \neq 0$

- The value of $\frac{f}{g}(x)$ at each x is obtained by dividing $f(x)$ by $g(x)$.
- Domain: All values of x that belong to both domains of f and g , excluding x where $g(x) = 0$

Example 1

Given, $f(x) = x^2 + 1$ and $g(x) = x - 2$, $\frac{f}{g}(x) = \frac{x^2 + 1}{x - 2}$, $x \neq 2$

Domain: $\mathbb{R} \setminus \{2\}$

Example 2

Let f and g be the functions; $f(x) = 3x^4 - 5x^3 + x - 4$ and $g(x) = 4x^3 - 3x^2 + 4x + 3$.

Find $k(x) = \frac{f(x)}{g(x)}$

Solution

$$k(x) = \frac{f(x)}{g(x)} = \frac{3x^4 - 5x^3 + x - 4}{4x^3 - 3x^2 + 4x + 3}$$



Application activity 3.10

1) Answer by True or False

- The product of two functions f and g is defined by $(f \cdot g)(x) = f(x)g(x)$.
- The quotient $\left(\frac{f}{g}\right)(x)$ is defined even when $g(x) = 0$.

2) Given the functions $f(x) = 3x^3 - 5x^2 + 7x - 4$ and $g(x) = 2x^2 - x + 3$.
 Find $(f \cdot g)(x)$

3.11. Composite function



Activity 3.11

A school organizes a trip for students. The cost of transport depends on the number of students. First, the number of buses needed depends on the number of students: $b(x) = \frac{x}{20}$; where x is the number of students and $b(x)$ is the number of buses needed (1 bus carries 20 students). The total transport cost depends on the number of buses: $c(b) = 150b$; where b is the number of buses and the cost is 150 thousand francs per bus. To find the total cost depending on the number of students, we substitute the first function into the second.

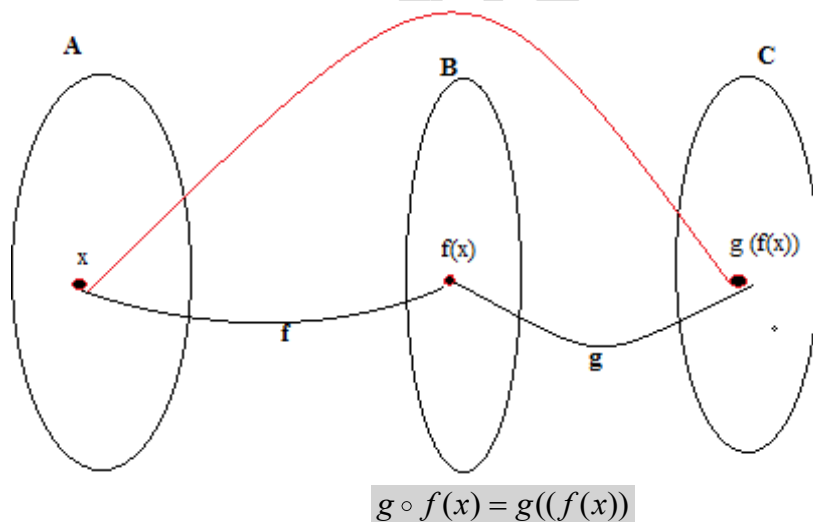
Answer the following questions

- 1) If there are 40 students, how many buses are needed?
- 2) Using the number of buses found, what will be the total cost of transport?
- 3) Instead of calculating in two steps, can we combine the two functions into one function?
- 4) What happens when the output of one function becomes the input of another function?



Content Summary 3.11

Consider the functions $f(x) = 2x - 1$ and $g(x) = x^2$.



This combined or composite function is written $(g \circ f)(x)$ or $g[f(x)]$, simply gf . The function f is performed first and so is written nearer to the variable x .

x	$y = f(x) = 2x - 1$	$g(f(x)) = g(y) = y^2$
-----	---------------------	------------------------

1	1	1
3	5	25
5	9	81
7	13	169

The set $\{1, 3, 5, 7\}$ is the **domain** for the composite function and $\{1, 25, 81, 169\}$ is the **range**.

Note that $(f \circ g)(x) \neq (g \circ f)(x)$.

Example 1

If $f(x) = 2x$ and $g(x) = 3x + 1$, express $g \circ f$ as a single function $h(x)$.

Solution

$f(x) = 2x$ so $(g \circ f)(x) = g(2x) = 3(2x) + 1 \leftrightarrow 6x + 1$. finally, $h(x) = 6x + 1$

Example 2

Let $f(x) = x - 1$ and $g(x) = \sqrt{x}$, find $(f \circ g)(x)$ and $(g \circ f)(x)$

- $g(x) = \sqrt{x}$, $fg(x) = f(\sqrt{x}) = \sqrt{x} - 1 \quad \therefore (f \circ g)(x) = \sqrt{x} - 1$
- $f(x) = x - 1$, $gf(x) = g(x - 1) = \sqrt{x - 1} \quad \therefore (g \circ f)(x) = \sqrt{x - 1}$



Application activity 3.11

1. Answer by True or False

- In a composite function, the output of one function becomes the input of another function.
- $f(g(x))$ means we first apply f and then apply g

2. Multiple Choice

i. Let $f(x) = 2x + 1$ and $g(x) = x + 3$. Find $f(g(x))$.

- a) $2x + 7$ b) $2x + 5$ c) $2x + 4$ d) $x + 7$

ii. Let $f(x) = x^2$ and $g(x) = x + 1$. find $g(f(x))$.

- a) $x^2 + 1$ b) $x^2 + x$ c) $x + 1$ d) $x^2 - 1$

3. Open Questions

- Let $f(x) = x^2$ and $g(x) = 2x + 1$. Find $(fg)(x)$ and $(gf)(x)$
- Given that $f(x) = x + 3$ and $g(x) = 2x$. Find a) $fg(x)$ and b) $gf(x)$

3.12. Inverse of a function



Activity 3.12

Find the value of x in function of y if:

- $y = x + 1$

$$2) \quad y = 3x - 2$$

$$3) \quad y = \frac{-x + 3}{2x - 1}$$



Content Summary

Consider a function f which maps each element x of the domain X onto its image y in the range Y that is $f : x \rightarrow y$ where $x \in X, y \in Y$. If this map can be reversed, i.e. $f^{-1} : y \rightarrow x$ and the resulting relationship is a function, it is called the **inverse of the original function**, and is denoted by f^{-1} . Only one-to-one functions can have an inverse function. To find the inverse of one-to-one functions, we change the subject of a formula.

Definition: Let $f(x)$ and $g(x)$ be two functions such that $f[g(x)] = x$ for each x in the domain of g and $g[f(x)] = x$ for each x in the domain of f . Under these conditions, the function g is the inverse of f . The function g is denoted by $f^{-1}(x)$, which is read as “ f – inverse”. So

$$f(f^{-1}(x)) = x \text{ and } f^{-1}(f(x)) = x$$

Note:

- If $f(x)$ is inverse of $g(x)$, then $g(x)$ is inverse of $f(x)$.
- The domain of f must be equal to the range of f^{-1} , and the range of f must be equal to the domain of f^{-1} .
- The notation $f^{-1}(x) \neq \frac{1}{f(x)}$

Steps to find inverse functions

Let $f : x \rightarrow \mathbb{R}$ be an injective function where $x \subseteq \mathbb{R}$. To find the inverse function of f means to find the domain of f^{-1} as well as a formula for $f^{-1}(y)$. If the formula for $f(x)$ is not very complicated, $\text{dom}(f^{-1})$ and $f^{-1}(y)$ can be found by solving the equation $y = f(x)$ for x .

(Step 1): Put $y = f(x)$.

(Step 2): Solve for x in terms of y . then exchange x and y .

(Step 3): From Step 2, the range of f will be determined and be the domain of f^{-1} .

Example 1

Find the inverse function of $f(x) = 2x + 3$.

Solution

$$y = 2x + 3 \text{ (Solve for the variable } x); \quad y - 3 = 2x, \quad x = \frac{y - 3}{2} \Rightarrow f^{-1}(x) = \frac{x - 3}{2}$$

Example 2

Find the inverse of the function $f(x) = 3x - 1$

Solution: $y = 3x - 1$, $x = \frac{y+1}{3}$. Hence, $f^{-1}(x) = \frac{x+1}{3}$



Application activity 3.12

1. Answer by True or False

- a) The inverse of a function is obtained by interchanging y and x and solving for y .
- b) If a function has an inverse, then applying the function, its inverse returns the original value.

2. Find the inverse of the following functions

- a) $f(x) = 5x + 2$
- b) $g(x) = -7x - 2$
- c) $h(x) = \frac{-2x+1}{x-2}$

3.13. Graphical representation and interpretation of linear function



Activity 3.13

A taxi company charges a fixed starting fee of 2000 RWF plus 500 RWF per kilometer traveled. Let x represent the number of kilometers traveled. The total cost $C(x)$ can be written as a linear function: $C(x) = 500x + 2000$

- 1) What is the cost when the distance is 0 km?
- 2) Find the cost when the distance is 1 km, 2 km, and 3 km.
- 3) Plot the points obtained on a Cartesian plane.
- 4) What type of graph do you obtain when the points are joined?



Content Summary

Any function of the form $f(x) = mx + b$, where $m \neq 0$; is called a linear function. The domain and range of this function is the set of all real numbers. The graph of f is a line with slope m and y intercept b .

Examples of linear functions

- a) $y = x + 1$, b) $y = 2x - 3$, c) $y = -3x + 4$, ...

Graphs of linear functions

When you are given the linear function $f(x) = mx + b$, you can dress a table of values

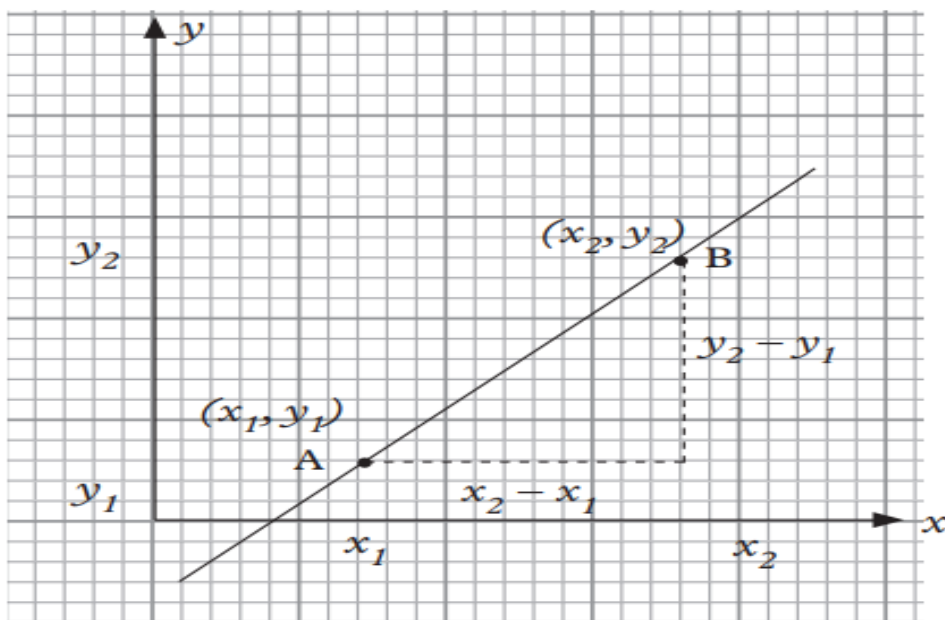
Set of x (domain)	0	1	$-b/m$	..
Set of $y =$ $f(x) = mx + b$ (range)	b	$m+b$	0	..
Ordered pairs (x,y)	$(0, b)$	$(1, m+b)$	$(-b/m, 0)$	..

Note on the point	y-intercept		x-intercept	
-------------------	-------------	--	-------------	--

The ordered pair (x, y) represents coordinates of any point on the Cartesian plane.

When the ordered pairs (x, y) are plotted and joined in the cartesian plane, the obtained graph is the line.

Consider a line passing through the points $A(x_1, y_1)$ and $B(x_2, y_2)$.



From A to B, the change in the x-coordinate (horizontal change) is $x_2 - x_1$ and the change in the y-coordinate (vertical change) is $y_2 - y_1$.

By definition, gradient / slope $(m) = \frac{y_2 - y_1}{x_2 - x_1}$.

In the Cartesian plane, the gradient of a line is the measure of its slope or inclination to the x-axis. It is defined as the ratio of the change in y-coordinate (vertical) to the change in the x-coordinate (horizontal).

When drawing a graph of a linear function, it is sufficient to plot only two points and these points may be chosen as the x-intercept $\left(\frac{-b}{m}, 0\right)$ and y-intercept $(0, b)$ of the graph.

To determine the x-intercept, set $f(x) = 0$ and solve for x .

For y intercept, set $x = 0$ to find $f(0)$.

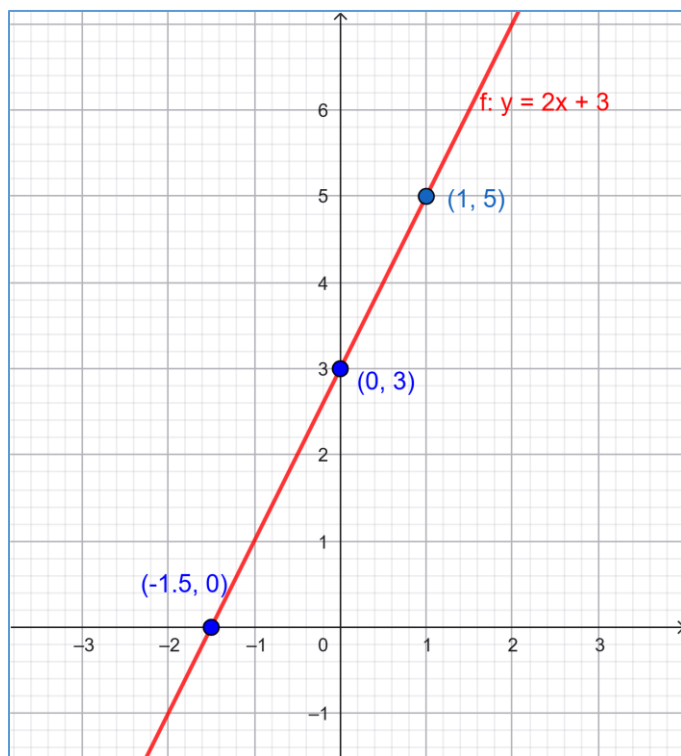
Example

Given $f(x) = 2x + 3$, Find x and y-intercept points.

$$\text{x-intercept point: } 2x + 3 = 0 \rightarrow x = \frac{-3}{2} \rightarrow \left(\frac{-3}{2}, 0\right)$$

$$\text{y-intercept point: } f(0) = 2 \times 0 + 3 \rightarrow f(0) = 3 \rightarrow (0, 3)$$

When drawing the graph, the dependent variable is marked on the vertical axis generally known as the y – axis. The independent variable is marked on the horizontal axis also known as the x – axis.



Application activity 3.13

- 1) Answer by True or False
 - a) The graph of a linear function
 - b) is always a straight line.
 - c) In the function $f(x) = mx + b$, the number b represents the slope of the line.
- 2) Multiple Choice

- a) The slope of the function: $f(x) = 3x + 2$ is: A.(4) B.3 C.2 D.-4
- b) The y-intercept point of the function: $f(x) = 4x - 1$ is:
A.(-1,0) B.(0,0) C.(0,-1) D.(2,3)

3.14. Graphical representation and interpretation of quadratic function



Activity 3.14

A student throws a ball upward. The height of the ball after x seconds can be modeled by the function: $h(x) = -x^2 + 4x + 1$

- 1) Calculate the value of $h(x)$ for $x = 0, 1, 2, 3$.

- 2) Plot the points obtained on a Cartesian plane.
- 3) Join the points smoothly. What shape do you observe?
- 4) Does the graph look like a straight line or a curve?



Content Summary

A polynomial equation in which the highest power of the variable is 2 is called a quadratic function. The expression $y = ax^2 + bx + c$, where a, b and c are constants and $a \neq 0$, is called a quadratic function of x or a function of the second degree (highest power of x is 2).

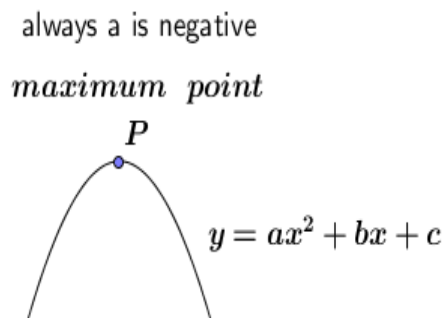
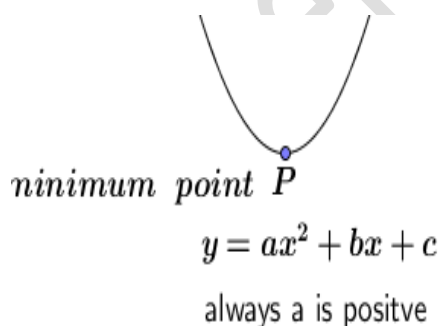
Table of values are used to determine the coordinates that are used to draw the graph of a quadratic function.

To get the table of values, we need to have the **domain** (values of an independent variable) and then the domain is replaced in a given quadratic function to find **range** (values of dependent variables). The values obtained are useful for plotting the graph of a quadratic function. All quadratic function graphs are parabolic in nature.

Any quadratic function has a graph which is symmetrical about a line which is parallel to the y -axis i.e. a line $x = h$ where h is constant value. This line is called **axis of symmetry**.

For any quadratic function $f(x) = ax^2 + bx + c$ whose axis of symmetry is the line $x = h = -\frac{b}{2a}$ the vertex is the point $(h, f(h))$.

The vertex of a quadratic function is the point where the function crosses its axis of symmetry. If the coefficient of the x^2 term is positive, the vertex will be the lowest point on the graph, the point at the bottom of the U-shape. If the coefficient of the term x^2 is negative, the vertex will be the highest point on the graph, the point at the top of the $\cap$ -shape. The shapes are as below.



Since, the quadratic function written as $f(x) = ax^2 + bx + c$, then we can get the y -coordinate of the vertex by substituting the x -coordinate which is $x = -\frac{b}{2a}$.

So the vertex becomes $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$. The axis of symmetry of a quadratic function is the

x – coordinate of the quadratic function and it is calculated from $x = -\frac{b}{2a}$.

Graph of a quadratic function

The graph of a quadratic function can be sketched using a **table of values** or the **following main points: vertex, x-intercept point and y-intercept point**.

Example

Find the vertex and axis of symmetry of the parabolic curve: $y = 2x^2 - 8x + 6$

Solution

- Coefficients : $a = 2$ $b = -8$ $c = 6$

- Vertex = $\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$, $\left(\frac{-b}{2a}\right) = \frac{-(-8)}{2 \times 2} \Leftrightarrow \left(\frac{-b}{2a}\right) = 2$

$$f\left(\frac{-b}{2a}\right) = 2(2^2) - 8(2) + 6 \Leftrightarrow f\left(\frac{-b}{2a}\right) = -2$$

Then, vertex = $(2, -2)$ and axis of symmetry = $\frac{-b}{2a} = 2$.

To sketch this function, find intercept points:

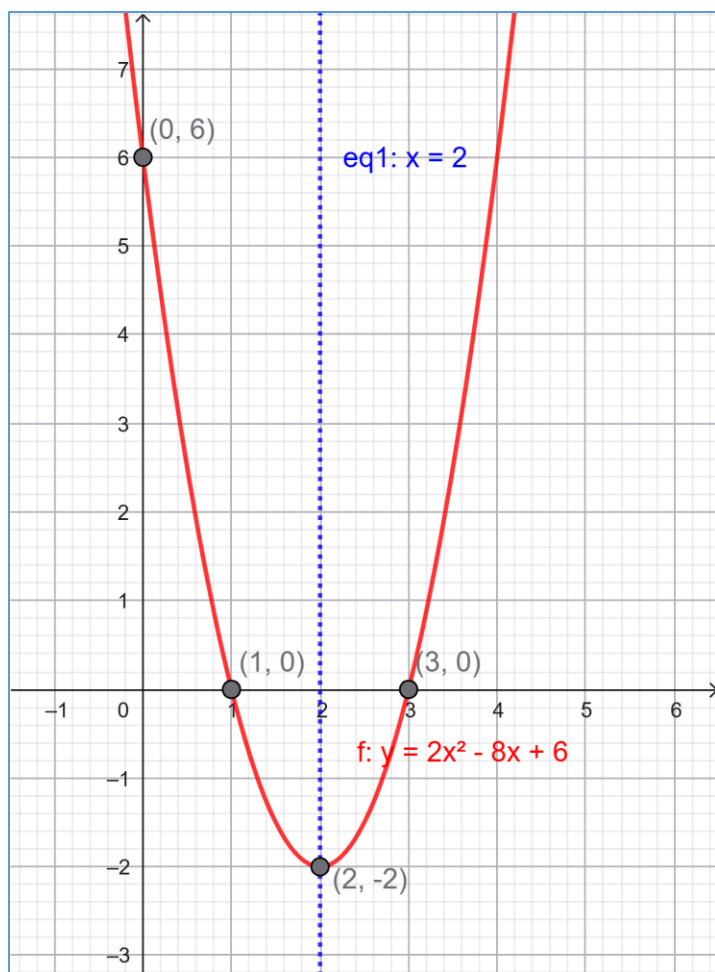
x – intercept point, $f(x) \equiv y = 0$, $2x^2 - 8x + 6 = 0$, $\Delta = b^2 - 4ac$, $\Delta = (-8)^2 - 4.2.6$, $\Delta = 16$

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}; \quad x_1 = \frac{-(-8) + \sqrt{16}}{2.2} = 3 \text{ and } x_2 = \frac{-(-8) - \sqrt{16}}{2.2} = 1$$

So, the x – intercept points are : $(3, 0)$ and $(1, 0)$.

y – intercept point: $x = 0$, $f(0) = 2.(0)^2 - 8.0 + 6$, $f(0) = 6$ so, y – intercept point is $(0, 6)$.

The graph is as below.



Application activity 3.14

1) Answer by True or False

- The graph of a quadratic function is called a parabola.
- If $a < 0$ in $f(x) = ax^2 + bx + c$, the parabola opens upward.

2) Multiple Choice

Consider the quadratic function : $f(x) = -x^2 + 4x + 1$. The vertex of the parabola represents:

- The minimum point of the graph.
- The maximum point of the graph.
- The y – intercept only.
- The x – intercept only.

3.15. Price as function of quantity supplied and consumption as function of income



Activity 3.15

A farmer brings bananas to the market. When he supplies few bananas, the price is low because customers do not see many bananas. When he supplies more bananas, the price changes depending on the amount supplied. At the same time, a family buys food depending on their income. When their income increases, they buy more goods.

- 1) What may happen to the price when the quantity supplied increases?
- 2) Do people consume the same amount of goods when their income increases? Explain.
- 3) Can we say that price depends on quantity supplied and consumption depends on income?



Content Summary

A. Price as function of quantity supplied

The quantity q of an item that is manufactured and sold depends on its price p .

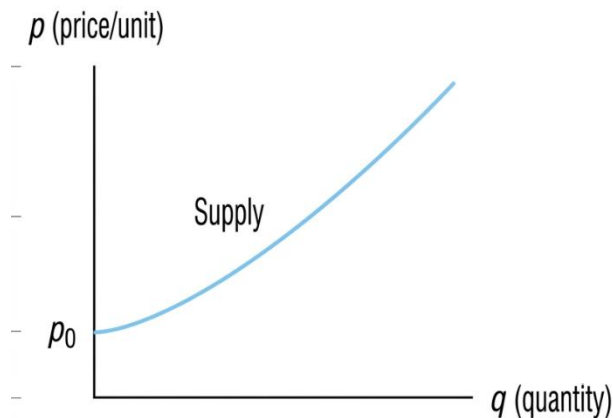
As the price increases, manufacturers are usually willing to supply more of the product, whereas the quantity demanded by consumers falls.

The supply function expresses the relationship between the market price p of a commodity and the quantity supplied q by producers during a specific period, assuming all other factors (like production costs, technology, and taxes) remain constant.

Example

Given $p = 200 + 2q$

The supply curve, for a given item, relates the quantity, q , of the item that manufacturers are willing to make per unit time to the price, p , for which the item can be sold.



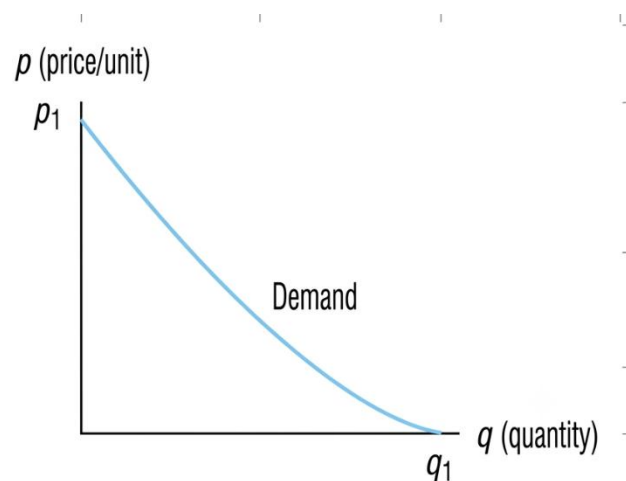
The demand function mathematically models the relationship between the market price p of a good and the quantity q that consumers are willing and able to purchase, holding all other economic factors constant.

Example

Given $p = 2000 - 4q$

The demand curve relates the quantity, q , of an item demanded by consumers per unit time to the price, p , of the item. Economists often think of the quantities supplied and demanded Q as functions of price p .

However, for historical reasons, the economists put price (the independent variable) on the vertical axis and quantity (the dependent variable) on the horizontal axis. (The reason for this state of affairs is that economists originally took price to be the dependent variable and put it on the vertical axis)



Example 1

A market for a specific textbook is governed by the following linear equations, where p represents the price in Rwandan Francs (RWF) and q represents the quantity of books:

Demand Function: $p = 2000 - 4q$

Supply Function: $p = 200 + 2q$

If market equilibrium occurs at the exact point where the quantity demanded equals the quantity supplied ($Q_d = Q_s$), find the equilibrium price (p^*) and equilibrium quantity (q^*) for the market.

Solution

Market equilibrium occurs at the exact point where the quantity demanded equals the quantity supplied ($Q_d = Q_s$). Consequently, the demand price equals the supply price.

Set the two functions equal to each other: $2000 - 4q = 200 + 2q$

By solving for q , we find: $q = \frac{1800}{6}$

$$q^* = 300 \text{ units}$$

Substitute q^* back into either original equation to find p^* :

Using the supply function: $p = 200 + 2(300)$

$$p^* = 800 \text{ RWF}$$

Conclusion: The equilibrium quantity is 300 books and the equilibrium market price is 800 RWF.

B. Consumption as function of income

The **consumption function** is an economic formula that models the relationship between total consumer spending (consumption) and disposable income. Introduced by British economist John Maynard Keynes, it demonstrates that a household's or nation's consumption behavior is directly driven by its income level. Mathematically, a linear consumption function is expressed as:

$$C = c_0 + c_1 Y_d \quad (\text{Or simply } C = a + bY).$$

Where: **C** = Total Consumption spending.

c_0 = **Autonomous Consumption**. This is the level of spending when disposable income is zero (e.g., spending on basic necessities like food and shelter financed through savings, borrowing, or aid). It represents the vertical intercept on a graph.

c_1 = **Marginal Propensity to Consume (MPC)**. This is the fraction of any additional income that is spent on consumption.

Y_d = **Disposable Income** (National income after taxes, or $Y - T$).

Example 2

An economy's consumption behavior is represented by the linear consumption function:

$$C = 500,000 + 0.75Y_d$$

Where C and Y_d are measured in Rwandan Francs (Frw).

- 1) Identify the value of autonomous consumption and explain the meaning of the Marginal Propensity to Consume (MPC) in this context.
- 2) Calculate the total consumption spending if the aggregate disposable income is 4,000,000 Frw.

- 3) Find the break-even level of disposable income where total consumption exactly equals total disposable income ($C = Y_d$).

Solution

1. Autonomous Consumption (c_0): 500,000 Frw . This means even if the community's disposable income drops to zero, they will still spend 500,000 Frw on basic survival needs by liquidating assets or borrowing.

Marginal Propensity to Consume (c_1): 0.75. This indicates that for every additional 1,000 Frw earned by consumers, they will spend 750 Frw (75%) on consumption and save the remaining 250 Frw (25%).

2. Calculating Total Consumption for $Y_d = 4,000,000$ Frw

Substitute the value of Y_d into the consumption function: $C = 500,000 + 0.75(4,000,000)$
 $C = 3,500,000$ Frw

When disposable income is 4,000,000Frw, total consumer spending is 3,500,000Frw.

3. Finding the Break-Even Level of Income

At the break-even point, consumption equals disposable income ($C = Y_d$). Replace C with Y_d in the formula: $Y_d = 500,000 + 0.75Y_d$

By solving for Y_d : $Y_d = \frac{500,000}{0.25}$

$Y_d = 2,000,000$ Frw

The break-even level of income is 2,000,000Frw. At this income level, households spend exactly what they earn, leaving personal savings at zero.



Application activity 3.15

- 1) Select the correct answer:

In the function $C = a + bY$, the variable Y represents:

- A. Consumption
- B. Income
- C. Price
- D. Quantity supplied

- 2) Given the consumption function: $C = 40 + 0.5Y$. Find the consumption when income $Y = 80$.
3) Explain in your own words why consumption increases when income increases

3.16. Price as function of quantity demanded and the cost function



Activity 3.16

A shop sells notebooks. When the price of notebooks is high, few students buy them. When the price decreases, many students buy them. At the same time, a company producing notebooks spends money on materials, workers, and electricity. The more notebooks they produce, the higher the cost of production.

- 1) What happens to quantity demanded when price decreases?
- 2) Do you think the cost of producing goods changes when the quantity produced increases?
- 3) Can we express price and cost using mathematical functions?



Content Summary

1. Price as a Function of Quantity Demanded

Sometimes the price of a product depends on how many units people want to buy.

This relationship can be written as a function. $P = f(Q_d)$; Where: P is the price of the product and Q_d is the quantity demanded.

Example 1

Given $P = 20 - 2Q_d$. If:

Quantity demanded (Q_d)	Price (P)
2	16
4	12
6	8

This shows that when quantity demanded increases, the price decreases. This follows the law of demand.

2. The cost function

In economics and business, the **cost function** mathematically models the relationship between the total cost of production and the quantity of units produced. It helps businesses determine the most efficient scale of production and calculate profit margins.

A standard linear cost function is expressed as: $C(q) = F + vq$

Where:

$C(q)$ = Total Cost (TC) of producing q units.

F = Total Fixed Cost (FC). These are expenses that do not change with the level of output, such as rent, salaries, insurance, and machinery depreciation. It represents the vertical intercept on a graph.

v = Variable Cost per unit. This is the cost incurred for every single additional unit produced (e.g., raw materials, direct labor).

vq = Total Variable Cost (VC). This represents the total costs that scale directly with production volume.

q = Quantity of output produced.

We define the **average Cost (AC)**: The cost per unit of output, calculated as:

$$AC = \frac{C(q)}{q} = \frac{F}{q} + v$$

Example 2

If a firm faces the total cost function $TC = 6 + x^2$, where x is output, what is TC when x is

a) 14 b) 1 c) 0

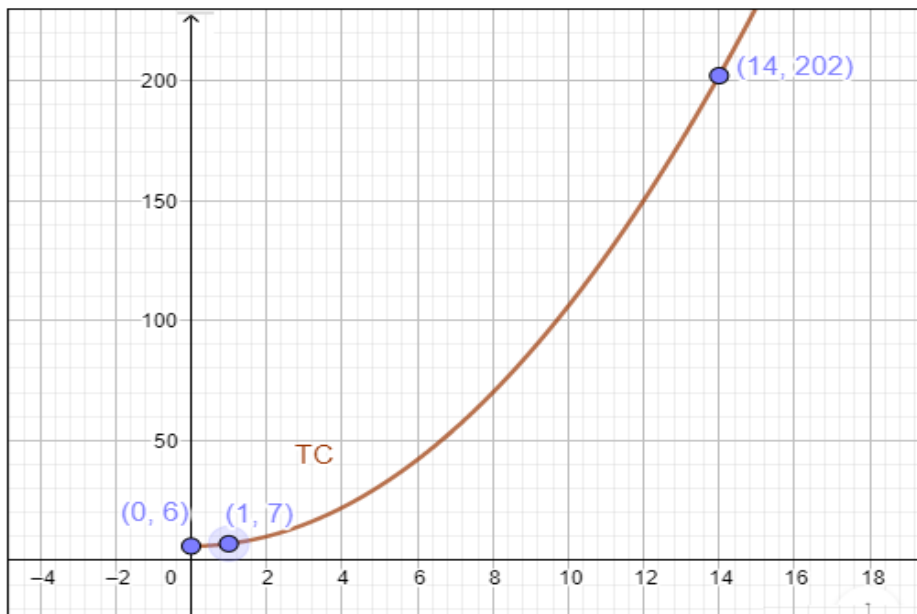
d) What restrictions on the domain of this function would it be reasonable to make?

Solution

a) When $x = 14$, the value of TC is 202

b) When $x = 1$, the value of TC is 7

c) When $x = 0$, the value of TC is 6,



Graph of the function $TC(x) = 6 + x^2$

(d) The restrictions on the domain to make the cost function $TC = 6 + x^2$ reasonable is to consider the values of $x \geq 0$

Example 3

A cooperative factory manufacturing school chalk sets up its operations. The monthly rent for the facility and machinery lease amounts to a fixed cost of 300,000 Frw . The raw materials (gypsum, water, pigments) and packaging required to produce a single box of chalk cost 400 Frw .

- 1) Formulate the linear total cost function, $C(q)$, for the factory.
- 2) Calculate the total cost of producing 1,500 boxes of chalk in a month.
- 3) Determine the average cost per box if 1,500 boxes are produced.
- 4) If the cooperative sells each box of chalk for 700Frw, calculate the **Break-even Quantity** (the number of boxes they must sell to cover all their costs).

Solution

- 1) Formulating the cost function

The Fixed Cost (F) = 300,000 and Variable Cost per unit (v) = 400

$$C(q) = 300,000 + 400q$$

- 2) Total Cost for $q = 1,500$

$$C(1,500) = 300,000 + 400(1,500)$$

$$C(1,500) = 900,000 \text{ Frw}$$

The total cost of production for 1,500 boxes is 900,000Frw.

- 3) Average Cost for 1,500 boxes

Divide the total cost by the number of units produced:

$$\text{Average Cost} = \frac{900,000}{1,500} = 600 \text{ Frw / box}$$

The average cost to produce one box drops to 600Frw due to the distribution of fixed costs across a larger volume.

- 4) Finding the Break-Even Quantity

To break even, Total Revenue (R) must equal Total Cost (C).

$$\text{Revenue Function: } R(q) = 700q$$

$$\text{Cost Function: } C(q) = 300,000 + 400q$$

$$\text{Set } R(q) = C(q) : 700q = 300,000 + 400q$$

Isolate q :

$$700q - 400q = 300,000$$

$$q = \frac{300,000}{300}$$

$$q = 1,000 \text{ boxes}$$

The cooperative must produce and sell exactly 1,000 boxes of chalk per month to cover all expenses without making a profit or loss.



Application activity 3.16

- 1) Write True or False and justify your answer.
 - a) In the function $P = 20 - 2Q_d$, the price decreases when the quantity demanded increases.
 - b) In a cost function $C = F + vQ$, the fixed cost changes when the quantity produced increases.
- 2) The cost function of producing a product is $C = 40 + 5Q$. Find the total cost when the quantity produced $Q = 6$.

3.17 The Revenue function and the profit function



Activity 3.17

A shop sells pens.

- The price of a pen and the number of pens sold determine the shop's revenue.
 - The shop also spends money to produce or buy the pens, which is the cost.
 - Profit is what is left after subtracting total cost from revenue.
- 1) How can we calculate the money a shop earns if it sells many pens at a certain price?
 - 2) What happens to profit if production costs are high?
 - 3) Can these relationships be written as mathematical functions?



Content Summary

(a) Revenue Function

Revenue is the total payment received from selling a good or performing a service. The **revenue function**, $R(q)$, gives the total revenue received by a firm from selling a quantity, q , of some good. If the good sells for a price of p per unit, and the quantity sold is q , then

Revenue = price $\times$ quantity or $R = pq$.

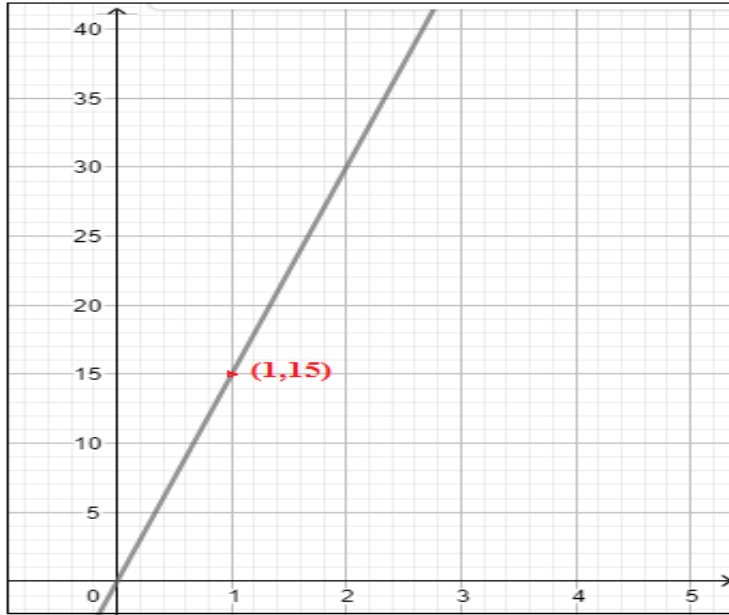
If the price does not depend on the quantity sold, so p is a constant, the graph of revenue as a function of q is a line through the origin, with slope equal to the price p .

Example 4

If radios sell for \$15 each, sketch the manufacturer's revenue function. Show the price of a radio on the graph.

Solution

Since $R(q) = pq = 15q$, the revenue graph is a line through the origin with a slope of 15. See the figure. The price is the slope of the line.



Example 5

Graph the cost function $C(q) = 24,000 + 7q$ and the revenue function $R(q) = 15q$ on the same axes. For what values of q does the company make money?

Solution

The company makes money whenever revenues are greater than costs, so we find the values of q for which the graph of $R(q)$ lies above the graph of $C(q)$.

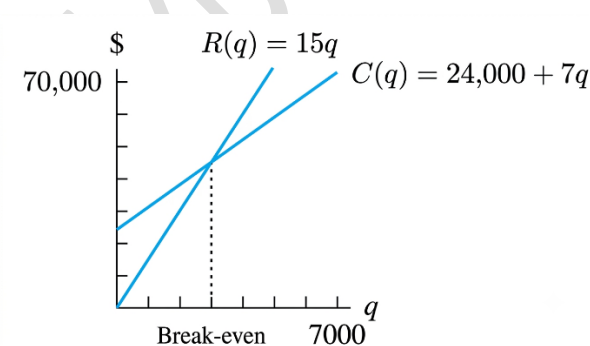
We find the point at which the graphs of $R(q)$ and $C(q)$ cross: **Revenue = Cost**

$$15q = 24000 + 7q$$

$$8q = 24000$$

$$q = 3000$$

The company makes a profit if it produces and sells more than 3 000 radios. The company loses money if it produces and sells fewer than 3 000 radios.



(b) Profit Function

The Profit Function $P(x)$ is the difference between the revenue function $R(x)$ and the total cost function $C(x)$. Thus $P(x) = R(x) - C(x)$

We have: **Profit = Revenue – Cost.**

The **break-even point** for a company is the point where the profit is zero and revenue equals cost.

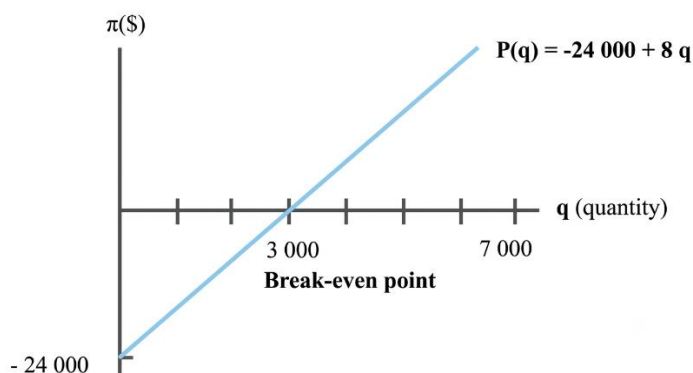
Example 6

Let's consider a company that makes radios. The factory and machinery needed to begin production with fixed costs which are incurred/earned even if no radios are made. The costs of labor and raw materials are variable costs since these quantities depend on how many radios are made. The fixed costs for this company are \$24,000 and the variable costs are \$7 per radio while each radio is sold for \$15. Find a formula for the profit function of the radio manufacturer. Graph it, marking the break-even point.

Solution

Since $R(q) = 15q$ and $C(q) = 24,000 + 7q$, we have

$$P(q) = R(q) - C(q) = 15q - (24,000 + 7q) = -2,400 + 8q$$



We see that the negative of the fixed costs is the vertical intercept and the break-even point is the horizontal intercept.



Application activity 3.17

1) Closed: Answer by True or False

- Revenue is calculated by multiplying the price by the quantity sold.
- Profit is calculated by subtracting revenue from cost.
- If total cost is greater than revenue, the business makes a loss.

2) Open Questions

- The cost function for producing these pens is $C = 50 + 5Q$. Find the profit if 8 pens are sold at 15 units each.
- In your own words, explain why profit depends on both revenue and cost.

3.18. The Marginal cost, Marginal revenue, and Marginal profit



Activity 3.18

A small bakery makes cakes.

- When the bakery produces one more cake, the cost, revenue, and profit change slightly.
 - Understanding how these change helps the bakery decide how many cakes to produce.
- 1) If producing one more cake costs more money, what do we call this extra cost?
 - 2) If selling one more cake brings extra revenue, what is that called?
 - 3) How can we calculate the extra profit from producing and selling one more unit?



Content Summary

(a) Marginal Cost (MC)

Marginal cost is the extra cost of producing one additional unit of a product.

$MC = \frac{\Delta C}{\Delta Q}$; Where, ΔC is the change in total cost and ΔQ is the change in quantity produced

Example

- 1) Total cost for 5 cakes = 50 units and Total cost for 6 cakes = 58 units

$MC = 58 - 50 = 8$. The marginal cost of the 6th cake is 8 units of money.

- 2) Imagine a cooperative that produces artisanal leather bags.

To produce **10 bags**, the total cost is **150,000 Frw**.

To produce **11 bags**, the total cost increases to **162,000 Frw**.

Marginal Cost = $162,000 - 150,000 = 12,000$ Frw

The marginal cost of the 11th bag is **12,000Frw**. This tells the producer exactly how much extra expenditure is required to increase production by one unit.

(b) Marginal Revenue (MR)

Marginal revenue is the extra money earned by selling one more unit.

$$MR = \frac{\Delta R}{\Delta Q}$$

Example

- 3) Revenue from selling 5 cakes = 100 units and Revenue from selling 6 cakes = 120 units

$MR = 120 - 100 = 20$.

The marginal revenue of the 6th cake is 20 units of money.

- 4) Using the same leather bag example, suppose the cooperative sells its products in a local market:

Selling **10 bags** generates a total revenue of **200,000Frw**.

Selling **11 bags** generates a total revenue of **220,000 Frw**.

$$\text{Marginal Revenue} = 220,000 - 200,000 = 20,000\text{Frw}$$

The marginal revenue of the 11th bag is **20,000 Frw** (which is also the market price per unit).

(c) Marginal Profit (MΠ)

Marginal profit is the extra profit from producing and selling one more unit.

$$M\Pi = \frac{\Delta\Pi}{\Delta Q} = MR - MC$$

Example:

3) Given, $MC = 8$ units and $MR = 20$ units. $M\Pi = 20 - 8 = 12$.

The extra profit from the 6th cake is 12 units.

4) Let's combine the figures calculated from the 11th leather bag:

Marginal Revenue (MR) = 20,000Hrw

Marginal Cost (MC) = 12,000 Frw

$$\text{Marginal Profit} = 20,000 - 12,000 = 8,000 \text{ Frw}$$

Since the marginal profit is positive (+8,000 Frw), producing and selling this 11th bag actively increases the cooperative's total accumulated profit.



Application activity 3.18

1) Closed questions: Answer by True or False

- Marginal cost is the extra cost of producing one more unit.
- Marginal revenue is the extra money earned from selling one more unit.
- Marginal profit equals marginal revenue minus marginal cost.

2) Open Questions

- A company produces 10 units. Total cost = 200, total revenue = 300. When it produces 11 units, total cost = 212, total revenue = 320. Calculate MC, MR
- Explain in your own words why marginal profit is important for deciding production level.



End of unit assessment

Assessing Knowledge and Understanding

Section A: True or False

1) A constant function has the same output for all input values.

- 2) The domain of a rational function includes all real numbers except where the denominator is zero.
- 3) In a linear function, the graph is always a straight line.
- 4) Marginal profit equals marginal revenue minus marginal cost.

Section B: Multiple Choice

- 1) If the quadratic function is $f(x) = x^2 - 4x + 3$, the value of $f(2)$ is:

A. -1 B. -2 C. 1 D. 0

- 2) Which of the following is an even function?

- a) $f(x) = x^3$
- b) $f(x) = x^2$
- c) $f(x) = x + 1$
- d) $f(x) = x^3 + x$

Assessing Skills

Section C: Open-Ended Questions

- 1) If a company has a cost function $C = 50 + 5Q$ and sells the product at 12 units each, the revenue from selling 10 units is : A. 50 B. 120 C. 100 D. 150
- 2) Given the function $f(x) = x^3 - x$
 - a) Determine whether the function is odd, even, or neither
 - b) Sketch a simple graph showing the general shape.

Assessing Attitudes and Values

- 1) Explain why

- 2) it is important for a business to calculate marginal cost and marginal revenue before making production decisions.
- 3) Explain how understanding functions (such as cost, revenue, and profit functions) helps in real-life decision making.



- Show accuracy, logical reasoning, and persistence when calculating terms and sums of sequences.
- Develop confidence and perseverance in solving both theoretical and applied sequence problems.

Introduction

In everyday life, many situations follow a pattern. For example, the arrangement of chairs in rows, the number of steps in a staircase, the amount of money saved regularly, or the growth of a population often follow a specific order. In mathematics, such ordered lists of numbers that follow a rule are called sequences. In this unit, learners will explore different types of number patterns, especially arithmetic sequences and geometric sequences. They will learn how to identify these patterns, determine the n^{th} term, calculate the sum of terms, and apply sequences to solve problems that arise in real-life situations.

Studying sequences helps learners develop logical thinking, problem-solving skills, and the ability to recognize patterns, which are important skills in mathematics and many real-world applications.



4.0. Introductory activity

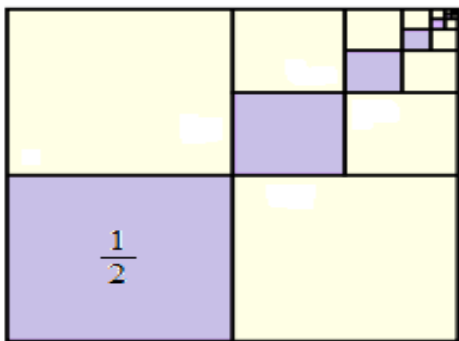
Suppose that an insect population is growing in such a way that each new generation is 2 times as large as the previous generation. If there are 126 insects in the first generation, on a piece of paper, write down the number of insects that will be there in second, third, fourth,... n^{th} generation. How can we name the list of the number of insects for different generations?

4.1. Definition of sequences



Activity 4.1

- 1) Fold the square paper once, what is the fraction that represents the coloured part?
- 2) Fold it twice. What is the fraction that represents the coloured part?
- 3) What is the fraction that represents the coloured part if you fold it ten times?
- 4) What is the fraction that represents the coloured part if you fold it n times?
- 5) Write a list of the fractions obtained starting from the first until the n^{th} fraction.



Content Summary

Let us consider the following list of numbers: $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n}, \dots$. The terms of this list are compared to the images of the function $f(x) = \frac{1}{x}$. The list never ends, as the ellipsis indicates.

The list is called a sequence, and the numbers in this ordered list are called the **terms** of the sequence. In dealing with sequences, we usually use subscripted letters, such as u_1 to represent the first term, u_2 for the second term, u_3 for the third term, and so on such as in the sequence

$f(n) = u_n = \frac{1}{n}$. However, in the sequence such as $\{u_n\} : u_n = \sqrt{n-3}$, the first term is u_3 as the previous are not possible, in the sequence $\{u_n\} : u_n = 2n-5$, the first term is u_0 .

Definition: A sequence is a function whose domain is a subset of the set of natural numbers. The terms of a sequence are the range elements of the function. A sequence is denoted $u_1, u_2, u_3, \dots, u_{n-1}, u_n$ shortly by $\{u_n\}$. We can also write $\{u_1, u_2, u_3, \dots, u_{n-1}, u_n\}$ for a finite sequence whose first term is u_1 . The numbers $u_1, u_2, u_3, \dots, u_{n-1}, u_n$ in a sequence are called **terms of the sequence**. The natural number n is called the **term number** and value u_n is called the **general term** of a sequence and the term u_1 is the **initial term** or **the first term**. If a sequence continues indefinitely, it can be denoted as $\{u_n\}_{n=1}^{\infty}$. The number of terms of a sequence (possibly infinite) is called the **length of the sequence**.

Notice:

- Sometimes, the term number, n , starts from 0. In this case, terms of a sequence are $u_0, u_1, u_2, \dots, u_{n-1}, u_n, \dots$ and this sequence is denoted by $\{u_n\}_{n=0}^{+\infty}$. In this case the initial term is u_0 .
- A sequence can be finite, like the sequence $2, 4, 8, 16, \dots, 256$.

A finite sequence can be empty. This particular case will not be considered in our study of sequences.

Usually a numerical sequence is given by some formula $u_n = f(n)$, permitting to find any term of the sequence in terms of its number n ; this formula is called a **general term formula**.

A second way of defining a sequence is to assign a value to the first (or the first few) term(s) and specify the n^{th} term by a formula or equation that involves one or more of the terms preceding it. Sequences defined this way are said to be defined **recursively**, and the rule or formula is called a **recursive formula**.

Example

The sequence $u_1 = 1, u_n = n.u_{n-1}$

Infinite and finite sequences

Consider the sequence of odd numbers less than 11: This is 1, 3, 5, 7, 9. This is a finite sequence as the list is limited and countable. However, the sequence made by all odd numbers is:

1, 3, 5, 7, 9, ..., $2n+1$. This suggests that an **infinite sequence** is a sequence whose terms are infinite.

Note that it is not always possible to give the numerical sequence by a general term formula; sometimes a sequence is given by description of its terms.

Examples

1) Give examples of numerical sequences.

Solution

0, 1, 2, 3, 4, 5, ... a sequence of natural numbers.

0, 2, 4, 6, 8, 10, ... a sequence of even numbers.

1.4, 1.41, 1.414, 1.4142, ... numerical sequence of approximate, defined more precisely values of $\sqrt{2}$.

For the last sequence it is impossible to give a general term formula, nevertheless this sequence is described completely.

2) List the first five terms of the sequence: $\{2^n\}_{n=1}^{+\infty}$

Solution

Here, we substitute $n = 1, 2, 3, 4, 5, \dots$ into the formula 2^n .

This gives $2^1, 2^2, 2^3, 2^4, 2^5, \dots$ or equivalently 2, 4, 8, 16, 32, ...

3) Express the following sequences in general notation

a) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots$

b) $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$

Solution

a) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots$

Beginning by comparing terms and term numbers:

Term number	1	2	3	4	...
Term	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{3}{4}$	$\frac{4}{5}$	...

In each term, the numerator is the same as the term number, and the denominator is one greater than the term number.

Thus, the n^{th} term is $\frac{n}{n+1}$ and the sequence may be written as $\left\{ \frac{n}{n+1} \right\}_{n=1}^{+\infty}$.

b) $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$

Beginning by comparing terms and term numbers:

Term number	1	2	3	4	...
Term	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	...

or

Term number	1	2	3	4	...
Term	$\frac{1}{2^1}$	$\frac{1}{2^2}$	$\frac{1}{2^3}$	$\frac{1}{2^4}$	...

In each term, the denominator is equal to the power of 2, where the exponent is the term number.

We observe that the n^{th} term is $\frac{1}{2^n}$ and the sequence may be written as: $\left\{ \frac{1}{2^n} \right\}_{n=1}^{+\infty}$.

4) A sequence is defined by $\{u_n\} : \begin{cases} u_0 = 1 \\ u_{n+1} = 3u_n + 2 \end{cases}$

Determine u_1, u_2 and u_3

Solution

Since $u_0 = 1$ and $u_{n+1} = 3u_n + 2$, replace n by 0, 1, 2 to obtain u_1, u_2, u_3 respectively.

$$\begin{array}{lll}
 n=0 & u_{0+1} = u_1 = 3u_0 + 2 & n=1 \quad u_{1+1} = u_2 = 3u_1 + 2 \quad n=2 \quad u_{2+1} = u_3 = 3u_2 + 2 \\
 & = 3 \times 1 + 2 & = 3 \times 5 + 2 \quad = 3 \times 17 + 2 \\
 & = 5 & = 17 \quad = 53
 \end{array}$$

Thus, $\begin{cases} u_1 = 5 \\ u_2 = 17 \\ u_3 = 53 \end{cases}$



Application activity 4.1

- 1) Consider the following sequence: 2, 4, 6, 8, 10,
 - (a) What pattern do you observe in this sequence?
 - (b) What will be the next two numbers in the sequence?
- 2) Which of the following sequences shows a regular pattern?

A. 2, 5, 9, 14, ... B. 4, 8, 12, 16, ... C. 7, 3, 10, 5, D. 9, 4, 9, 7,

Choose the correct answer and explain your choice.

- 3) Look at the following sequence: 3, 6, 12, 24,
 - a) How is each term obtained from the previous one?
 - b) Write the next term of the sequence

4.2. Arithmetic sequences



Activity 4.2

A school organizes a competition where the winner receives money in the following pattern:

- Day 1: 1000 RWF
- Day 2: 1500 RWF
- Day 3: 2000 RWF
- Day 4: 2500 RWF
- Day 5: 3,000 RWF

The reward increases by the same amount every day.

Answer the following questions

- 1) Write the amounts received during the first five days.
- 2) What is the difference between the reward of two consecutive days?
- 3) Predict the amount received on Day 6.
- 4) What do you notice about how the numbers change?



Content Summary

Let u_1 be an initial term of a sequence. If we add d successively to the initial term to find other terms, the difference between successive terms of a sequence is always the same number and the sequence is called **arithmetic**.

This sequence has the following term u_1 , $u_2 = u_1 + d$, $u_3 = u_2 + d = u_1 + 2d$,

$u_4 = u_3 + d = u_1 + 3d$, ..., $u_n = u_{n-1} + d = u_1 + (n-1)d$, ...

An **arithmetic sequence** may be defined recursively as $u_n = u_1 + (n-1)d$ where u_1 and d are real numbers. The number u_1 is the first term, and the number d is called the **common difference**.

Examples

The following sequences are arithmetic sequences:

Sequence $\{u_n\}$: 5, 8, 11, 14, 17, ...

Sequence $\{v_n\}$: 26, 31, 36, 41, 46, ...

Sequence $\{w_n\}$: 20, 18, 16, 14, 12

Common difference

The fixed number that bind each sequence together is called the **common difference**. Sometimes mathematicians use the letter d when referring to these types of sequences.

d can be calculated by subtracting any two consecutive terms in an arithmetic sequence. That is $d = u_{n+1} - u_n$ or $d = u_n - u_{n-1}$.

Note: If three consecutive terms are in an arithmetic sequence, the twice of the middle term is equal to the sum of the extreme terms. That is for an arithmetic sequence u_{n-1}, u_n, u_{n+1} , we have

$$2u_n = u_{n-1} + u_{n+1}.$$

Proof:

If u_{n-1}, u_n, u_{n+1} form an arithmetic sequence. Then, $u_{n+1} = u_n + d$ and $u_{n-1} = u_n - d$

Adding two equations, you get $u_{n+1} + u_{n-1} = 2u_n$.



Application activity 4.2

2) Answer by True or False

- The sequence 3, 6, 9, 12 is an arithmetic sequence.
- The sequence 2, 5, 10, 17 is an arithmetic sequence.

3) Multiple Choice

What is the common difference of the sequence: 7, 11, 15, 19 ?

A.2 B.3 C.4 D.5

4) Open Questions

- Find the 10^{th} term of the sequence: 6, 9, 12, 15, ...
- The first term of an arithmetic sequence is 8 and the common difference is 5.
Find the 12^{th} term.

4.3. n^{th} term of an arithmetic sequence



Activity 4.3

Write the following pattern on the board: 3, 7, 11, 15, 19,

Answer the following questions

- Write the next three terms of the sequence.
- What is the difference between consecutive terms?
- Which term will appear in the 10th position?



Content Summary

General term of an arithmetic sequence

The n^{th} term, u_n of an arithmetic sequence $\{u_n\}$ with common difference d and initial term u_1 is given by $u_n = u_1 + (n-1)d$ which is **the general term of an arithmetic sequence**. If the initial term is u_0 , then the general term of an arithmetic sequence becomes $u_n = u_0 + nd$.

Generally, if u_p is any p^{th} term of a sequence then the n^{th} term is given by $u_n = u_p + (n-p)d$

Examples

- 4, 6, 8 are three consecutive terms of an arithmetic sequence because $2 \times 6 = 4 + 8 \Leftrightarrow 12 = 12$
- If the first and tenth terms of an arithmetic sequence are 3 and 10, respectively, find the fiftieth term of the sequence.

Solution

$$u_1 = 3 \text{ and } u_{10} = 30 \text{ and } u_n = u_1 + (n-1)d,$$

$$u_{10} = u_1 + (10-1)d. \text{ Then, } 30 = 3 + 9d \Rightarrow d = 3$$

$$\text{Now, } u_{50} = u_1 + (50-1)d = 3 + 49 \times 3 = 150$$

Thus, The fiftieth term of the sequence is 150.

3) If the 3^{rd} term and the 8^{th} term of an arithmetic sequence are 5 and 15 respectively, find the common difference.

Solution

$$u_3 = 5, u_8 = 15$$

Using the general formula: $u_n = u_p + (n - p)d$

$$u_3 = u_8 + (3 - 8)d$$

$$5 = 15 - 5d \Rightarrow 5 - 15 = -5d$$

$$-10 = -5d \Leftrightarrow 5d = 10$$

$$d = \frac{10}{5} \Rightarrow d = 2$$

The common difference is 2 .

4) Consider the sequence 5, 8, 11, 14, 17, ..., 17 . Find the number of terms in this sequence .

Solution

We see that $u_1 = 5, u_n = 47$ and $d = 3$.

But we know that $u_n = u_1 + (n - 1)d$. This gives

$$47 = 5 + (n - 1)3 \Leftrightarrow 42 = 3n - 3 \Rightarrow n = 15$$

This means that there are 15 terms in the sequence.

5) Consider the sequence which went from 20 to -26 with -2 as common difference. Find the number of terms.

Solution

We have

$$-26 = 20 + (n - 1)(-2) \Leftrightarrow -46 = -2n + 2 \Rightarrow n = 24$$

This means that there are 24 terms in the sequence.

6) Show that the following sequence is arithmetic. Find the first term and the common difference. $\{s_n\} = \{3n + 5\}$

Solution:

The first term is $s_1 = 8$. The n^{th} term and the $(n - 1)^{st}$ term of the sequence $\{s_n\}$ are

The first term is $s_1 = 8$.

$$s_n = 3n + 5 \text{ and } s_{n-1} = 3(n - 1) + 5 = 3n + 2$$

Their difference d is $s_n - s_{n-1} = (3n + 5) - (3n + 2) = 3$.

Since, the difference of any two successive terms is the constant 3 , the sequence $\{s_n\}$ is arithmetic and the common difference is 3 .



Application activity 4.3

1) Answer by True or False

- a) The formula used to find the n^{th} term of an arithmetic sequence is

$$u_n = u_1 + (n-1)d$$

- b) The sequence 5, 10, 15, 20 has a common difference of 10.

2) Open Questions

- d) Find the 12th term of the arithmetic sequence

$$4, 8, 12, 16, \dots$$

- e) The first term of an arithmetic sequence is 6 and the common difference is 4.

Find the 15th term.

- f) Form an arithmetic progression of three numbers such that the sum of its terms is 30 and the sum of the squares of its terms is 332

4.4. Arithmetic mean



Activity 4.4

Write three numbers on the board: 4, , 10. Find the number that should be placed in the blank so that the numbers form a regular pattern with equal spacing.

Answer the following questions

- 1) What number should replace the blank so that the difference between consecutive numbers is the same?
- 2) What is the difference between the first and last number?
- 3) How can you find the middle number without guessing?



Content Summary

If three or more than three numbers form an arithmetic sequence, then all terms lying between the first and the last numbers are called **arithmetic means**. If B is arithmetic mean between A

and C , then $B = \frac{A+C}{2}$.

Let us see how to insert the k rms between two terms u_1 and u_n to form an arithmetic sequence:

u_1	...	...	...	...	u_n
-------	-----	-----	-----	-----	-------

The terms to be inserted are called **arithmetic means** between two terms u_1 and u_n .

This requires to form an arithmetic sequence of $n = k + 2$ terms whose the first term is u_1 and the last term is u_n .

While there are several methods, we will use our n^{th} term formula $u_n = u_1 + (n - 1)d$.

As u_1 and u_n are known, we need to find the common difference d taking $n = k + 2$ where k is the number of terms to be inserted and 2 stands for the first and last terms.

Example 1

Insert three arithmetic means between 7 and 23.

Solution

Here $k = 3$ and then $n = k + 2 = 5$, $u_1 = 7$ and $u_n = u_5 = 23$.

Then ; $u_5 = u_1 + (5 - 1)d \Leftrightarrow 23 = 7 + 4d \Rightarrow d = 4$

Now, insert the terms using $d = 4$, the sequence is 7, 11, 15, 19, 23.

Example 2

Insert five arithmetic means between 2 and 20.

Solution

Here $k = 5$ and then $n = k + 2 = 7$, $u_1 = 2$ and $u_n = u_7 = 20$.

Then; $u_7 = u_1 + (7 - 1)d \Leftrightarrow 20 = 2 + 6d \Rightarrow d = 3$

Now, insert the terms using $d = 3$, the sequence is 2, 5, 8, 11, 14, 17, 20.



Application activity 4.4

- 1) Insert 4 arithmetic means between -3 and 7
- 2) Insert 9 arithmetic means between 2 and 32
- 3) Between 3 and 54 , n terms have been inserted in such a way that the ratio of 8^{th} mean and $(n - 2)^{\text{th}}$ mean is $\frac{3}{5}$. Find the value of n .

4.5. Sum of n first terms of an arithmetic sequence



Activity 4.5

Consider a finite arithmetic sequence 2, 5, 8, 11, 14,

- a) What is its first term, the common difference d and the general term?
- b) Determine the sum s_6 (in function of 6 and the first term 2) of the first 6 terms for this sequence taking that for each $u_n = u_1 + (n - 1)d$ where for example $u_3 = u_1 + 2d$.

c) Try to generalize your results to determine the sum s_n for the first n terms of the arithmetic sequence $\{u_n\}$.



Content Summary

The sum of n first terms of a finite arithmetic sequence denoted by S_n , with initial term u_1 is given by

$$\begin{aligned} s_n &= u_1 + u_2 + u_3 + \dots + u_n \\ &= u_1 + (u_1 + d) + (u_1 + 2d) + \dots + (u_1 + (n-1)d) \\ &= \underbrace{(u_1 + u_1 + \dots + u_1)}_{n \text{ terms}} + (d + 2d + \dots + (n-1)d) \\ &= nu_1 + d[1 + 2 + 3 + \dots + (n-1)] = nu_1 + d\left[\frac{n(n-1)}{2}\right] \\ s_n &= nu_1 + \frac{n}{2}(n-1)d = \frac{n}{2}[2u_1 + (n-1)d] \\ &= \frac{n}{2}[u_1 + u_1 + (n-1)d] = \frac{n}{2}(u_1 + u_n) \quad \text{or} \quad s_n = \frac{n}{2}(u_1 + u_n) \end{aligned}$$

If the initial term is u_0 , the formula becomes $S_n = \frac{n+1}{2}(u_0 + u_n)$

Example 1

Calculate the sum of first 100 terms of the sequence

Solution

We see that the common difference is 2 and the initial term is $u_1 = 2$. We need to find $u_n = 100$.

$$u_{100} = 2 + (100-1)2 = 2 + 198 = 200$$

$$\text{Now, } S_{100} = \frac{100}{2}(u_1 + u_{100}) = 50(2 + 200) = 10100$$

Example 2

Find the sum of first k even integers ($k \neq 0$).

Solution

$$u_1 = 2 \text{ and } d = 2$$

$$u_n = u_k, \quad u_k = 2 + (k-1)2, \quad u_k = 2k$$

$$S_n = S_k, \quad S_k = \frac{k}{2}(2 + 2k), \quad S_k = k(1 + k)$$

Example 3

Find the sum: $60 + 64 + 68 + 72 + \dots + 120$

Solution

This is the sum s_n of an arithmetic sequence u_n whose first term is $u_1 = 60$ and whose common difference is $d = 4$. The n th term is u_n . We have $u_n = u_1 + (n-1)d$ and

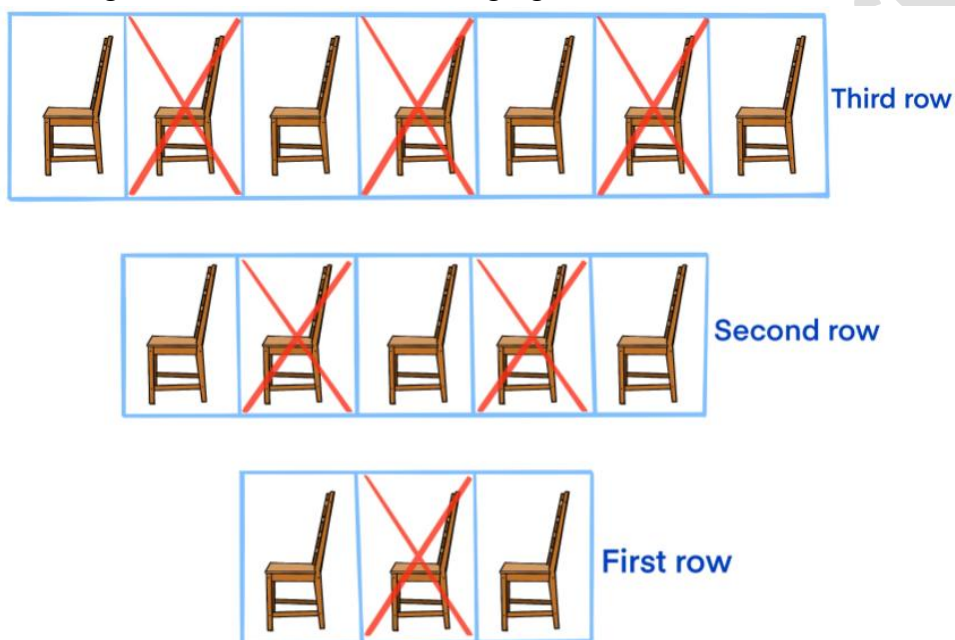
$$120 = 60 + (n-1)4, \quad 60 = 4(n-1), \quad n = 16$$

Now, the sum is

$$u_{16} = 60 + 64 + 68 + \dots + 120 = \frac{16}{2}(60 + 120) = 1440$$

Example 4

One of the measures for preventing the spread of Covid-19 is physical distancing. Consider a conference hall where people in the meeting are arranged in a way that chairs in each row are increasing by 2 from the previous line and between two people there must be a chair for social distancing as illustrated in the following figure.



Solution

Order of line (n)	Number of authorized seats (U_n)	Total number of authorized seats from the first line (S_n)	S_n in terms of U_1 and d
1	2	$S_1 = U_1 = 2$	$S_1 = U_1$
2	3	$S_2 = S_1 + U_2 = 5$	$S_2 = S_1 + U_2 = U_1 + U_1 + d = 2U_1 + d$
3	4	$S_3 = S_2 + U_3 = 9$	$S_3 = S_2 + U_3 = 2U_1 + d + U_1 + 2d = 3U_1 + 3d$
4	5	$S_4 = S_3 + U_4 = 14$	$S_4 = S_3 + U_4 = 3U_1 + 3d + U_1 + 3d = 4U_1 + 6d$
5	6	$S_5 = S_4 + U_5 = 20$	$S_5 = S_4 + U_5 = 4U_1 + 6d + U_1 + 4d = 5U_1 + 10d$

$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	$U_1 + 2(n-1)$	$S_n = S_{n-1} + U_n$	$S_n = u_1 + u_2 + u_3 + \dots + u_n$ $= u_1 + (u_1 + d) + (u_1 + 2d) + \dots + (u_1 + (n-1)d)$ $= \underbrace{(u_1 + u_1 + \dots + u_1)}_{n \text{ terms}} + (d + 2d + \dots + (n-1)d)$ $= nu_1 + d[1 + 2 + 3 + \dots + (n-1)] = nu_1 + d\left[\frac{n(n-1)}{2}\right]$ $= nu_1 + \frac{n}{2}(n-1)d = \frac{n}{2}[2u_1 + (n-1)d]$ $= \frac{n}{2}[u_1 + u_1 + (n-1)d]$ $= \frac{n}{2}(u_1 + u_n)$

Example 5

Odd numbers are numbers that are not multiples of 2. Assuming we take the natural numbers, the odd numbers among them would be 1, 3, 5, 7, etc. Find the sum of n first odd numbers.

Solution

There is a need to know the total number of authorized chairs if there are n rows in the room.

Position of odd number (n)	1	2	3	4	5	6	7	8	9
Odd number as generated from the figure	1	3	5	7	9	11	13	15	17
Sum of odd number	1^2								
	$1 + 3 = 2^2$								
	$1 + 3 + 5 = 3^2$								
	$1 + 3 + 5 + 7 = 4^2$								

	$1+3+5+7+9=5^2$				
	$1+3+5+7+9+11=6^2$				
	$1+3+5+7+9+11+13=7^2$				
	$1+3+5+7+9+11+13+15=8^2$				
	$1+3+5+7+9+11+13+15+17=9^2$				
The sum of n first odd number	$1+3+5+7+\dots+(2n-1)=n^2$				



Application activity 4.5

1. Answer by True or False

- The formula $S_n = \frac{n}{2}(u_1 + u_n)$ is used to find the sum of the first n terms of an arithmetic sequence.
- The sum of the first 4 terms of the sequence 2, 4, 6, 8 is 20.

2. Multiple Choice

- Find the sum of the first 5 terms of the sequence: 2, 4, 6, 8, ...
A. 20 B. 25 C. 30 D. 35
- In an arithmetic sequence, $u_1 = 5, d = 3, n = 4$.
the correct value of S_4 is: A. 26 B. 32 C. 38 D. 44

3. Open Questions

- Consider the arithmetic sequence 8, 12, 16, 20, Find the expression for S_n
- Find Sum the first twenty terms of the sequence 5, 9, 13,

4.6. Geometric sequence



Activity 4.6

A farmer plants trees in rows. The number of trees in each row follows this pattern:
2, 4, 8, 16, 32, ... Each row has twice as many trees as the previous row.

Answer the following questions

- 1) Write the next two terms of the sequence.
- 2) How is each term obtained from the previous term?
- 3) Is the difference between consecutive terms constant?
- 4) What operation is used to move from one term to the next?



Content Summary

Sequences of numbers that follow a pattern of multiplying a term by a fixed number r , from one term u_1 to the next, are called **geometric sequences**.

The following are examples of geometric sequences:

Sequence $\{u_n\}$: 5, 10, 20, 40, 80, ...

Sequence $\{v_n\}$: $2, 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$

Sequence $\{w_n\}$: 1, -2, 4, -8, 16

Common ratio

We can examine these sequences to greater depth, we must know that the fixed numbers that bind each sequence together are called the **common ratios**, denoted by the letter r . This means if u_1 is the first term, $u_2 = ru_1$; $u_3 = r^2u_1$; $u_4 = r^3u_1$; $u_n = r^{n-1}u_1$; ...

$u_2 = ru_1$; $u_3 = r^2u_1$; $u_4 = r^3u_1$; ... $u_n = r^{n-1}u_1$; ...

Example 1

Sequence: 3, 6, 12, 24, 48, ... ; Common ratio: $6 \div 3 = 2$, $12 \div 6 = 2$, $24 \div 12 = 2$ Therefore: $r = 2$

Example 2

Find the 6th term of the sequence: 2, 6, 18, 54, ...

Solution

First term $a_1 = 2$; Common ratio: $r = \frac{6}{2} = 3$; $a_6 = 2(3)^{6-1} \Rightarrow a_6 = 2(3)^5 \Rightarrow a_6 = 486$ Therefore,

the 6th term is 486 .



Application activity 4.6

1. Answer by True or False

- In a geometric sequence, each term is obtained by multiplying the previous term by a constant number.
- The sequence 4, 8, 16, 32 is a geometric sequence.

2. Multiple Choice

- The correct common ratio of the sequence: 5, 10, 20, 40. is: A. 2 B. 3 C. 4 D. 5
- Find the term of the sequence: 3, 9, 27,

3. Open Questions

- Find 7th term of the geometric sequence 2, 4, 8, 16,
- The first term of a geometric sequence is 5 and the common ratio is 3. Find the 6th term.

4.7. Nth term of a geometric sequence



Activity 4.7

A rabbit population doubles every month:

- Month 1: 3 rabbits
- Month 2: 6 rabbits
- Month 3: 12 rabbits
- Month 4: 24 rabbits

The farmer wants to know how many rabbits will be present in Month 7 without counting month by month.

Answer the following questions

- Write the sequence of the number of rabbits.
- How is each term obtained from the previous term?
- Can you find a pattern or formula to determine any month's population directly?
- How can we predict the 7th term without listing all previous terms?



Content Summary

The n^{th} term or the general term of a geometric sequence becomes $u_n = r^{n-1}u_1$.

If the first term is u_0 , then the general term of an geometric sequence becomes $u_n = r^n u_0$
 $u_n = r^n u_0$. The common ration r can be calculated by dividing any two consecutive terms in a
 geometric sequence. That is $r = \frac{u_{n+1}}{u_n}$ or $r = \frac{u_n}{u_{n-1}}$ or $u_n = r u_{n-1}$.

Generally, if u_p is the P^{th} term of the sequence, then the n^{th} term is given by $u_n = u_p r^{n-p}$.

Example 1

6,12,24 are consecutive terms of a geometric sequence because $(12)^2 = 6 \times 24 \Leftrightarrow 144 = 144$

Find b such that 8, b ,18 will be in geometric sequence.

Solution

$$b^2 = 8 \times 18 = 144$$

$$b = \pm \sqrt{144} = \pm 12$$

Thus, 8,12,18 or 8,-12,18 are in geometric sequence.

Example 2

The product of three consecutive numbers in geometric progression is 27. The sum of the first two and nine times the third is -79. Find the numbers.

Solution

Let the three terms be $\frac{x}{a}, x, ax$.

The product of the numbers is 27. So, $\frac{x}{a} x ax = 27 \Rightarrow x^3 = 27 \Rightarrow x = 3$

The sum of the first two and nine times the third is -79:

$$\frac{x}{a} + x + ax = -79 \Rightarrow \frac{3}{a} + 3 + 27a = -79$$

$$27a^2 + 82a + 3 = 0 \Rightarrow a = -3 \text{ or } a = -\frac{1}{27}$$

The numbers are: -1,3,-9 or -81,3,- $\frac{1}{9}$

Example 3

If the first and tenth terms of a geometric sequence are 1 and 4, respectively, find the nineteenth term.

Solution

$$u_1 = 1 \text{ and } u_{10} = 4$$

But $u_n = u_1 r^{n-1}$, then $4 = 1r^9 \Leftrightarrow r = \sqrt[9]{4} \text{ or } r = 4^{\frac{1}{9}}$

Now,

$$u_{19} = u_1 r^{19-1} = 1 \left(4^{\frac{1}{9}} \right)^{18} = 16$$

Thus, the nineteenth term of the sequence is 16.

Example 4

If the 2nd term and the 9th term of a geometric sequence are 2 and $-\frac{1}{64}$ respectively, find the common ratio.

Solution

$$u_2 = 2, u_9 = -\frac{1}{64}$$

Using the general formula: $u_n = u_1 r^{n-1}$

$$u_2 = u_1 r^{2-1}$$

$$\Rightarrow 2 = -\frac{1}{64} r^{-7} \Leftrightarrow 128 = -\frac{1}{r^7} \Leftrightarrow r^7 = -\frac{1}{128} \Leftrightarrow r = \sqrt[7]{-\frac{1}{128}} \Rightarrow r = -\frac{1}{2}$$

The common ratio is $r = -\frac{1}{2}$.

Example 5

Find the number of terms in sequence 2, 4, 8, 16, ..., 256.

Solution

This sequence is geometric with common ratio 2, $u_1 = 2$ and $u_n = 256$

But $u_n = u_1 r^{n-1}$, then, $256 = 2 \times 2^{n-1} \Leftrightarrow 256 = 2^n$ or $2^8 = 2^n \Rightarrow n = 8$.

Thus, the number of terms in the given sequence is 8.



Application activity 4.7

1) Answer by True or False

- The formula for the n th term of a geometric sequence is $a_n = a_1 r^{n-1}$.
- In the sequence 5, 15, 45, 135, the 4th term is 135.

2) Open Questions

- If the second and fifth terms of a geometric sequence are 6 and 48, respectively, find the sixteenth term.
- If the third term and the 8th term of a geometric sequence are $\frac{1}{2}$ and $\frac{1}{128}$ respectively, find the common ratio.
- The 4th term of a geometric sequence is square of its 2nd term, and the first term is -3 . Determine its 7th term.

4.8. Geometric mean



Activity 4.8

Write the following pattern on the board and form a geometric sequence: 2, __, 18

Answer the following questions

- 1) What number should replace the blank so that the numbers form a geometric sequence?
- 2) What number multiplies 2 to obtain the next term?
- 3) How can you check if the ratio between consecutive terms is the same?
- 4) What do we call the number between two numbers in a geometric sequence?



Content Summary

To insert k terms called **geometric means** between two terms u_1 and u_n is to form a geometric sequence of $n = k + 2$ terms whose the first term is u_1 and the last term is u_n .

While there are several methods, we will use our n^{th} term formula $u_n = u_1 r^{n-1}$.

As u_1 and u_n are known, we need to find the common ratio r taking $n = k + 2$ where k is the number of terms to be inserted.

Example 1

Insert three geometric means between 3 and 48.

Solution

Here $k = 3$, then $n = 5$, $u_1 = 3$ and $u_n = u_5 = 48$

$$u_5 = u_1 r^{n-1} \Leftrightarrow 48 = 3r^4 \Rightarrow r = 2$$

Inserting three terms using common ratio $r = 2$ gives 3, 6, 12, 24, 48

Example 2

Insert 6 geometric means between 1 and $-\frac{1}{128}$.

Solution

Here, $k = 6$, then $n = 8$, $u_1 = 1$ and $u_n = u_8 = -\frac{1}{128}$

$$u_8 = u_1 r^{n-1} \Leftrightarrow -\frac{1}{128} = 1r^7 \Leftrightarrow r^7 = -\frac{1}{128} \Rightarrow r^7 = -\frac{1}{(2)^7}$$

$$r = \sqrt[7]{-\frac{1}{(2)^7}} \rightarrow r = \left(-\frac{1}{(2)^7}\right)^{\frac{1}{7}} \rightarrow r = -\frac{1}{2}$$

Inserting 6 terms using common ratio $r = -\frac{1}{2}$ gives $1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16}, -\frac{1}{32}, \frac{1}{64}, -\frac{1}{128}$.



Application activity 4.8

1) Answer by True or False

- The geometric mean between two numbers a and b is $G = \sqrt{ab}$.
- The geometric mean between 5 and 20 is 10.

2) Multiple Choice

- The correct geometric mean between 9 and 25 is: A.12 B.15 C.18 D.20
- The correct geometric mean between 2 and 32 is: A.4 B.6 C.8 D.10

3) Open Questions

- Insert 5 geometric means between $\frac{1}{4}$ and $\frac{1}{256}$.
- Insert 5 geometric means between 2 and $\frac{2}{729}$.
- Find the geometric mean between 2 and 98.

4.9. Sum of n first terms of a geometric sequence



Activity 4.9

A science experiment shows the growth of bacteria:

- Day 1: 2 bacteria
- Day 2: 4 bacteria
- Day 3: 8 bacteria
- Day 4: 16 bacteria

The number of bacteria doubles every day. The scientist wants to know the total number of bacteria after 5 days.

Answer the following questions

- Write the sequence of bacteria counts.
- How is each term obtained from the previous term?
- How would you calculate the total number of bacteria after 5 days without adding each term one by one?
- What kind of formula can help you find this sum?



Content Summary

The sum S_n of the first n terms of a geometric sequence $\{u_n\} = \{u_1 r^{n-1}\}$ is

$$S_n = u_1 + ru_1 + \dots + r^{n-1}u_1 \quad (1)$$

Multiply each side by r to obtain $rS_n = ru_1 + r^2u_1 + \dots + r^nu_1 \quad (2)$

Subtracting equation (2) from equation (1) we obtain

$$S_n - rS_n = u_1 - u_1 r^n$$

$$(1-r)S_n = u_1(1-r^n)$$

Since, $r \neq 1$, we can solve for S_n and find $S_n = u_1 \frac{(1-r^n)}{(1-r)}$

If the initial term u_1 and the common ratio r are given, the sum $S_n = \frac{u_1(1-r^n)}{1-r}$ with $r \neq 1$

If the initial term is u_0 , then the formula is $S_n = \frac{u_0(1-r^{n+1})}{1-r}$ with $r \neq 1$

If $r = 1$, $S_n = nu_1$

Also, the product of first n terms of a geometric sequence with initial term u_1 and common ratio

r is given by $P_n = (u_1)^n r^{\frac{n(n-1)}{2}}$. If the initial term is u_0 then $P_n = (u_0)^{n+1} r^{\frac{n}{2}(n+1)}$

Example 1

Find the sum of the first 20 terms of a geometric sequence if the first term is 1 and common ratio is 2.

Solution

Here $u_1 = 1, r = 2, n = 20$

$$\text{Then, } S_{20} = \frac{1(1-2^{20})}{1-2} = \frac{1-2^{20}}{-1} = 1048575$$

Example 2

Consider the sequence $\{u_n\}$ defined by $u_0 = 0$ and $u_{n+1} = u_n + \frac{1}{2^n}$.

Consider another sequence $\{v_n\}$ defined by $v_n = u_{n+1} - u_n$.

- Show that $\{v_n\}$ is a geometric sequence and find its first term and common ratio.
- Express $\{v_n\}$ in term of n .

Solution

$$\text{a) } u_0 = 0 \quad v_0 = u_1 - u_0 = 1$$

$$u_1 = u_0 + \frac{1}{2^0} = 1, \quad u_2 = u_1 + \frac{1}{2^1} = \frac{3}{2};$$

$$v_1 = u_2 - u_1 = \frac{1}{2}, \quad v_2 = u_3 - u_2 = \frac{1}{4}$$

$\{v_n\}$ is a geometric sequence if $v_1^2 = v_0 \cdot v_2$. $v_1^2 = \frac{1}{4}$ and $v_0 \cdot v_2 = \frac{1}{4}$. Thus, $\{v_n\}$ is a geometric sequence. First term is $v_0 = 1$, Common ratio is $r = \frac{v_1}{v_0} = \frac{1}{2}$

b) General term: $v_n = v_0 r^n = \frac{1}{2^n}$

Thus, $\{v_n\}$ is defined by $v_n = \frac{1}{2^n}$

Example 3

Find the product of the first 20 terms of a geometric sequence if the first term is 1 and common ratio is 2.

Solution

$$P_n = (u_1)^n r^{\frac{n(n-1)}{2}}. \text{ Here, } u_1 = 1, r = 2, n = 20,$$

$$\text{Then, } P_{20} = (1)^{20} 2^{\frac{20(19)}{2}} = 2^{190}.$$



Application activity 4.9

1) Answer by True or False

- The formula for the sum of the first n terms of a geometric sequence is $s_n = u_1 \left(\frac{r^n - 1}{r - 1} \right)$.
- The sum of the first 3 terms of 2, 6, 18 is 24.

2) Multiple Choice

- Find the sum of the first 5 terms of the sequence 3, 6, 12, 24, 48.
A. 90 B. 93 C. 96 D. 99
- The first term of a geometric sequence is 2, the common ratio is 3, and $n = 4$. Find s_4 .

A. 80 B. 78 C. 90 D. 84

3) Open Question

Find the first term and the common ratio of the geometric sequence : $S_n = \frac{5^n - 4^n}{4^{n-1}}$

4.10. Application of sequences in solving real life problems



Activity 4.10

Carry out a research in the library or on internet and find out at least 3 problems or scenarios of the real life where sequences and series are applied.



Content summary 4.9

There are many applications of sequences. Sequences are useful in our daily lives as well as in higher mathematics. For example; the monthly payments made to pay off an automobile or home loan with interest portion, the list of maximum daily temperatures in one area for a month are sequences. Sequences are used in calculating interest, population growth, half-life and decay in radioactivity ...

In economics and Finance, sequences and series can be used for example in solving problems related to:

a) Final sum, the initial sum, the time period and the interest rate for an investment

The amount A after t year due to a principal P invested at an annual interest rate r

compounded n times per year is $A = P \cdot \left(1 + \frac{r}{n}\right)^{n \cdot t}$.

5) If the compounding takes place Annually, $n = 1$ and $A = P \cdot \left(1 + \frac{r}{1}\right)^t$

6) If the compounding takes place Monthly, $n = 12$ and $A = P \cdot \left(1 + \frac{r}{12}\right)^{12t}$

7) If the compounding takes place Daily, $n = 365$ and $A = P \cdot \left(1 + \frac{r}{365}\right)^{365t}$

In each case, the Interest due is $A - P$.

Note that From the compound interest formula $A = P \left(1 + \frac{r}{n}\right)^{nt}$ If we let $k = \frac{n}{r}$,

then $n = kr$ and $nt = krt$, and we may write the formula as $A = P \left(1 + \frac{1}{k}\right)^{krt} = P \left[\left(1 + \frac{1}{k}\right)^k\right]^{rt}$.

For continuously compound interest, we may let n (the number of interest period per year) increase without bound towards infinity ($n \rightarrow \infty$), equivalently, by $k \rightarrow \infty$.

Using the definition of the number e , we see that $P \left[\left(1 + \frac{1}{k}\right)^k\right]^{rt} \rightarrow P[e]^{rt}$ as $k \rightarrow \infty$.

This result gives us the following formula: $A = Pe^{rt}$ where $P = \text{Principal or initial value at } t = 0$; r is the Interest rate expressed as a decimal; t is the number of years p is invested; A is the amount after t years.

The amount A after t year due to a principal p invested at an annual interest rate r compounded continuously is $A = P \cdot e^{rt}$.

b) Annual Equivalent Rate for part year investments and the nominal annual rate of return

For loan repayments the annual equivalent rate is usually referred to as the annual percentage rate (APR). If you take out a bank loan you will usually be quoted an APR even though you will be asked to make monthly repayments.

The corresponding AER for any given monthly rate of interest i_m can be found using the formula : $AER = (1 + i_m)^{12} - 1$.

The APR on loans is the same thing as the annual equivalent rate and so the same formula applies.

The relationship between the daily interest rate i_d on a deposit account and the AER can be formulated as $AER = (1 + i_d)^{365} - 1$.

c) The Present value of investment

The present value P of A money to be received after t years, assuming a per annum interest rate r compounded n times per year, is $P = A \cdot \left(1 + \frac{r}{n}\right)^{-n \cdot t}$.

If the interest is compounded continuously, $P = A \cdot e^{-n \cdot t}$

d) Monthly repayments and the Annuity.

Suppose P is the deposit made at the end of each payment period for an annuity paying an interest rate of i per payment period.

The amount A of the annuity after n deposits is $A = P \cdot \left[\frac{(1+i)^n - 1}{i} \right]$

Example 1

A child building a tower with blocks uses 15 for the bottom row. Each row has 2 fewer blocks than the previous row. Suppose that there are 8 rows in the tower.

- How many blocks are used for the top row?
- What is the total number of blocks in the tower?

Solution

- The number of blocks in each row forms an arithmetic sequence with $u_1 = 15$ and $d = -2$
 $n = 8, u_8 = u_1 + (8-1)(-2)$. There is just one block in the top row.

- b) Here we must find the sum of the terms of the arithmetic sequence formed with

$$u_1 = 15, n = 8, u_8 = 1$$

$$S_8 = \frac{8}{2}(15+1) = 64$$

There are 64 blocks in the tower.

Example 2

An insect population is growing in such a way that each new generation is 1.5 times as large as the previous generation. Suppose there are 100 insects in the first generation.

- a) How many will there be in the fifth generation?
b) What will be the total number of insects in the five generations?

Solution

- a) The population can be written as a geometric sequence with $u_1 = 100$ as the first-generation population and common ratio $r = 1.5$. Then the fifth-generation population will be

$u_5 = 100(1.5)^{5-1} = 506.25$. In the fifth-generation, the population will number about 506 insects.

- b) The sum of the first five terms using the formula for the sum of the first n terms of a

geometric sequence.
$$S_5 = \frac{100(1 - (1.5)^5)}{1 - 1.5} = 1318.75$$

The total population for the five generations will be about 1319 insects.

Example 3

Find the accumulated value of \$15,000 at 5% per year for 18 years using simple interest.

Solution

$$P = 15000, r = 0.05, t = 18$$

$$I = 15000(0.05)(18) = 13500$$

A total of \$13,500 in interest will be earned.

Hence, the accumulated value in the account will be $13,500 + 15,000 = \$28,500$.

Example 4

Suppose 20,000 *frw* is deposit in a bank account that pays interest at a rate of 8% per year compound Continuously. Determine the balance in the account in 5 years.

Solution

Applying the formula for Continuously compound interest with $P = 20,000$; $r = 0.08$; and $t = 5$, we have $A = Pe^{rt} = 20,000e^{0.08(5)} = 20,000e^{0.4} = 29,836.49 \text{ Frw}$.

Example 5

Find the amount of an annuity after 5 deposits if a deposit of \$100 is made each year, at 4% compounded annually. How much interest is earned?

Solution

The deposit is $P = \$100$. The number of deposits is $n = 5$ and the interest per payment period is $i = 0.04$. Using the formula, the amount A after 5 deposits is

$$A = P \cdot \left[\frac{(1+i)^n - 1}{i} \right] = 100 \left[\frac{((1+0.04)^5 - 1)}{0.04} \right] = 541.63$$

The interest earned is the amount after 5 deposits less the 5 annual payments of \$100 each:

Interest earned = $A - 500 = 543.63 - 500 = 41.63$. It is \$41.63.

Example 6

Mary decides to put aside \$100 every month in a credit union that pays 5% compounded monthly. After making 8 deposits, how much money does Mary have?

Solution

This is an annuity with $P = \$100$, $n = 8$ deposits, and interest $i = \frac{0.05}{12}$ per payment period. Using the formula, the amount A after 8 deposits is

$$A = P \cdot \left[\frac{(1+i)^n - 1}{i} \right] = 100 \left[\frac{\left(\left(1 + \frac{0.05}{12} \right)^8 - 1 \right)}{\frac{0.05}{12}} \right] = 811.76$$

Mary has \$811.76 after making 8 deposits.

Example 7

To save for her daughter's college education, Martha decides to put \$50 aside every month in a bank guaranteed-interest account paying 4% interest compounded monthly.

She begins this savings program when her daughter is 3 years old. How much will she have saved by the time she makes the 180th deposit? How old is her daughter at this time?

Solution

This is an annuity with $P = \$50$, $n = 180$ deposits, and $i = \frac{0.04}{12}$. The amount A saved is

$$A = P \cdot \left[\frac{(1+i)^n - 1}{i} \right] = 50 \left[\frac{\left(\left(1 + \frac{0.04}{12} \right)^{180} - 1 \right)}{\frac{0.04}{12}} \right] = 12,304.52$$

Since there are 12 deposits per year, when the 180th deposit is made $\frac{180}{12} = 15$ have passed and

Martha's daughter is 18 years old.



Application activity 4.10

1) If Linda deposits \$1300 in a bank at 7% interest compounded annually, how much will be in the bank 17 years later?

- 2) The population of a city in 1970 was 153,800. Assuming that the population increases continuously at a rate of 5% per year, predict the population of the city in the year 2000.
- 3) To save for retirement, Manasseh, at age 35, decides to place 2000 *frw* into an Individual Retirement Account (*IRA*) each year for the next 30 years. What will the value of the *IRA* be when Manasseh makes his 30th deposit? Assume that the rate of return of the *IRA* is 4% per annum compounded annually.
- 4) A private school leader received permission to issue 4,000,000 *frw* in bonds to build a new high school. The leader is required to make payments every 6 months into a sinking fund paying 4% compounded semi-annually. At the end of 12 years the bond obligation will be retired. What should each payment be?



END OF UNIT ASSESSMENT

Assessing Knowledge and Understanding:

Section A: True or False

- 1) Arithmetic sequences always have a constant difference between consecutive terms.
- 2) The n th term of a geometric sequence is given by $a_n = a_1 + (n-1)r$.
- 3) The geometric mean between two numbers a and b is $\sqrt{ab}$

Section B: Multiple Choice Questions

- 1) Find the 8th term of the arithmetic sequence: 3, 7, 11, 15, ...
 A. 30 B. 31 C. 32 D. 33
- 2) A population triples every month. If initially there are 2 individuals, how many will there be in the 5th month?
 A. 32 B. 48 C. 162 D. 242
- 3) The sum of the first 6 terms of the arithmetic sequence 5, 8, 11, ... is:
 A. 69 B. 63 C. 66 D. 75

Assessing Skills:

- 1) Find first four terms of the sequence

a) $\left\{ \frac{1-n}{n^2} \right\}$ b) $\left\{ \frac{(-1)^{n+1}}{2n-1} \right\}$ c) $\{2 + (-1)^n\}$

2) Find the formula for the n^{th} term of the sequence

a) $1, -1, 1, -1, \dots$ b) $0, 3, 8, 15, 24, \dots$ c) $1, 5, 9, 13, 17, \dots$

3) Which of the following sequences converge, and which diverge? Find the limit of each

convergent sequence.: a) $\left\{ \sqrt{\frac{2n}{n+1}} \right\}$ b) $\left\{ \frac{n}{2^n} \right\}$ c) $\left\{ 8^{\frac{1}{n}} \right\}$

Assessing Attitudes and Values:

- 1) Imagine a savings plan where your money grows as a geometric sequence versus a plan where it grows as an arithmetic sequence. How does understanding these patterns help you make better financial decisions for your future?
- 2) While finding the n^{th} term of a sequence, why is it important to test your formula against multiple terms (like $n = 1, n = 2$ and $n = 3$ instead of just the first one)?

BIBLIOGRAPHY

1. J. Sadler, D. W. S. Thorning (1987). *Understanding Pure Mathematics*, Oxford University.
2. Adam, A., Goossens, F., & Lousberg, F. (1991). *Mathématisons 65* (3e éd.). De Boeck.
3. Hodgman, C. D., Selby, S. M., & Weast, R. C. (1959). *Standard mathematical table*. Chemical Rubber Publishing Company.
4. Rayner, D. (2000). *Higher GCSE mathematics*. Oxford University Press.
5. Direction des Programmes de l'Enseignement Secondaire. (1988). *Géométrie de l'espace* (1er fascicule). Kigali.
6. Direction des Programmes de l'Enseignement Secondaire. (1988). *Géométrie de l'espace* (2ème fascicule). Kigali.
7. Ebos, F., Hamaguchi, D., Morrison, B., & Klassen, J. (1990). *Mathematics principles and process*. Nelson Canada.
8. Thomas, G. B., Weir, M. D., & Hass, J. R. (2010). *Thomas' calculus* (12th ed.). Pearson Education.
9. Mannall, G., & Kenwood, M. (1995). *Pure mathematics 2*. Heinemann Educational Publishers.
10. Dass, H. K. (2007). *Engineering mathematics* (13th rev. ed.). S. Chand & Company.
11. Carnec, H., Haye, G., Nouet, M., Seroux, R., & Venard, J. (1994). *Mathématiques TS: Enseignement obligatoire*. Bordas.
12. McClave, J. T., & Benson, P. G. (1988). *Statistics for business and economics*. Dellen Publishing.
13. Crawshaw, J., & Chambers, J. (1984). *A concise course in A-level statistics with worked examples*. Stanley Thornes.
14. Beltramonde, J. P., Brun, V., Felloneau, C., Misset, L., & Talamoni, C. (2005). *Declic Ire S: Mathématiques*. Hachette-Éducation.
15. Talber, J. F., & Heing, H. H. (1995). *Additional mathematics* (6th ed.). Pearson Education South Asia.
16. Backhouse, J. K., Houldsworth, S. P. T., Copper, B. E. D., & Horril, P. J. F. (1985). *Pure mathematics 2* (3rd ed.). Longman.

17. Mukasonga, S. (2006). *Mathématiques 12: Analyse numérique*. Kigali Institute of Education(KIE).
18. Piskounov, N (1980) *Calcul Différentiel et Integral tom II* (9ème éd.). Editions MIR. Moscou.
19. Faure, P., & Bouchon, B. (1992). *Mathématiques terminales F*. Éditions Nathan.
20. Smythe, P. (2005). *Mathematics HL & SL with HL options* (Rev. ed.). Mathematics Publishing Pty. Limited.
21. Adams, R. A., & Essex, C. (2010). *Calculus: A complete course* (7th ed.). Pearson Canada.
22. Lipschutz, S. (1966). *Schaum's outline of theory and problems of finite mathematics*. Schaum Publishing.
23. "Lipschutz, S. (1968). *Schaum's outline of theory and problems of linear algebra*. McGraw-Hill."
24. Shampiyona, A. (2005). *Mathématiques 6*. Kigali.
25. Noirot, Y., Parisot, J.-P., & Brouillet, N. (1997). *Cours de physique mathématiques pour la physique*. Dunod.
26. Swokowski, E. W. (1994). *Pre-calculus: Functions and graphs* (7th ed.). PWS Publishing Company.
27. Allan, G. B. (2007). *Elementary statistics: A step by step approach* (7th ed.). Von Hoffmann Press.
28. "David, R. (2000). *Higher GCSE mathematics: Revision and practice*. Oxford University Press."
29. Ngezahayo, E. (2016). *Subsidiary mathematics for Rwanda secondary schools: Learner's book 4*. Fountain Publishers.
30. Rwanda Basic Education Board (REB). (2015). *Subsidiary mathematics syllabus*. MINEDUC.
31. "Rwanda Basic Education Board (REB). (2019). *Mathematics syllabus for TTC-Option of LE*. MINEDUC."
32. Peter, S. (2005). *Mathematics HL & SL with HL options* (Rev. ed.). Mathematics Publishing Pty. Limited.

33. Elliot, M. (1998). *Schaum's outline series of calculus*. McGraw-Hill Companies.
34. Frank, E., et al. (1990). *Mathematics*. Nelson Canada.
35. Gilbert, J. C., et al. (2006). *Glencoe advanced mathematical concepts*. McGraw-Hill Companies.
36. "Robert, A. (2006). *Calculus: A complete course* (6th ed.). Pearson Education Canada."
37. Sadler, A. J., & Thorning, D. W. (1997). *Understanding pure mathematics*. Oxford University Press.
38. Crawshaw, J., & Chambers, J. (2001). *A concise course in advanced level statistics with worked examples* (4th ed.). Nelson Thornes.
39. Larson, R., & Falvo, D. C. (2009). *Brief calculus: An applied approach*. Houghton Mifflin.
40. Sullivan, M. (2012). *Algebra and trigonometry* (9th ed.). Pearson Education.
41. "Swokowski, E. W., & Cole, J. A. (1992). *Precalculus: Functions and graphs* (7th ed.).
42. Glencoe. (2006). *Advanced mathematical concepts: Precalculus with applications*.
43. Lipschutz, S., & Lipson, M. (2007). *Discrete mathematics* (3rd ed.).
44. Stroud, K. A. (2001). *Engineering mathematics* (5th ed.). Industrial Press.
45. Bird, J. (2005). *Basic engineering mathematics* (4th ed.). Linacre House.